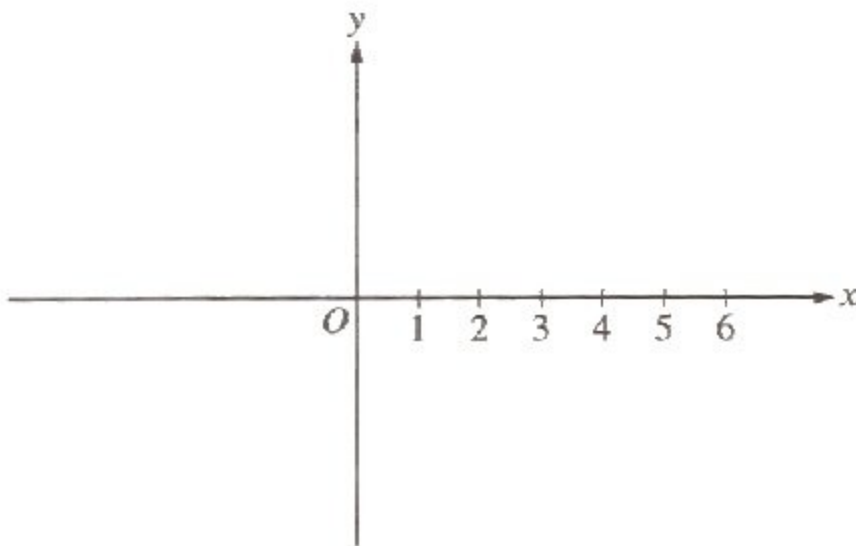


8-4b

1.  Let f be the function given by $f(x) = \sqrt{x - 3}$.

(a) On the axes provided below, sketch the graph of f and shade the region R enclosed by the graph of f , the x -axis, and the vertical line $x = 6$.



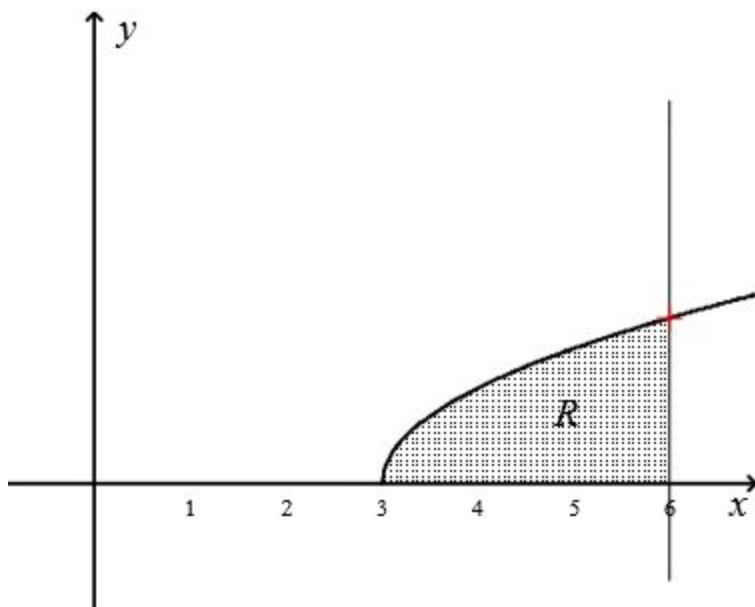
- (b) Find the area of the region R described in part (a).
- (c) Rather than using the line $x = 6$ as in part (a), consider the line $x = w$, where w can be any number greater than 3. Let $A(w)$ be the area of the region enclosed by the graph of f , the x -axis, and the vertical line $x = w$. Write an integral expression for $A(w)$.
- (d) Let $A(w)$ be as described in part (c). Find the rate of change of A with respect to w when $w = 6$.

Part A

1 point is earned for the correct graph of f , (domain is $x \geq 3$ goes through $(3,0)$, is increasing, positive, and concave down)

1 point is earned for the correct region relative to graph of f

8-4b



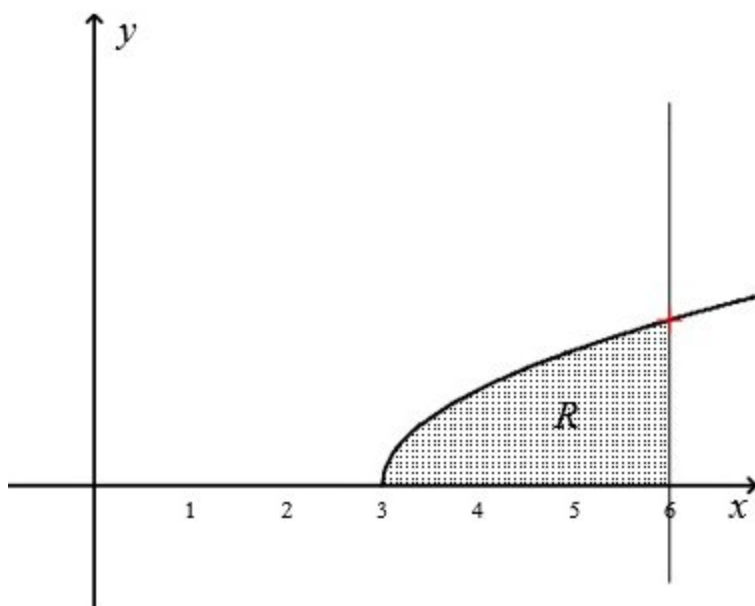
✓

0	1	2
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The student response earns all of the following points:

1 point is earned for the correct graph of f , (domain is $x \geq 3$ goes through (3,0), is increasing, positive, and concave down)

1 point is earned for the correct region relative to graph of f

**Part B**

8-4b

1 point is earned for the correct limits

1 point is earned for the correct integrand

$$\text{area} = \int_3^6 \sqrt{x-3} \, dx = \left. \frac{2}{3}(x-3)^{3/2} \right|_3^6$$

1 point is earned for the correct answer

0/1 if second point is not earned

$$= 2\sqrt{3} = 3.464$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct limits

1 point is earned for the correct integrand

$$\text{area} = \int_3^6 \sqrt{x-3} \, dx = \left. \frac{2}{3}(x-3)^{3/2} \right|_3^6$$

1 point is earned for the correct answer

0/1 if second point is not earned

$$= 2\sqrt{3} = 3.464$$

Part C

1 point is earned for the correct limits

1 point is earned for the correct integrand

$$A(w) = \int_3^w \sqrt{x-3} \, dx$$

8-4b



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct limits

1 point is earned for the correct integrand

$$A(w) = \int_3^w \sqrt{x-3} \, dx$$

Part D

1 point is earned for the correct answer for $\frac{dA}{dw}$

$$\frac{dA}{dw} = \sqrt{w-3}$$

1 point is earned for the correct evaluation at 6

0/2 if $\frac{dA}{dw}$ is constant

$$\left. \frac{dA}{dw} \right|_{w=6} = \sqrt{3} = 1.732$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct answer for $\frac{dA}{dw}$

$$\frac{dA}{dw} = \sqrt{w-3}$$

1 point is earned for the correct evaluation at 6

0/2 if $\frac{dA}{dw}$ is constant

8-4b

$$\left. \frac{dA}{dw} \right|_{w=6} = \sqrt{3} = 1.732$$

Let f be the function given by $f(x) = \sqrt{x-3}$.

2. Find the area of the region R described in part (a).

Part B

1 point is earned for the correct limits

1 point is earned for the correct integrand

$$\text{area} = \int_3^6 \sqrt{x-3} dx = \frac{2}{3} (x-3)^{3/2} \Big|_3^6$$

1 point is earned for the correct answer

0/1 if second point is not earned

$$= 2\sqrt{3} = 3.464$$



0	1	2	3
---	---	---	---

The student response earns none of the following points:

1 point is earned for the correct limits

1 point is earned for the correct integrand

$$\text{area} = \int_3^6 \sqrt{x-3} dx = \frac{2}{3} (x-3)^{3/2} \Big|_3^6$$

1 point is earned for the correct answer

0/1 if second point is not earned

$$= 2\sqrt{3} = 3.464$$

8-4b

3. Rather than using the line $x = 6$ as in part (a), consider the line $x = w$, where w can be any number greater than 3. Let $A(w)$ be the area of the region enclosed by the graph of f , the x -axis, and the vertical line $x = w$. Write an integral expression for $A(w)$.
-

Part C

1 point is earned for the correct limits

1 point is earned for the correct integrand

$$A(w) = \int_3^w \sqrt{x-3} dx$$



0	1	2
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The student response earns two of the following points:

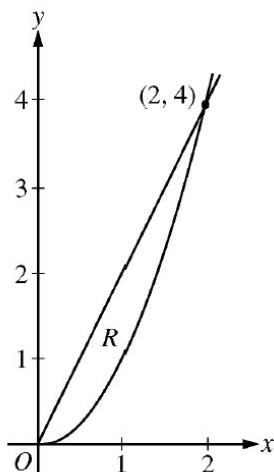
1 point is earned for the correct limits

1 point is earned for the correct integrand

$$A(w) = \int_3^w \sqrt{x-3} dx$$

8-4b

4.



Let R be the region in the first quadrant enclosed by the graphs of $y=2x$ and $y=x^2$, as shown in the figure above.

Find the area of R .

Part A

1 point is earned for the integrand

1 point is earned for the anti derivative

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^2 (2x - x^2) dx \\ &= x^2 - \frac{1}{3}x^3 \Big|_{x=0}^{x=2} \\ &= \frac{4}{3} \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for the integrand

1 point is earned for the anti derivative

1 point is earned for the answer

8-4b

$$\begin{aligned} \text{Area} &= \int_0^2 (2x - x^2) dx \\ &= x^2 - \frac{1}{3}x^3 \Big|_{x=0}^{x=2} \\ &= \frac{4}{3} \end{aligned}$$

8-4b

5. Let R be the region in the first quadrant enclosed by the graph of $y = \sqrt{6x + 4}$, the line $y = 2x$, and the y -axis.

Find the area of R .

Part A

1 point is earned for limit

2 point are earned for integrand

1 point is earned for antiderivative

1 point is earned for $x=3$

$$\begin{aligned} \text{Area} &= \int_0^2 \sqrt{6x + 4} - 2x dx \\ &= \frac{1}{6} \cdot \frac{2}{3} (6x + 4)^{3/2} - x^2 \Big|_0^2 \\ &= \left(\frac{64}{9} - 4 \right) - \frac{8}{9} = \frac{20}{9} \end{aligned}$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns five of the following points:

1 point is earned for limit

2 point are earned for integrand

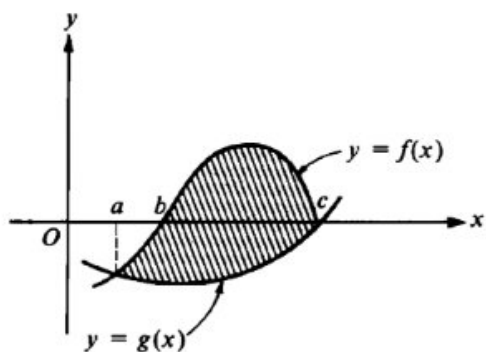
1 point is earned for antiderivative

1 point is earned for $x=3$

8-4b

$$\begin{aligned} \text{Area} &= \int_0^2 \sqrt{6x+4} - 2x \, dx \\ &= \frac{1}{6} \cdot \frac{2}{3} (6x+4)^{3/2} - x^2 \Big|_0^2 \\ &= \left(\frac{64}{9} - 4 \right) - \frac{8}{9} = \frac{20}{9} \end{aligned}$$

6.

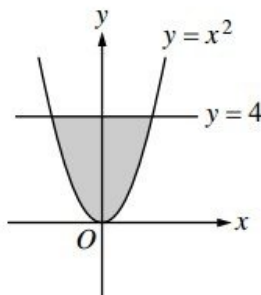


The area of the shaded region in the figure above is represented by which of the following integrals?

- (A) $\int_a^c (|f(x)| - |g(x)|) dx$
- (B) $\int_b^c f(x) dx - \int_a^c g(x) dx$
- (C) $\int_a^c (g(x) - f(x)) dx$
- (D) $\int_a^c (f(x) - g(x)) dx$ ✓
- (E) $\int_a^b (g(x) - f(x)) dx + \int_b^c (f(x) - g(x)) dx$

8-4b

7.



The shaded region, R , is bounded by the graph of $y = x^2$ and the line $y = 4$, as shown in the figure above. Find the area of R .

Part A

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned}
 \text{Area} &= \int_{-2}^2 (4 - x^2) dx \\
 &= 2 \int_0^2 (4 - x^2) dx \\
 &= 2 \left[4x - \frac{x^3}{3} \right]_0^2 \\
 &= \frac{32}{3} = 10.66 \text{ or } 10.667
 \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\text{Area} = \int_{-2}^2 (4 - x^2) dx$$

8-4b

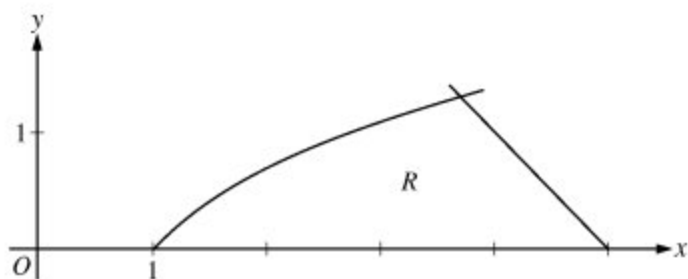
$$= 2 \int_0^2 (4 - x^2) dx$$

$$= 2 \left[4x - \frac{x^3}{3} \right]_0^2$$

$$= \frac{32}{3} = 10.66 \text{ or } 10.667$$

8-4b

8.



Let R be the region in the first quadrant bounded by the x -axis and the graphs of $y = \ln x$ and $y = 5 - x$, as shown in the figure above.



Find the area of R .

Part A

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^B (5 - y - e^y) dy \\ &= 2.986 \text{ (or } 2.985) \end{aligned}$$

OR

$$\begin{aligned} \text{Area} &= \int_1^A \ln x dx + \int_A^5 (5 - x) dx \\ &= 2.986 \text{ (or } 2.985) \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

8-4b

$$\begin{aligned} \text{Area} &= \int_0^B (5 - y - e^y) dy \\ &= 2.986 (\text{or } 2.985) \end{aligned}$$

OR

$$\begin{aligned} \text{Area} &= \int_1^A \ln x dx + \int_A^5 (5 - x) dx \\ &= 2.986 (\text{or } 2.985) \end{aligned}$$

9. The area of the region between the graph of $y = 4x^3 + 2$ and the x -axis from $x = 1$ to $x = 2$ is

(A) 36

(B) 23

(C) 20

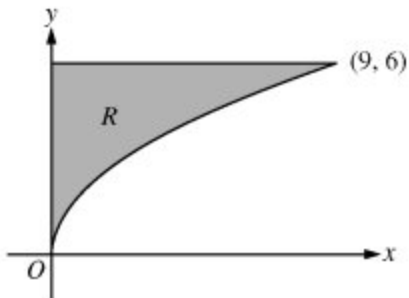
(D) 17

(E) 9



8-4b

10.



Let R be the region in the first quadrant bounded by the graph of $y = 2\sqrt{x}$, the horizontal line $y = 6$, and the y -axis, as shown in the figure above.

Find the area of R .

Part A

1 point is earned for integrand

1 point is earned for the antiderivative

1 point is earned for the answer

$$\text{Area} = \int_0^9 (6 - 2\sqrt{x}) dx = \left(6x - \frac{4}{3}x^{\frac{3}{2}} \right) \bigg|_{x=0}^{x=9} = 18$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for the antiderivative

1 point is earned for the answer

$$\text{Area} = \int_0^9 (6 - 2\sqrt{x}) dx = \left(6x - \frac{4}{3}x^{\frac{3}{2}} \right) \bigg|_{x=0}^{x=9} = 18$$

8-4b

11. Let R be the region in the first quadrant bounded by the graphs of $y = \sqrt{x}$ and $y = \frac{x}{3}$.

Find the area of R .

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

The graphs of $y = \sqrt{x}$ and $y = \frac{x}{3}$ intersect at the points $(0,0)$ and $(9, 3)$.

$$\int_0^9 \left(\sqrt{x} - \frac{x}{3} \right) dx = 4.5$$

OR

$$\int_0^3 (3y - y^2) dy = 4.5$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

The graphs of $y = \sqrt{x}$ and $y = \frac{x}{3}$ intersect at the points $(0,0)$ and $(9, 3)$.

$$\int_0^9 \left(\sqrt{x} - \frac{x}{3} \right) dx = 4.5$$

OR

$$\int_0^3 (3y - y^2) dy = 4.5$$

8-4b

12. The area of the region enclosed by the graphs of $y=x$ and $y = x^2 - 3x + 3$ is

- (A) $\frac{2}{3}$
 (B) 1
 (C) $\frac{4}{3}$
 (D) 2
 (E) $\frac{14}{3}$



13. Let R be the region in the first and second quadrants bounded above by the graph of $y = \frac{20}{1+x^2}$ and below by the horizontal line $y = 2$.



Find the area of R .

Part A

1 point is earned for the correct limits in an integral in (a), (b), or (c)

1 point is earned for the integrand

1 point is earned for the answer

$$\frac{20}{1+x^2} = 2, \text{ when, } x = \pm 3$$

$$\text{Area} = \int_{-3}^3 \left(\frac{20}{1+x^2} - 2 \right) dx = 37.961, \text{ or, } 37.962$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct limits in an integral in (a), (b), or (c)

1 point is earned for the integrand

1 point is earned for the answer

$$\frac{20}{1+x^2} = 2, \text{ when, } x = \pm 3$$

$$\text{Area} = \int_{-3}^3 \left(\frac{20}{1+x^2} - 2 \right) dx = 37.961, \text{ or, } 37.962$$

8-4b

Let R be the region in the first quadrant under the graph of $y = \frac{1}{\sqrt{x}}$ for $4 \leq x \leq 9$.

14. Find the area of R .

Part A

1 point is earned for the correct limits in x

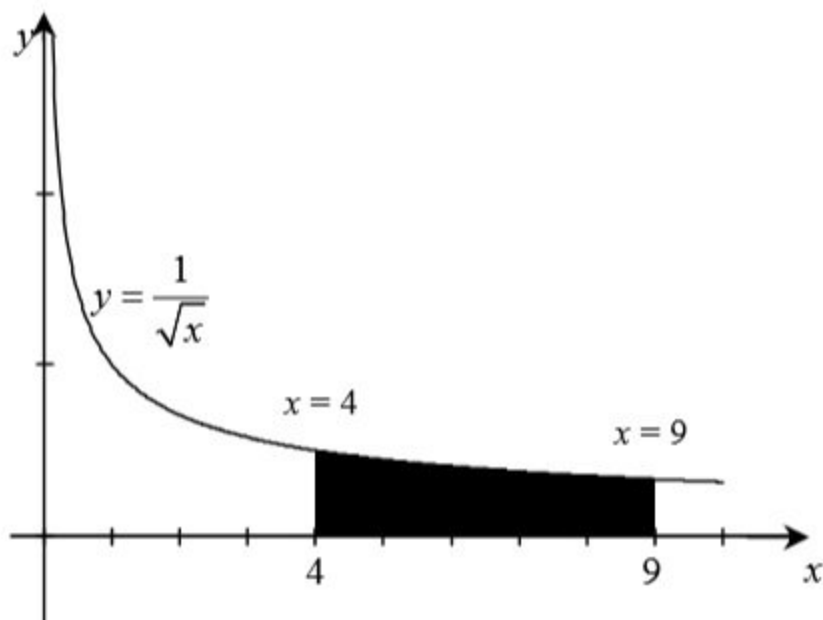
1 point is earned for the correct integrand including constant

must be $\frac{1}{\sqrt{x}}$

1 point is earned for the correct answer

0/1 if the integrand is incorrect, wrong limits are read

$$\int_4^9 \frac{dx}{\sqrt{x}} = 2$$



✓

0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for the correct limits in x

1 point is earned for the correct integrand including constant

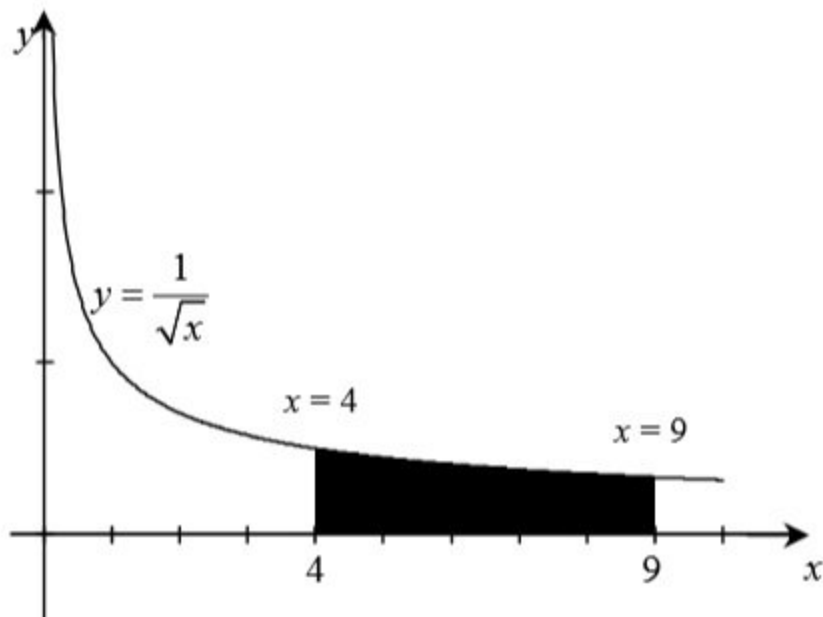
8-4b

must be $\frac{1}{\sqrt{x}}$

1 point is earned for the correct answer

0/1 if the integrand is incorrect, wrong limits are read

$$\int_4^9 \frac{dx}{\sqrt{x}} = 2$$



8-4b

15. If the line $x=k$ divides the region R into two regions of equal area, what is the value of k ?
-

Part B

1 point is earned for the correct demonstration of variable limit using k as upper or lower limit with 4 or 9 (or imported limit)

$$\int_4^k \frac{dx}{\sqrt{x}} \text{ or } \int_k^9 \frac{dx}{\sqrt{x}}$$

1 point is earned for the correct equation involving the two halves of R that when solved will give answer with demonstrated algorithm that knows when to stop

1 point is earned for the correct answer

0/1 if answer from equation not involving relevant areas

1/3 bald answer

$$\int_4^k \frac{dx}{\sqrt{x}} = 1$$

$$2\sqrt{x} \Big|_4^k = 1$$

$$2\sqrt{k} - 2\sqrt{4} = 1$$

$$k = \frac{25}{4}$$

$$\left(\text{or } \int_k^9 \frac{dx}{\sqrt{x}} = 1 \text{ or } \int_4^k \frac{dx}{\sqrt{x}} = \int_k^9 \frac{dx}{\sqrt{x}} \right)$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for the correct demonstration of variable limit using k as upper or lower limit with 4 or 9 (or imported limit)

$$\int_4^k \frac{dx}{\sqrt{x}} \text{ or } \int_k^9 \frac{dx}{\sqrt{x}}$$

1 point is earned for the correct equation involving the two halves of R that when solved will give answer with demonstrated algorithm that knows when to stop

1 point is earned for the correct answer

8-4b

0/1 if answer from equation not involving relevant areas

1/3 bald answer

$$\int_4^k \frac{dx}{\sqrt{x}} = 1$$

$$2\sqrt{x}]_4^k = 1$$

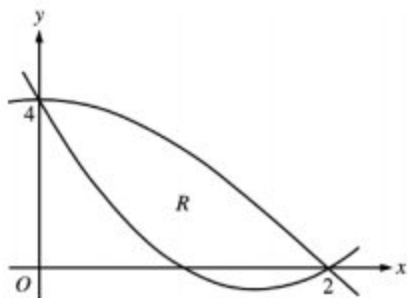
$$2\sqrt{k} - 2\sqrt{4} = 1$$

$$k = \frac{25}{4}$$

$$\left(\text{or } \int_k^9 \frac{dx}{\sqrt{x}} = 1 \text{ or } \int_4^k \frac{dx}{\sqrt{x}} = \int_k^9 \frac{dx}{\sqrt{x}} \right)$$

8-4b

16.



Let $f(x) = 2x^2 - 6x + 4$ and $g(x) = 4\cos(1/4 \pi x)$. Let R be the region bounded by the graphs of f and g , as shown in the figure above.

Find the area of R .

Part A

1 point is earned for the integrand

2 points are earned for the antiderivative

1 point is earned for the answer

$$\text{Area} = \int_0^2 [g(x) - f(x)] dx$$

$$\int_0^2 \left[4\cos\left(\frac{\pi}{4}x\right) - (2x^2 - 6x + 4) \right] dx$$

$$\left[4 \cdot \frac{4}{\pi} \sin\left(\frac{\pi}{4}x\right) - \left(\frac{2x^3}{3} - 3x^2 + 4x\right) \right]_0^2$$

$$= \frac{16}{\pi} - \left(\frac{16}{3} - 12 + 8\right) = \frac{16}{\pi} - \frac{4}{3}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

2 points are earned for the antiderivative

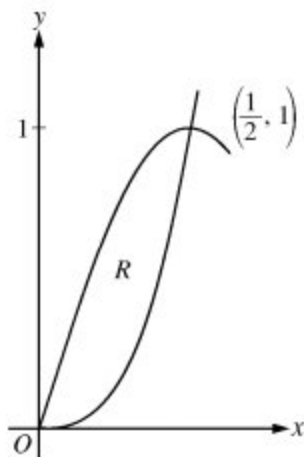
1 point is earned for the answer

8-4b

$$\begin{aligned}\text{Area} &= \int_0^2 [g(x) - f(x)] dx \\&= \int_0^2 \left[4\cos\left(\frac{\pi}{4}x\right) - (2x^2 - 6x + 4) \right] dx \\&= \left[4 \cdot \frac{4}{\pi} \sin\left(\frac{\pi}{4}x\right) - \left(\frac{2x^3}{3} - 3x^2 + 4x\right) \right]_0^2 \\&= \frac{16}{\pi} - \left(\frac{16}{3} - 12 + 8\right) = \frac{16}{\pi} - \frac{4}{3}\end{aligned}$$

8-4b

17.



Let R be the region in the first quadrant enclosed by the graphs of $f(x) = 8x^3$ and $g(x) = \sin(\pi x)$, as shown in the figure above.

Find the area of R .

Part B

1 point is earned for the integrand

2 points are earned the antiderivative

1 point is earned for the answer

$$\text{Area} = \int_0^{\frac{1}{2}} (g(x) - f(x)) dx$$

$$= \int_0^{\frac{1}{2}} (\sin(\pi x) - 8x^3) dx$$

$$= \left[-\frac{1}{\pi} \cos(\pi x) - 2x^4 \right]_{x=0}^{x=\frac{1}{2}}$$

$$= -\frac{1}{8} + \frac{1}{\pi}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

2 points are earned the antiderivative

1 point is earned for the answer

8-4b

$$\text{Area} = \int_0^{\frac{1}{2}} (g(x) - f(x)) dx$$

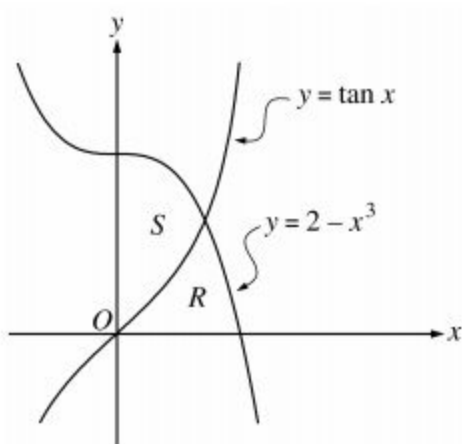
$$= \int_0^{\frac{1}{2}} (\sin(\pi x) - 8x^3) dx$$

$$= \left[-\frac{1}{\pi} \cos(\pi x) - 2x^4 \right]_{x=0}^{x=\frac{1}{2}}$$

$$= -\frac{1}{8} + \frac{1}{\pi}$$

8-4b

18.



Let R and S be the regions in the first quadrant shown in the figure above. The region R is bounded by the x -axis and the graphs of $y = 2 - x^3$ and $y = \tan x$. The region S is bounded by the y -axis and the graphs of $y = 2 - x^3$ and $y = \tan x$.



Find the area of S .

Part B

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Area, } S = \int_0^A (2 - x^3 - \tan x) dx = 1.160 \text{ or } 1.161$$

or

$$\text{Area, } S = \int_0^B \tan^{-1} y dy + \int_B^2 (2 - y)^{1/3} dy = 1.160 \text{ or } 1.161$$

or

Area, S

$$= \int_0^2 (2 - y)^{1/3} dy - \int_0^B ((2 - y)^{1/3} - \tan^{-1} y) dy$$

$$= 1.160 \text{ or } 1.161$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

8-4b

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Area, } S = \int_0^A (2 - x^3 - \tan x) dx = 1.160 \text{ or } 1.161$$

or

$$\text{Area, } S = \int_0^B \tan^{-1} y dy + \int_B^2 (2 - y)^{1/3} dy = 1.160 \text{ or } 1.161$$

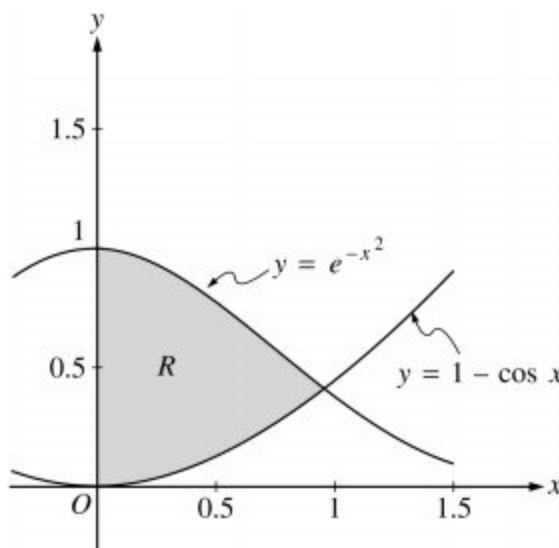
or

Area, S

$$\begin{aligned} &= \int_0^2 (2 - y)^{1/3} dy - \int_0^B \left((2 - y)^{1/3} - \tan^{-1} y \right) dy \\ &= 1.160 \text{ or } 1.161 \end{aligned}$$

8-4b

19.



Let R be the shaded region in the first quadrant enclosed by the graphs of $y = e^{-x^2}$, $y = 1 - \cos x$, and the y -axis, as shown in the figure above.



Find the area of the region R .

Part A

1 point is earned for the correct limits in an integral in (a), (b), or (c).

Region R

$$e^{-x^2} = 1 - \cos x, \text{ at } x = 0.941944 = A$$

1 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^A (e^{-x^2} - (1 - \cos x)) dx \\ &= 0.590, \text{ or } 0.591 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct limits in an integral in (a), (b), or (c).

Region R

8-4b

$$e^{-x^2} = 1 - \cos x, \text{ at } x = 0.941944 = A$$

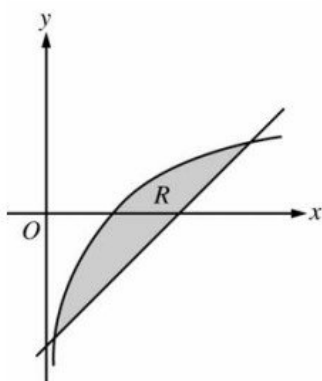
1 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^A (e^{-x^2} - (1 - \cos x)) dx \\ &= 0.590, \text{ or } 0.591 \end{aligned}$$

8-4b

20.



Let R be the shaded region bounded by the graph of $y = \ln x$ and the line $y = x - 2$, as show above.



Find the area of R .

Part A

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$\ln(x) = x - 2$ when $x = 0.15859$ and 3.14619 .

Let $S = 0.15859$ and $T = 3.14619$

$$\text{Area of } R = \int_S^T (\ln(x) - (x - 2)) dx = 1.949$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$\ln(x) = x - 2$ when $x = 0.15859$ and 3.14619 .

Let $S = 0.15859$ and $T = 3.14619$

8-4b

$$\text{Area of } R = \int_S^T (\ln(x) - (x - 2))dx = 1.949$$

21.  Let R be the region enclosed by the graph of $y = 1 + \ln(\cos^4 x)$, the x -axis, and the lines $x = -2/3$ and $x = 2/3$. The closest integer approximation of the area of R is

(A) 0

(B) 1



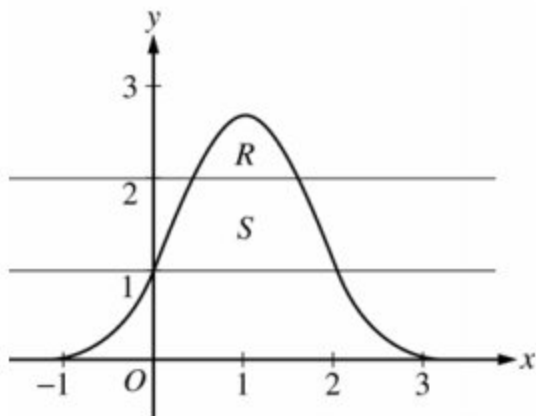
(C) 2

(D) 3

(E) 4

8-4b

22.



Let R be the region bounded by the graph of $y=e^{2x-x^2}$ and the horizontal line $y=2$, and let S be the region bounded by the graph of $y=e^{2x-x^2}$ and the horizontal lines $y=1$ and $y=2$, as shown above.



Find the area of R .

Part A

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$e^{2x-x^2}=2 \text{ when } x=0.446057, 1.553943$$

Let $P=0.446057$ and $Q=1.553943$

$$\text{Area of } R = \int_P^Q (e^{2x-x^2} - 2) dx = 0.514$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

8-4b

$$e^{2x-x^2}=2 \text{ when } x=0.446057, 1.553943$$

Let $P=0.446057$ and $Q=1.553943$

$$\text{Area of } R = \int_P^Q (e^{2x-x^2} - 2) dx = 0.514$$

23. What is the area of the region between the graphs of $y=x^2$ and $y=-x$ from $x=0$ to $x=2$?

(A) $\frac{2}{3}$

(B) $\frac{8}{3}$

(C) 4

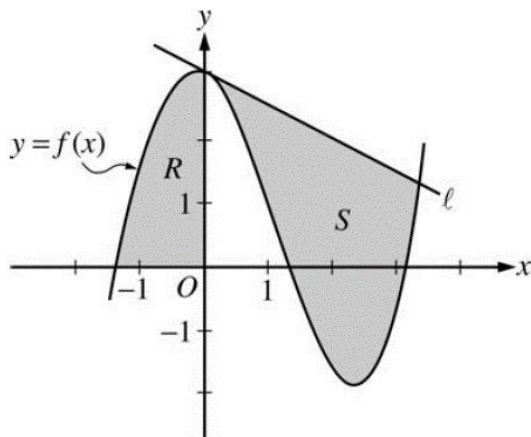
(D) $\frac{14}{3}$

(E) $\frac{16}{3}$



8-4b

24.



Let f be the function given by $f(x) = \frac{x^3}{4} - \frac{x^2}{3} - \frac{x}{2} + 3 \cos x$. Let R be the shaded region in the second quadrant bounded by the graph of f , and let S be the shaded region bounded by the graph of f and line ℓ , the line tangent to the graph of f at $x = 0$, as shown above.



Write, but do not evaluate, an integral expression that can be used to find the area of S .

Part C

1 point is earned for the tangent line

1 point is earned for the integrand

1 point is earned for the limits

The equation of the tangent line ℓ is $y = 3 - \frac{1}{2}x$.

The graph of f and line ℓ intersect at $A=3.38987$.

$$\text{Area of } S = \int_0^A \left(\left(3 - \frac{1}{2}x \right) - f(x) \right) dx$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the tangent line

1 point is earned for the integrand

1 point is earned for the limits

8-4b

The equation of the tangent line l is $y = 3 - \frac{1}{2}x$.

The graph of f and line l intersect at $A=3.38987$.

$$\text{Area of } S = \int_0^A \left(\left(3 - \frac{1}{2}x \right) - f(x) \right) dx$$

25. Let R be the region enclosed by the graph of $y = \sqrt{x-1}$, the vertical line $x=10$, and the x -axis. Find the area of R .

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Area} = \int_1^{10} \sqrt{x-1} dx = 18$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Area} = \int_1^{10} \sqrt{x-1} dx = 18$$

26. The region bounded by the x -axis and the part of the graph of $y = \cos x$ between $x = -\pi/2$ and $x = \pi/2$ is separated into two regions by the line $x = k$. If the area of the region for $-\pi/2 \leq x \leq k$ is three times the area of the region for $k \leq x$, then $k =$
- (A) $\arcsin(1/4)$
- (B) $\arcsin(1/3)$
- (C) $\pi/6$ ✓
- (D) $\pi/4$
- (E) $\pi/3$

8-4b

27.  If $0 \leq k < \frac{\pi}{2}$ and the area under the curve $y = \cos x$ from $x = k$ to $x = \frac{\pi}{2}$ is 0.1, then $k =$

(A) 1.471

(B) 1.414

(C) 1.277

(D) 1.120

(E) 0.436



8-4b

28. Let R be the region enclosed by the graphs of $y=e^x$, $y=(x-1)^2$, and the line $x=1$.
Find the area of R .

Part A

1 point is earned for definite integral

1 point is **deducted** for incorrect limits

1 point is **deducted** for reversal

1 point is earned for anti derivative

1 point is earned for evaluation

$$\begin{aligned}
 A &= \int_0^1 e^x - (x-1)^2 dx \\
 &= \int_0^1 e^x - x^2 + 2x - 1 dx \\
 &= e^x \Big|_0^1 - \frac{1}{3}(x-1)^3 \Big|_0^1 \\
 &= (e-1) - \frac{1}{3} = e - \frac{4}{3}
 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for definite integral

1 point is **deducted** for incorrect limits

1 point is **deducted** for reversal

1 point is earned for anti derivative

1 point is earned for evaluation

8-4b

$$\begin{aligned} A &= \int_0^1 e^x - (x-1)^2 dx \\ &= \int_0^1 e^x - x^2 + 2x - 1 dx \\ &= e^x \Big|_0^1 - \frac{1}{3}(x-1)^3 \Big|_0^1 \\ &= (e-1) - \frac{1}{3} = e - \frac{4}{3} \end{aligned}$$

8-4b

29. Let f be the function given by $f(x)=e^{-x}$, and let g be the function given by $g(x)=kx$, where k is the nonzero constant such that the graph of f is tangent to the graph of g .

Let R be the region enclosed by the y -axis and the graphs of f and g . Using the results found in part (a), determine the area of R .

Part B

2 point is earned for integral

1 point is earned if anti-differentiates

$$\begin{aligned}\int_{-1}^0 (e^{-x} - (-ex)) dx &= \int_{-1}^0 (e^{-x} + ex) dx \\ &= \left(-e^{-x} + \frac{ex^2}{2}\right) \Big|_{-1}^0 \\ &= (-1 + 0) - \left(-e + \frac{e}{2}\right) \\ &= \frac{e}{2} - 1\end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

2 point is earned for integral

1 point is earned if anti-differentiates

$$\begin{aligned}\int_{-1}^0 (e^{-x} - (-ex)) dx &= \int_{-1}^0 (e^{-x} + ex) dx \\ &= \left(-e^{-x} + \frac{ex^2}{2}\right) \Big|_{-1}^0 \\ &= (-1 + 0) - \left(-e + \frac{e}{2}\right) \\ &= \frac{e}{2} - 1\end{aligned}$$

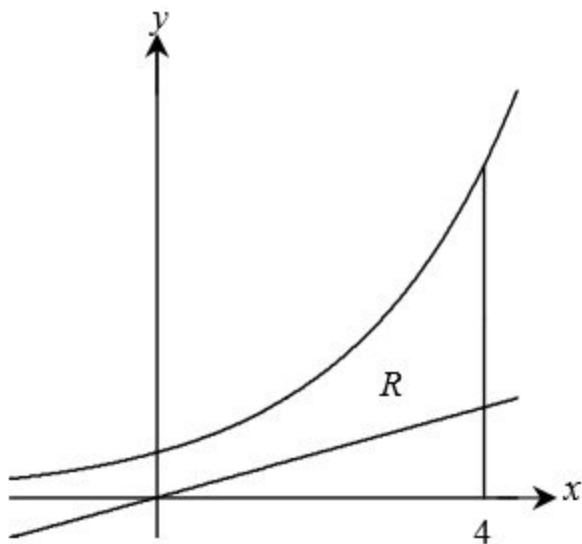
8-4b

30. Let R be the region enclosed by the graphs of $y=e^x$, $y=x$ and the lines $x=0$ and $x=4$.
Find the area of R .

Part A

Using Geometry (Area of Triangle) 3/3:

2 points are earned for the correct integral where one point is earned for the correct limits and one point is earned for the correct integrand



1 point is earned for the correct antidifferentiation and evaluation

0/1 if integrand is not $ex - x$ or $x - ex$

$$\begin{aligned}
 \text{Area} &= \int_0^4 e^x - x \, dx \\
 &= e^x - \frac{x^2}{2} \Big|_0^4 \\
 &= \left(e^4 - \frac{16}{2} \right) - (e^0 - 0) \\
 &= e^4 - 9 \\
 &\approx 45.598
 \end{aligned}$$

OR

Using y-axis 3/3:

2 points are earned for the correct integral where one point is earned for the correct limits and one point is earned for answering all 3 integrands correct

Note: 0/2 if any limits incorrect

8-4b

$$\int_0^1 y dy + \int_1^4 y - \ln y dy + \int_4^{e^4} 4 - \ln y dy$$

1 point is earned for the correct answer

≈ 45.598

Note: $k \int_0^4 e^x - x dx, k \neq 1$

Student scores 1 - 0 on first two points and is eligible for third point

Note: Student who believes $y = ex$ meets $y = x$

$$\int_0^k e^x - x dx + \int_k^4 x - e^x dx, 0/2 \text{ Integral}$$

Eligible for third point whether k is a number or parameter

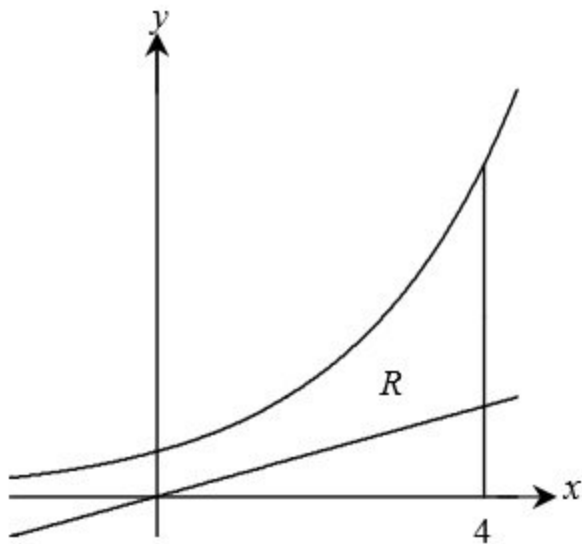


0	1	2	3
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The student response earns three of the following points:

Using Geometry (Area of Triangle) 3/3:

2 points are earned for the correct integral where one point is earned for the correct limits and one point is earned for the correct integrand



1 point is earned for the correct antidifferentiation and evaluation

8-4b

0/1 if integrand is not $ex - x$ or $x - ex$

$$\begin{aligned}
 \text{Area} &= \int_0^4 e^x - x dx \\
 &= e^x - \frac{x^2}{2} \Big|_0^4 \\
 &= \left(e^4 - \frac{16}{2}\right) - (e^0 - 0) \\
 &= e^4 - 9 \\
 &\approx 45.598
 \end{aligned}$$

OR

Using y -axis 3/3:

2 points are earned for the correct integral where one point is earned for the correct limits and one point is earned for answering all 3 integrands correct

Note: 0/2 if any limits incorrect

$$\int_0^1 y dy + \int_1^4 y - \ln y dy + \int_4^{e^4} 4 - \ln y dy$$

1 point is earned for the correct answer

$$\approx 45.598$$

Note: $k \int_0^4 e^x - x dx, k \neq 1$

Student scores 1 - 0 on first two points and is eligible for third point

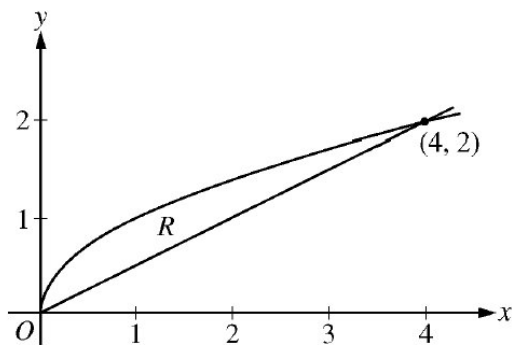
Note: Student who believes $y = ex$ meets $y = x$

$$\int_0^k e^x - x dx + \int_k^4 x - e^x dx, 0/2 \text{ Integral}$$

Eligible for third point whether k is a number or parameter

8-4b

31.



Let R be the region bounded by the graphs of $y = \sqrt{x}$ and $y = \frac{x}{2}$, as shown in the figure above. Find the area of R .

Part A

1 point is earned for integrand

1 point is earned for antiderivative

1 point is earned for answer

$$\text{Area} = \int_0^4 \left(\sqrt{x} - \frac{x}{2} \right) dx = \frac{2}{3} x^{\frac{3}{2}} - \frac{x^2}{4} \bigg|_{x=0}^{x=4} = \frac{4}{3}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrand

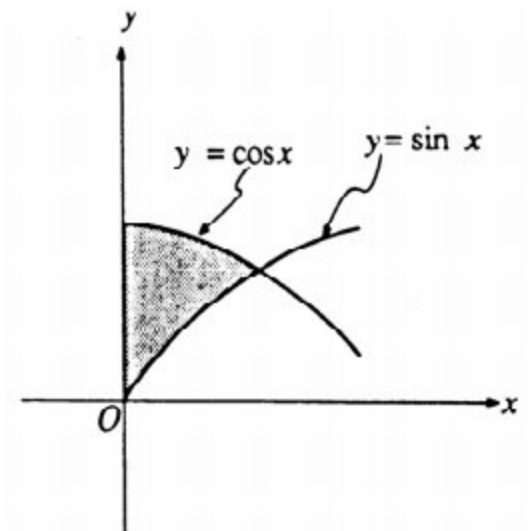
1 point is earned for antiderivative

1 point is earned for answer

$$\text{Area} = \int_0^4 \left(\sqrt{x} - \frac{x}{2} \right) dx = \frac{2}{3} x^{\frac{3}{2}} - \frac{x^2}{4} \bigg|_{x=0}^{x=4} = \frac{4}{3}$$

8-4b

32.



Let R be the shaded region in the first quadrant enclosed by the y -axis and the graphs of $y = \sin x$ and $y = \cos x$, as shown in the figure above.

Find the area of R

Part A

1 point is earned for limits

1 point is earned for integrand

1 point is earned for antiderivative and evaluation

$$\begin{aligned}
 \text{Area} &= \int_0^{\pi/4} (\cos x - \sin x) dx \\
 &= (\sin x + \cos x) \Big|_0^{\pi/4} \\
 &= \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \right) - (0 + 1) \\
 &= \sqrt{2} - 1
 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for limits

1 point is earned for integrand

1 point is earned for antiderivative and evaluation

8-4b

$$\begin{aligned} \text{Area} &= \int_0^4 \cos x - \sin x dx \\ &= (\sin x + \cos x) \Big|_0^{\frac{\pi}{4}} \\ &= \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \right) - (0 + 1) \\ &= \sqrt{2} - 1 \end{aligned}$$

33. The area of the region in the first quadrant that is enclosed by the graphs of $y = x^3 + 8$ and $y = x + 8$ is

(A) 1/4



(B) 1/2

(C) 3/4

(D) 1

(E) 65/4

34. The area of the region enclosed by the curve $y = 1/x - 1$, the x -axis, and the lines $x = 3$ and $x = 4$ is

(A) 5/36

(B) $\ln \frac{2}{3}$

(C) $\ln 4/3$

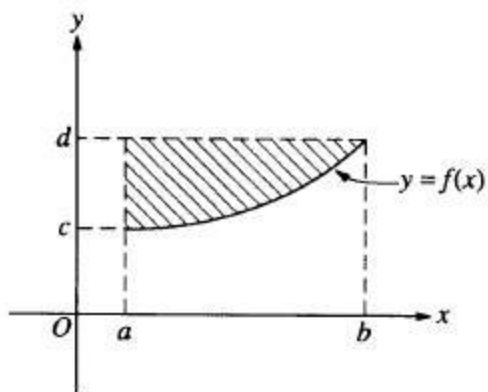
(D) $\ln 3/2$



(E) $\ln 6$

8-4b

35.



Which of the following represents the area of the shaded region in the figure above?

(A) $\int_c^d f(y)dy$

(B) $\int_a^b (d - f(x))dx$



(C) $f'(b) - f'(a)$

(D) $(b - a)[f(b) - f(a)]$

(E) $(d - c)[f(b) - f(a)]$

8-4b

36. Let R be the region enclosed by the graph of $y = \frac{x^2}{x^2+1}$, the line $x=1$, and the x -axis.
Find the area of R .

Part A

1 point is earned for bounds

1 point is earned for integrand

2 points are earned for antiderivative

2 points **deducted** for trig calculation error

1 point is earned for evaluation

$$\begin{aligned} \text{Area} &= \int_0^1 \frac{x^2}{x^2+1} dx \\ &= \int_0^1 1 - \frac{1}{x^2+1} dx \\ &= x - \arctan x \Big|_0^1 \\ &= 1 - \frac{\pi}{4} \end{aligned}$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns five of the following points:

1 point is earned for bounds

1 point is earned for integrand

2 points are earned for antiderivative

2 points **deducted** for trig calculation error

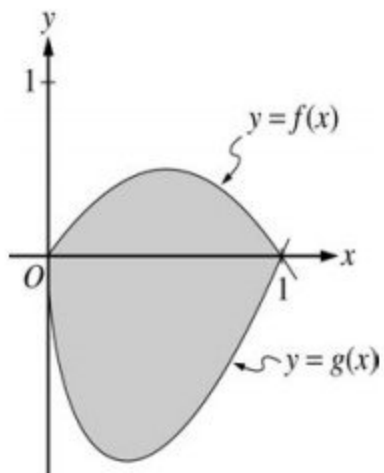
1 point is earned for evaluation

8-4b

$$\begin{aligned} \text{Area} &= \int_0^1 \frac{x^2}{x^2 + 1} dx \\ &= \int_0^1 1 - \frac{1}{x^2 + 1} dx \\ &= x - \arctan x \Big|_0^1 \\ &= 1 - \frac{\pi}{4} \end{aligned}$$

8-4b

37.



Let f and g be the functions given by $f(x) = 2x(1 - x)$ and $g(x) = 3(x - 1)\sqrt{x}$ for $0 \leq x \leq 1$. The graphs of f and g are shown in the figure above.

Find the area of the shaded region enclosed by the graphs of f and g .

Part A

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^1 (f(x) - g(x)) dx \\ &= \int_0^1 (2x(1 - x) - 3(x - 1)\sqrt{x}) dx = 1.133 \end{aligned}$$



0	1	2
---	---	---

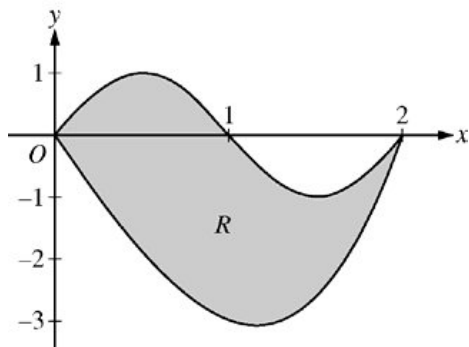
The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^1 (f(x) - g(x)) dx \\ &= \int_0^1 (2x(1 - x) - 3(x - 1)\sqrt{x}) dx = 1.133 \end{aligned}$$

8-4b



Let R be the region bounded by the graphs of $y = \sin(\pi x)$ and $y = x^3 - 4x$, as shown in the figure above.

38.  Find the area of R .

Part A

The response can earn up to 3 points:

1 point: For the correct limits

$$\sin(\pi x) = x^3 - 4x \text{ at } x = 0 \text{ and } x = 2$$

1 point: For the correct integrand

1 point: For the correct answer

$$\text{Area} = \int_0^2 (\sin(\pi x) - (x^3 - 4x)) dx = 4$$



0	1	2	3
---	---	---	---

The response earns all three of the following points:

1 point: For the correct limits

$$\sin(\pi x) = x^3 - 4x \text{ at } x = 0 \text{ and } x = 2$$

1 point: For the correct integrand

1 point: For the correct answer

$$\text{Area} = \int_0^2 (\sin(\pi x) - (x^3 - 4x)) dx = 4$$

8-4b

39.  The horizontal line $y = -2$ splits the region R into two parts. Write, but do not evaluate, an integral expression for the area of the part of R that is below this horizontal line.

Part B

The response can earn up to 2 points:

1 point: For the correct limits

$$x^3 - 4x = -2 \text{ at } r = 0.5391889 \text{ and } s = 1.6751309$$

1 point: For the correct integrand

The area of the shaded region is $\int_s^r (-2 - (x^3 - 4x))dx$



0	1	2
---	---	---

The response earns two of the following points:

1 point: For the correct limits

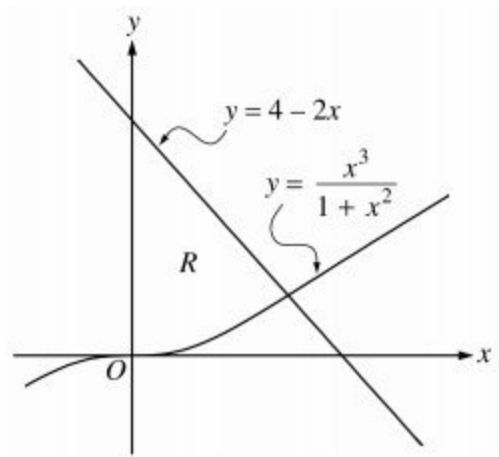
$$x^3 - 4x = -2 \text{ at } r = 0.5391889 \text{ and } s = 1.6751309$$

1 point: For the correct integrand

The area of the shaded region is $\int_s^r (-2 - (x^3 - 4x))dx$

8-4b

40.



Let R be the region bounded by the y -axis and the graphs of $y = \frac{x^3}{1+x^2}$ and $y = 4 - 2x$, as shown in the figure above.



Find the area of R .

Part A

1 point is earned for the correct limits in an integral in (a), (b), or (c).

Region R

$$\frac{x^3}{1+x^2} = 4 - 2x \text{ at } x = 1.487664 = A$$

1 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^A \left(4 - 2x - \frac{x^3}{1+x^2} \right) dx \\ &= 3.214 \text{ or } 3.215 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

The student response earns one of the following points:

1 point is earned for the correct limits in an integral in (a), (b), or (c).

Region R

8-4b

$$\frac{x^3}{1+x^2} = 4 - 2x \text{ at } x = 1.487664 = A$$

1 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^A \left(4 - 2x - \frac{x^3}{1+x^2} \right) dx \\ &= 3.214 \text{ or } 3.215 \end{aligned}$$

8-4b

41. Let R be the region enclosed by the graphs of $y = \ln(x^2 + 1)$ and $y = \cos x$.

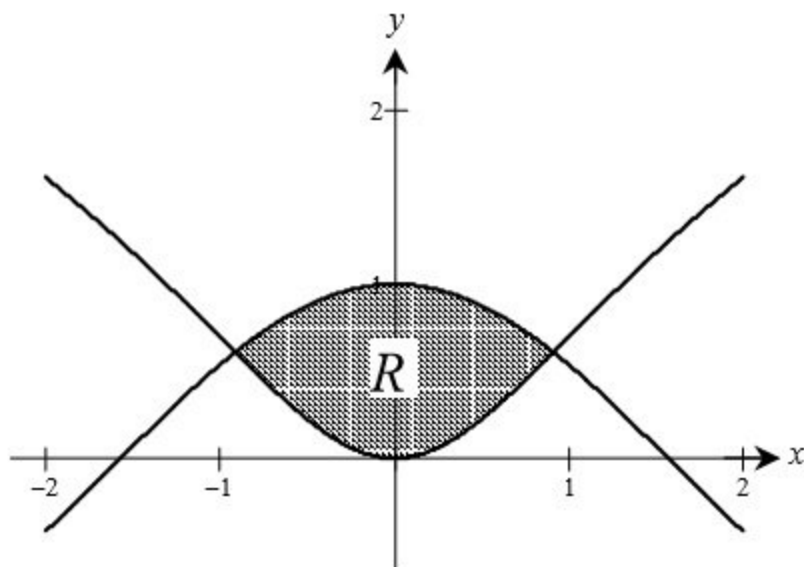


Find the area of R .

Part A

2 points are earned for the correct integral where 1 point is earned for the correct limits and 1 point is earned for the correct integrand

0/1 if integrand not $\cos x - \ln(x^2 + 1)$ or $\ln(x^2 + 1) - \cos x$



$$\ln(x^2 + 1) = \cos x$$

$$x = \pm 0.91586$$

$$\text{Let } B = 0.91586$$

1 point is earned for the correct answer

$$\begin{aligned} \text{area} &= \int_{-B}^B [\cos x - \ln(x^2 + 1)] dx \\ &= 1.168 \end{aligned}$$

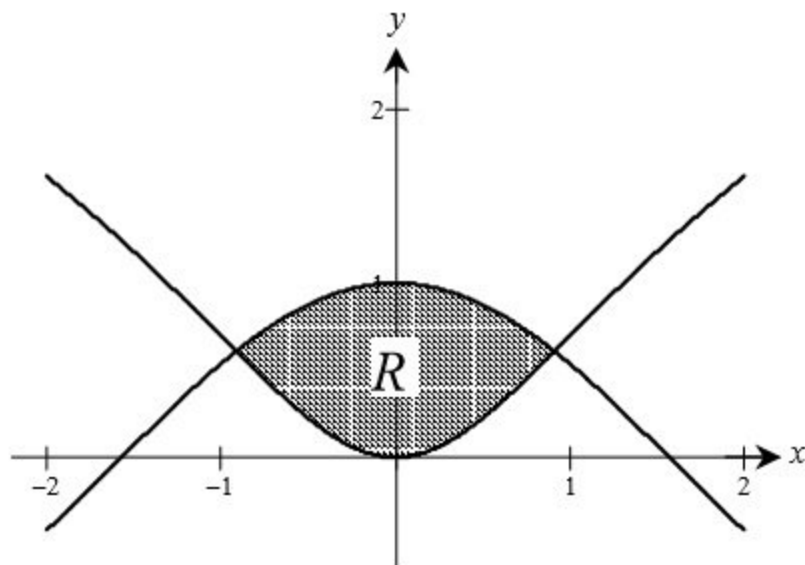


0	1	2	3
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The student response earns three of the following points:

2 points are earned for the correct integral where 1 point is earned for the correct limits and 1 point is earned for the correct integrand

8-4b

0/1 if integrand not $\cos x - \ln(x^2 + 1)$ or $\ln(x^2 + 1) - \cos x$ 

$$\ln(x^2 + 1) = \cos x$$

$$x = \pm 0.91586$$

Let $B = 0.91586$

1 point is earned for the correct answer

$$\begin{aligned}\text{area} &= \int_{-B}^B [\cos x - \ln(x^2 + 1)] dx \\ &= 1.168\end{aligned}$$

42. The area of the region enclosed by the graph of $y = x^2 + 1$ and the line $y = 5$ is

(A) $14/3$

(B) $16/3$

(C) $28/3$

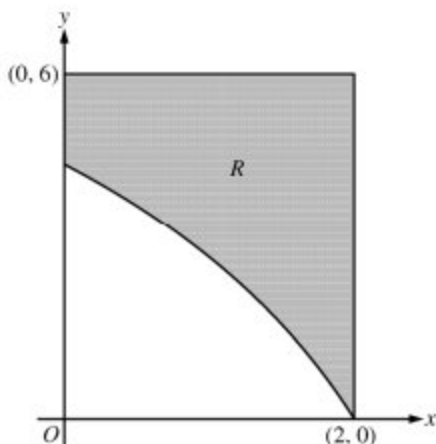
(D) $32/3$

(E) 8π



8-4b

43.



In the figure above, R is the shaded region in the first quadrant bounded by the graph of $y=4\ln(3-x)$, the horizontal line $y=6$, and the vertical line $x=2$.



Find the area of R .

Part A

1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^2 (6 - 4\ln(3-x)) dx = 6.816 \text{ or } 6.817$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^2 (6 - 4\ln(3-x)) dx = 6.816 \text{ or } 6.817$$

8-4b

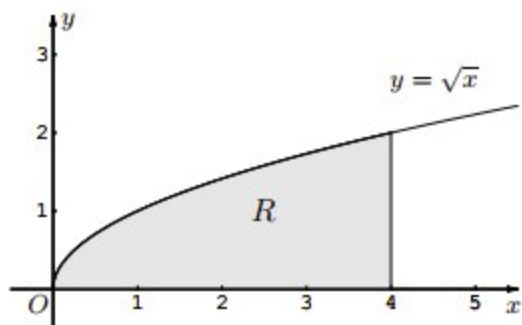
Let R be the region bounded by the x -axis, the graph of $y = \sqrt{x}$, and the line $x = 4$.

44. Find the area of the region R

Part A

1 point is earned for equation of area $A = \int_0^4 \sqrt{x} dx$

1 point is earned for solving equation



$$A = \int_0^4 \sqrt{x} dx = \left. \frac{2}{3} x^{3/2} \right|_0^4 = \frac{16}{3} \text{ or } 5.333$$

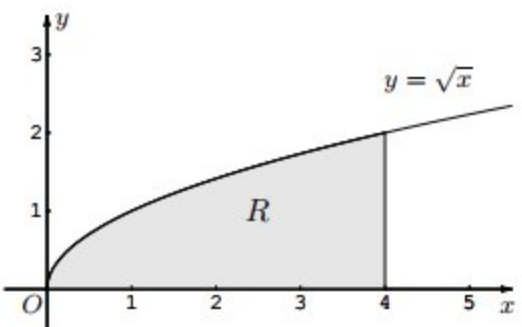


0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for equation of area $A = \int_0^4 \sqrt{x} dx$

1 point is earned for solving equation



$$A = \int_0^4 \sqrt{x} dx = \left. \frac{2}{3} x^{3/2} \right|_0^4 = \frac{16}{3} \text{ or } 5.333$$

8-4b

45. Find the value of h such that the vertical line $x = h$ divides the region R into two regions of equal area

Part B

1 point is earned for equation in h

1 point is earned for answer

$$\int_0^h \sqrt{x} dx = \frac{8}{3} \quad \int_0^h \sqrt{x} dx = \int_h^4 \sqrt{x} dx$$

or

$$\frac{2}{3}h^{3/2} = \frac{8}{3} \quad \frac{2}{3}h^{3/2} = \frac{16}{3} - \frac{2}{3}h^{3/2}$$

$$h = \sqrt[3]{16} \text{ or } 2.520 \text{ or } 2.519$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for equation in h

1 point is earned for answer

$$\int_0^h \sqrt{x} dx = \frac{8}{3} \quad \int_0^h \sqrt{x} dx = \int_h^4 \sqrt{x} dx$$

or

$$\frac{2}{3}h^{3/2} = \frac{8}{3} \quad \frac{2}{3}h^{3/2} = \frac{16}{3} - \frac{2}{3}h^{3/2}$$

$$h = \sqrt[3]{16} \text{ or } 2.520 \text{ or } 2.519$$

46. What is the area of the region in the first quadrant bounded by the graph of $y = e^{x/2}$ and the line $x = 2$?

(A) $2e - 2$

(B) $2e$

(C) $\frac{e}{2} - 1$

(D) $\frac{e-1}{2}$

(E) $e - 1$



8-4b

47. The area of the region enclosed by the graphs of $y = x^2$ and $y = x$ is

(A) $1/6$



(B) $1/3$

(C) $1/2$

(D) $5/6$

(E) 1

8-4b

Let R be the region in the first quadrant under the graph of $y = \frac{x}{x^2+2}$ for $0 \leq x \leq \sqrt{6}$.

48. Find the area of R .

Part A

1 point is earned for correct integral

1 point is earned for antiderivative

1 point is earned for evaluation

$$\begin{aligned} A &= \int_0^{\sqrt{6}} \frac{x}{x^2+2} dx \\ &= \frac{1}{2} \ln(x^2+2) \Big|_0^{\sqrt{6}} \\ &= \frac{1}{2} \ln 8 - \frac{1}{2} \ln 2 = \ln 2 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for correct integral

1 point is earned for antiderivative

1 point is earned for evaluation

$$\begin{aligned} A &= \int_0^{\sqrt{6}} \frac{x}{x^2+2} dx \\ &= \frac{1}{2} \ln(x^2+2) \Big|_0^{\sqrt{6}} \\ &= \frac{1}{2} \ln 8 - \frac{1}{2} \ln 2 = \ln 2 \end{aligned}$$

8-4b

49. If the line $x = k$ divides R into two regions of equal area, what is the value of k ?
-

Part B

1 point is earned for a correct equation involving integral(s)

1 point is earned for antiderivative

1 point is earned for evaluation of integral(s)

1 point is earned for finding the value of k

$$\begin{aligned}\frac{1}{2} \text{In} &= \int_0^k \frac{x}{x^2 + 2} dx \\ &= \frac{1}{2} \text{In}(x^2 + 2) \Big|_0^k \\ &= \frac{1}{2} \text{In}(k^2 + 2) - \frac{1}{2} \text{In}2 \\ \therefore \frac{1}{2} \text{In}(k^2 + 2) &= \frac{1}{2} \text{In}2 + \frac{1}{2} \text{In}2\end{aligned}$$

$$\text{or } \ln(k^2 + 2) = \ln 4$$

$$\therefore k^2 + 2 = 4 \text{ and } k = \sqrt{2}$$



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

1 point is earned for a correct equation involving integral(s)

1 point is earned for antiderivative

1 point is earned for evaluation of integral(s)

1 point is earned for finding the value of k

8-4b

$$\begin{aligned}
 \frac{1}{2} \ln &= \int_0^k \frac{x}{x^2 + 2} dx \\
 &= \frac{1}{2} \ln(x^2 + 2) \Big|_0^k \\
 &= \frac{1}{2} \ln(k^2 + 2) - \frac{1}{2} \ln 2 \\
 \therefore \frac{1}{2} \ln(k^2 + 2) &= \frac{1}{2} \ln 2 + \frac{1}{2} \ln 2
 \end{aligned}$$

$$\text{or } \ln(k^2 + 2) = \ln 4$$

$$\therefore k^2 + 2 = 4 \text{ and } k = \sqrt{2}$$

50. Let f and g be the functions given by $f(x) = e^x$ and $g(x) = \ln x$.



Find the area of the region enclosed by the graphs of f and g between $x = \frac{1}{2}$ and $x = 1$.

Part A

1 point is earned for integral

1 point is earned for the answer

$$\text{Area} = \int_{1/2}^1 (e^x - \ln x) dx = 1.222 \text{ or } 1.223$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

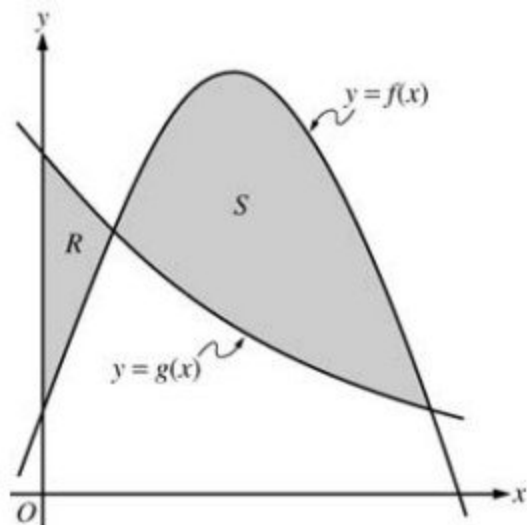
$$\text{Area} = \int_{1/2}^1 (e^x - \ln x) dx = 1.222 \text{ or } 1.223$$

8-4b

51. A region in the plane is bounded by the graph of $y = 1/x$, the x -axis, the line $x = m$, and the line $x = 2m$, $m > 0$. The area of this region

- (A) is independent of m . ✓
- (B) increases as m increases.
- (C) decreases as m increases.
- (D) decreases as m increases when $m < 1/2$; increases as m increases when $m > 1/2$.
- (E) increases as m increases when $m < 1/2$; decreases as m increases when $m > 1/2$.

8-4b



Let f and g be the functions given by $f(x) = \frac{1}{4} + \sin(\pi x)$ and $g(x) = 4^{-x}$. Let R be the shaded region in the first quadrant enclosed by the y-axis and the graphs of f and g , and let S be the shaded region in the first quadrant enclosed by the graphs of f and g , as shown in the figure above.

52.  Find the area of R .

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$f(x) = g(x) \text{ when } \frac{1}{4} + \sin(\pi x) = 4^{-x}.$$

f and g intersect when $x = 0.178218$ and when $x = 1$.

Let $a = 0.178218$

$$\int_0^a (g(x) - f(x)) dx = 0.064 \text{ or } 0.065$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

8-4b

1 point is earned for the integrand

1 point is earned for the answer

$$f(x) = g(x) \text{ when } \frac{1}{4} + \sin(\pi x) = 4^{-x}.$$

f and g intersect when $x = 0.178218$ and when $x = 1$.

Let $a = 0.178218$

$$\int_0^a (g(x) - f(x)) dx = 0.064 \text{ or } 0.065$$

53.  Find the area of S .

Part B

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^1 (g(x) - f(x)) dx = 0.410$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

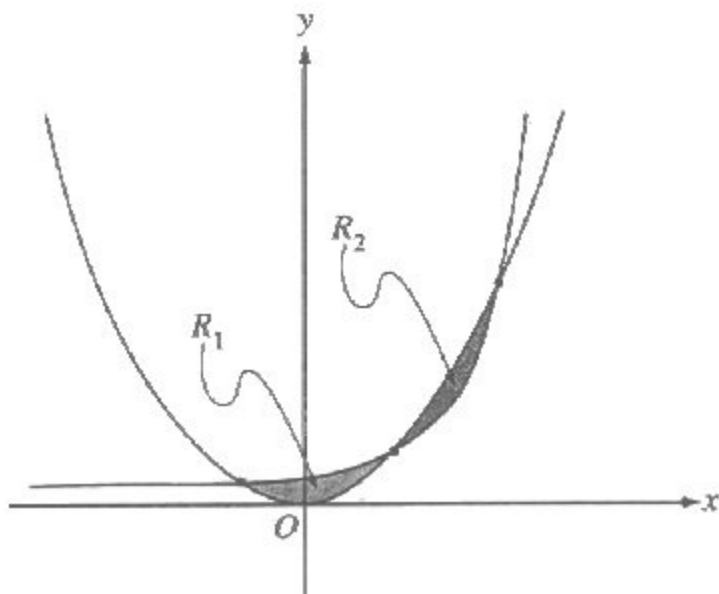
1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^1 (g(x) - f(x)) dx = 0.410$$

8-4b

54.



Note: Figure not drawn to scale.

The shaded regions R_1 and R_2 shown above are enclosed by the graphs of $f(x)=x^2$ and $g(x)=2^x$.

Without using absolute value, set up an expression involving one or more integrals that gives the total area enclosed by the graphs of f and g . Do not evaluate.

Part B

2 points are earned for each correct definite integral

1 point is earned for the correct limits

1 point is earned for the correct integrand, without absolute value

< -1 > Integrals considered but not added

Note: 1/4 for $\int_{-0.767}^4 |x^2 - 2^x| dx$

Note: max 1/4 dealing only with one of R_1 or R_2

$$\int_{-0.767}^2 (2^x - x^2) dx + \int_2^4 (x^2 - 2^x) dx$$

OR

2 points are earned for the correct area of R_1 where one point is earned for the correct limits and one point is earned for the correct integrand

8-4b

2 points are earned for the correct area of R_2 where one point is earned for the correct limits and one point is earned for the correct integrand

$$\int_0^{0.588} 2\sqrt{y} dy + \int_{0.588}^4 \left(\sqrt{y} - \frac{\ln y}{\ln 2} \right) dy + \int_4^{16} \left(\frac{\ln y}{\ln 2} - \sqrt{y} \right) dy$$



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

2 points are earned for each correct definite integral

1 point is earned for the correct limits

1 point is earned for the correct integrand, without absolute value

< -1 > Integrals considered but not added

Note: 1/4 for $\int_{-0.767}^4 |x^2 - 2^x| dx$

Note: max 1/4 dealing only with one of R_1 or R_2

$$\int_{-0.767}^2 (2^x - x^2) dx + \int_2^4 (x^2 - 2^x) dx$$

OR

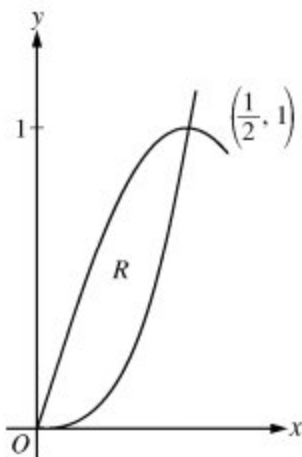
2 points are earned for the correct area of R_1 where one point is earned for the correct limits and one point is earned for the correct integrand

2 points are earned for the correct area of R_2 where one point is earned for the correct limits and one point is earned for the correct integrand

$$\int_0^{0.588} 2\sqrt{y} dy + \int_{0.588}^4 \left(\sqrt{y} - \frac{\ln y}{\ln 2} \right) dy + \int_4^{16} \left(\frac{\ln y}{\ln 2} - \sqrt{y} \right) dy$$

8-4b

55.



Let R be the region in the first quadrant enclosed by the graphs of $f(x) = 8x^3$ and $g(x) = \sin(\pi x)$, as shown in the figure above.

(a) Write an equation for the line tangent to the graph of f at $x = 1/2$.

(b) Find the area of R .

(c) Write, but do not evaluate, an integral expression for the volume of the solid generated when R is rotated about the horizontal line $y = 1$.

Part A

1 point is earned for $f'(\frac{1}{2})$

1 point is earned for the answer

$$f\left(\frac{1}{2}\right) = 1$$

$$f'(x) = 24x^2, \text{ so } f'\left(\frac{1}{2}\right) = 6$$

An equation for the tangent line is $y = 1 + 6\left(x - \frac{1}{2}\right)$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $f'(\frac{1}{2})$

1 point is earned for the answer

8-4b

$$f\left(\frac{1}{2}\right) = 1$$

$$f'(x) = 24x^2, \text{ so } f'\left(\frac{1}{2}\right) = 6$$

An equation for the tangent line is $y = 1 + 6\left(x - \frac{1}{2}\right)$.

Part B

1 point is earned for the integrand

2 points are earned the antiderivative

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^{1/2} (g(x) - f(x))dx \\ &= \int_0^{1/2} (\sin(\pi x) - 8x^3)dx \\ &= \left[-\frac{1}{\pi}\cos(\pi x) - 2x^4\right]_{x=0}^{x=1/2} \\ &= -\frac{1}{8} + \frac{1}{\pi} \end{aligned}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

2 points are earned the antiderivative

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^{1/2} (g(x) - f(x))dx \\ &= \int_0^{1/2} (\sin(\pi x) - 8x^3)dx \\ &= \left[-\frac{1}{\pi}\cos(\pi x) - 2x^4\right]_{x=0}^{x=1/2} \\ &= -\frac{1}{8} + \frac{1}{\pi} \end{aligned}$$

8-4b

Part C

1 point is earned for limits and constant

2 points are earned for integrand

$$\begin{aligned} & \pi \int_0^{1/2} \left((1 - f(x))^2 - (1 - g(x))^2 \right) dx \\ &= \pi \int_0^{1/2} \left((1 - 8x^3)^2 - (1 - \sin(\pi x))^2 \right) dx \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for limits and constant

2 points are earned for integrand

$$\begin{aligned} & \pi \int_0^{1/2} \left((1 - f(x))^2 - (1 - g(x))^2 \right) dx \\ &= \pi \int_0^{1/2} \left((1 - 8x^3)^2 - (1 - \sin(\pi x))^2 \right) dx \end{aligned}$$

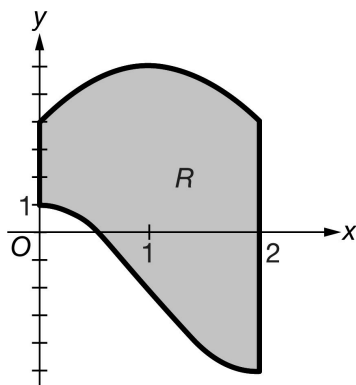
8-4b

56. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Let R be the region enclosed by the graphs of $g(x) = -2 + 3 \cos\left(\frac{\pi}{2}x\right)$ and $h(x) = 6 - 2(x-1)^2$, the y -axis, and the vertical line $x = 2$, as shown in the figure above.

- Find the area of R .
- Region R is the base of a solid. For the solid, at each x the cross section perpendicular to the x -axis has area $A(x) = \frac{1}{x+3}$. Find the volume of the solid.
- Write, but do not evaluate, an integral expression that gives the volume of the solid generated when R is rotated about the horizontal line $y = 6$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for an integrand that contains a form of $h(x) - g(x)$ or $g(x) - h(x)$.

The second point is earned for the antiderivative $\frac{6}{\pi} \sin\left(\frac{\pi}{2}x\right)$.

A response is eligible for the fourth point if (1) the first point is earned, (2) at least one of the second or third points is earned, (3) the integrand contains no copy errors or arithmetic errors. Simplification is not required to earn the fourth point. The limits of integration are part of the fourth point.

If the limits of integration are $x = 1$ to $x = 2$ in all parts of the question, the response is not eligible for the fourth point in Part A but eligible for points in Part B and Part C.

8-4b



0	1	2	3	4
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The student response accurately includes **all four** of the criteria below.

- ☐ integrand
- ☐ antiderivative of $3 \cos\left(\frac{\pi}{2}x\right)$
- ☐ antiderivative of remaining terms
- ☐ answer

Solution:

$$\begin{aligned}
 \int_0^2 (h(x) - g(x)) \, dx &= \int_0^2 \left((6 - 2(x-1)^2) - \left(-2 + 3 \cos\left(\frac{\pi}{2}x\right)\right) \right) \, dx \\
 &= \left[\left(6x - \frac{2}{3}(x-1)^3\right) - \left(-2x + \frac{6}{\pi} \sin\left(\frac{\pi}{2}x\right)\right) \right]_{x=0}^{x=2} \\
 &= \left(\left(12 - \frac{2}{3}\right) - (-4 + 0)\right) - \left(\left(0 + \frac{2}{3}\right) - (0 + 0)\right) \\
 &= 12 - \frac{2}{3} + 4 - \frac{2}{3} = \frac{44}{3}
 \end{aligned}$$

The area of R is $\frac{44}{3}$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the definite integral $\int_0^2 A(x) \, dx$ or $\int_0^2 \frac{1}{x+3} \, dx$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ integral
- ☐ answer

Solution:

8-4b

$$\int_0^2 A(x) dx = \int_0^2 \frac{1}{x+3} dx$$

$$= [\ln(x+3)]_{x=0}^{x=2} = \ln 5 - \ln 3$$

The volume of the solid is $\ln 5 - \ln 3$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for a definite integral, multiplied by π , with 0 and 2 as the lower and upper limits of integration, respectively. The π may appear in the integrand, but it must be correct to earn the first point.

The second point is earned with an integrand that is a difference of squares with a correct or consistent axis of rotation.

The third point is earned for the correct integrand. Note that a reversal of order within the squared terms, e.g.

$\pi \int_0^2 \left((g(x) - 6)^2 - (h(x) - 6)^2 \right) dx$, is equivalent to the given solution and thus earns the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ limits and constant
- ☐ form of integrand
- ☐ integrand

Solution:

$$\pi \int_0^2 \left((6 - g(x))^2 - (6 - h(x))^2 \right) dx$$

8-4b

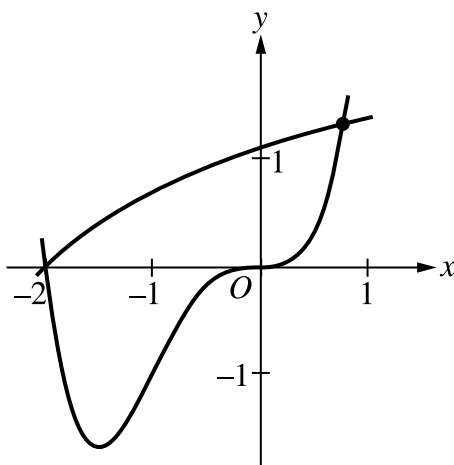
57.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Let f and g be the functions defined by $f(x) = \ln(x+3)$ and $g(x) = x^4 + 2x^3$. The graphs of f and g , shown in the figure, intersect at $x = -2$ and $x = B$, where $B > 0$.

- Find the area of the region enclosed by the graphs of f and g .
- For $-2 \leq x \leq B$, let $h(x)$ be the vertical distance between the graphs of f and g . Is h increasing or decreasing at $x = -0.5$? Give a reason for your answer.
- The region enclosed by the graphs of f and g is the base of a solid. Cross sections of the solid taken perpendicular to the x -axis are squares. Find the volume of the solid.
- A vertical line in the xy -plane travels from left to right along the base of the solid described in part (c). The vertical line is moving at a constant rate of 7 units per second. Find the rate of change of the area of the cross section above the vertical line with respect to time when the vertical line is at position $x = -0.5$.

Part A

8-4b

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Other forms of the integrand in a definite integral, e.g., $|f(x) - g(x)|$, $|g(x) - f(x)|$, or $g(x) - f(x)$, earn the first point.

To earn the second point, the response must have a lower limit of -2 and an upper limit expressed as either the letter B with no value attached, or a number that is correct to the number of digits presented, with at least one and up to three decimal places.

- Case 1: If the response did not earn the second point because of an incorrect value of B , $0 < B < 1$, but used a lower limit of -2 , the response earns the third point only for a consistent answer.
- Case 2: If the response did not earn the second point because the lower limit used was $x = 0$, but the response used a correct upper limit of B the response earns the third point for a consistent answer of 0.708 (or 0.707).
- Case 3: If a response uses any other incorrect limits it does not earn the second or third points.

A response containing the integrand $g(x) - f(x)$ must interpret the value of the resulting integral correctly to earn the third point. For example, the following response earns all 3 points: $\int_{-2}^B (g(x) - f(x)) dx = -3.604$ so the area is 3.604. However, the response “Area = $\int_{-2}^B (g(x) - f(x)) dx = 3.604$ ” presents an untrue statement and earns the first and second points but not the third point.

A response must earn the first point in order to be eligible for the third point. If the response has earned the second point, then only the correct answer will earn the third point.

Instructions for scoring a response that presents an integrand of $\ln(x + 3) - x^4 + 2x^3$:

- The first time a response implicitly presents $f(x) - g(x)$ as $j(x) = \ln(x + 3) - x^4 + 2x^3$ (with no explicit connection) in any part of this question, the response loses a point. The response is then eligible for all remaining points for a correct or consistent answer.
- In part (a) the consistent answer using $j(x)$ is negative and will not earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Integrand
- ☐ Limits of integration
- ☐ Answer

8-4b

Solution:

$$\ln(x+3) = x^4 + 2x^3 \Rightarrow x = -2, x = B = 0.781975$$

$$\int_{-2}^B (f(x) - g(x)) dx = 3.603548$$

The area of the region is 3.604 (or 3.603).

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response need not present the value of $h'(-0.5)$. The last line earns both points. However, if a value is presented it must be correct for the digits reported up to three decimal places.

A response that reports an incorrect value of $h'(-0.5)$ earns only the first point.

A response that presents only $h'(x)$ does not earn either point.

A response that compares the values of $f'(x)$ and $g'(x)$ at $x = -0.5$ earns the first point and is eligible for the second point. This comparison can be made symbolically or verbally; for example, the response “the rate of change of $f(x)$ is less than the rate of change of $g(x)$ at $x = -0.5$ ” earns the first point.

Instructions for scoring a response that presents $h(x)$ as $\ln(x+3) - x^4 + 2x^3$:

- If the first time the response presents $f(x) - g(x)$ as $j(x) = \ln(x+3) - x^4 + 2x^3$ occurs in part (b), the response loses a point. If the response already lost a point for this error in part (a), then the response is eligible for all remaining points for a correct or consistent answer.

- In part (b), the consistent answer using $j(x)$ is that $j'(-0.5) = 2.4 > 0$, so h is increasing at $x = -0.5$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers $h'(-0.5)$ – OR – $f'(x) - g'(x)$
- ☐ Answer with reason

Solution:

$$h(x) = f(x) - g(x)$$

$$h'(x) = f'(x) - g'(x)$$

8-4b

$$h'(-0.5) = f'(-0.5) - g'(-0.5) = -0.6 \text{ (or } -0.599)$$

Since $h'(-0.5) < 0$, h is decreasing at $x = -0.5$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for an integrand of $k(f(x) - g(x))^2$ or its equivalent with $k \neq 0$ in any definite integral. If $k \neq 1$, then the response is not eligible for the second point.

A response that does not earn the first point is ineligible to earn the second point, with the following exceptions:

- A response which has a presentation error in the integrand (for example, mismatched or missing parentheses, misplaced exponent) does not earn the first point but would earn the second point for the correct answer. A response which has a presentation error in the integrand and which reports an incorrect answer earns no points.
- A response that presents an integrand of $(\ln(x + 3) - x^4 + 2x^3)^2$.

If the first time the response presents $f(x) - g(x)$ as $j(x) = \ln(x + 3) - x^4 + 2x^3$ occurs in part (c), the response loses a point. If the response already lost a point for this error in part (a) or part (b), then the response is eligible for all remaining points for a correct or consistent answer.

In part (c), the consistent answer using $j(x)$ is 252.187 252.187 (or 252.188).

A response that uses incorrect limits is only eligible for the second point, provided the limits are imported from part (a) in Case 1 or Case 2. In both of these situations, the second point is earned only for answers consistent with the imported limits.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Integrand
- ☐ Answer

Solution:

$$\int_{-2}^B (f(x) - g(x))^2 dx = 5.340102$$

The volume of the solid is 5.340.

Part D

8-4b

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point may be earned by presenting $\frac{dA}{dx} \cdot \frac{dx}{dt}$, $A' \cdot \frac{dx}{dt}$, $A'(x) \cdot \frac{dx}{dt}$, $A'(x) \cdot x'$, $A'(-0.5) \cdot 7$, or $k \cdot 7$, where k is a declared value of $A'(-0.5)$, or any equivalent expression, including

$$2(f(x) - g(x))(f'(x) - g'(x))\frac{dx}{dt}.$$

If a response defines $f(x) - g(x)$ as a function in parts (b) or (c) (for example,

$h(x) = f(x) - g(x)$), then a correct expression for $\frac{dA}{dt}$ (for example, $2h\frac{dh}{dt}$) earns the first point.

A response that imports a function $A(x)$ declared in part (c) is eligible for both points (the answer must be consistent with the imported function $A(x)$).

A response that presents an incorrect function for $A(x)$ that is not imported from part (c) is eligible only for the first point.

Except when $A(x)$ is imported from part (c), the second point is earned only for the correct answer.

A response that does not earn the first point is ineligible to earn the second point except in the special case noted below.

· If the first time the response presents $f(x) - g(x)$ as $j(x) = \ln(x + 3) - x^4 + 2x^3$ occurs in part (d), the response loses a point. If the response already lost a point for this error in part (a), part (b), or part (c), then the response is eligible for all remaining points for a correct or consistent answer. In part (d) the consistent answer using $j(x)$ is 20.287.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ $\frac{dA}{dx} \cdot \frac{dx}{dt}$
- ☐ Answer

Solution:

The cross section has area $A(x) = (f(x) - g(x))^2$.

$$\frac{d}{dt}[A(x)] = \frac{dA}{dx} \cdot \frac{dx}{dt}$$

$$\frac{d}{dt}[A(x)]|_{x=-0.5} = A'(-0.5) \cdot 7 = -9.271842$$

At $x = -0.5$, the area of the cross section above the line is changing at a rate of -9.272 (or -9.271) square units per second.

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58. Let R be the region in the first quadrant under the graph of $y = \frac{x}{x^2+2}$ for $0 \leq x \leq \sqrt{6}$.

(a) Find the area of R .

(b) If the line $x = k$ divides R into two regions of equal area, what is the value of k ?

(c) What is the average value of $y = \frac{x}{x^2+2}$ on the interval $0 \leq x \leq \sqrt{6}$?

Part A

1 point is earned for correct integral

1 point is earned for antiderivative

1 point is earned for evaluation

$$\begin{aligned} A &= \int_0^{\sqrt{6}} \frac{x}{x^2+2} dx \\ &= \frac{1}{2} \ln(x^2+2) \Big|_0^{\sqrt{6}} \\ &= \frac{1}{2} \ln 8 - \frac{1}{2} \ln 2 = \ln 2 \end{aligned}$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for correct integral

1 point is earned for antiderivative

1 point is earned for evaluation

$$\begin{aligned} A &= \int_0^{\sqrt{6}} \frac{x}{x^2+2} dx \\ &= \frac{1}{2} \ln(x^2+2) \Big|_0^{\sqrt{6}} \\ &= \frac{1}{2} \ln 8 - \frac{1}{2} \ln 2 = \ln 2 \end{aligned}$$

Part B

1 point is earned for a correct equation involving integral (s)

1 point is earned for antiderivative

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1 point is earned for evaluation of integral (s)

1 point is earned for finding the value of k

$$\begin{aligned}\frac{1}{2} \ln 2 &= \int_0^k \frac{x}{x^2 + 2} dx \\ &= \frac{1}{2} \ln(x^2 + 2) \Big|_0^k \\ &= \frac{1}{2} \ln(k^2 + 2) - \frac{1}{2} \ln 2\end{aligned}$$

$$\therefore \frac{1}{2} \ln(k^2 + 2) = \frac{1}{2} \ln 2 + \frac{1}{2} \ln 2$$

$$\text{or } \ln(k^2 + 2) = \ln 4$$

$$\therefore k^2 + 2 = 4 \text{ and } k = \sqrt{2}$$



0	1	2	3	4
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The student response earns four of the following points:

1 point is earned for a correct equation involving integral (s)

1 point is earned for antiderivative

1 point is earned for evaluation of integral (s)

1 point is earned for finding the value of k

$$\begin{aligned}\frac{1}{2} \ln 2 &= \int_0^k \frac{x}{x^2 + 2} dx \\ &= \frac{1}{2} \ln(x^2 + 2) \Big|_0^k \\ &= \frac{1}{2} \ln(k^2 + 2) - \frac{1}{2} \ln 2\end{aligned}$$

$$\therefore \frac{1}{2} \ln(k^2 + 2) = \frac{1}{2} \ln 2 + \frac{1}{2} \ln 2$$

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$$\text{or } \ln(k^2 + 2) = \ln 4$$

$$\therefore k^2 + 2 = 4 \text{ and } k = \sqrt{2}$$

Part C

1 point is earned for correct integral

1 point is earned for evaluation

$$\begin{aligned} \text{Average value} &= \frac{1}{\sqrt{6}-0} \int_0^{\sqrt{6}} \frac{x}{x^2 + 2} dx \\ &= \frac{1}{\sqrt{6}} \ln 2 \end{aligned}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for correct integral

1 point is earned for evaluation

$$\begin{aligned} \text{Average value} &= \frac{1}{\sqrt{6}-0} \int_0^{\sqrt{6}} \frac{x}{x^2 + 2} dx \\ &= \frac{1}{\sqrt{6}} \ln 2 \end{aligned}$$

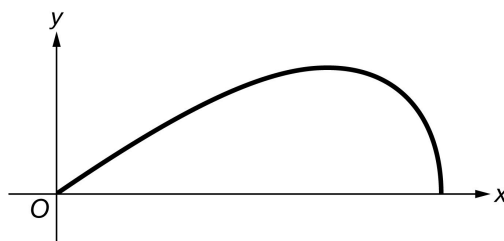
8-4b

59. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



A company designs spinning toys using the family of functions $y = cx\sqrt{4 - x^2}$, where c is a positive constant. The figure above shows the region in the first quadrant bounded by the x -axis and the graph of $y = cx\sqrt{4 - x^2}$, for some c . Each spinning toy is in the shape of the solid generated when such a region is revolved about the x -axis. Both x and y are measured in inches.

(a) Find the area of the region in the first quadrant bounded by the x -axis and the graph of $y = cx\sqrt{4 - x^2}$ for $c = 6$.

(b) It is known that, for $y = cx\sqrt{4 - x^2}$, $\frac{dy}{dx} = \frac{c(4-2x^2)}{\sqrt{4-x^2}}$. For a particular spinning toy, the radius of the largest cross-sectional circular slice is 1.2 inches. What is the value of c for this spinning toy?

(c) For another spinning toy, the volume is 2π cubic inches. What is the value of c for this spinning toy?

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Units are not required for any points in this question and are not read if presented (correctly or incorrectly) in any part of the response.

The first point is earned for presenting $cx\sqrt{4 - x^2}$ or $6x\sqrt{4 - x^2}$ as the integrand in a definite integral. Limits of integration (numeric or alphanumeric) must be presented (as part of the definite integral) but do not need to be correct in order to earn the first point.

If an indefinite integral is presented with an integrand of the correct form, the first point can be earned if the antiderivative (correct or incorrect) is eventually evaluated using the correct limits of integration.

The second point can be earned without the first point. The second point is earned for the presentation of a correct antiderivative of a function of the form $Ax\sqrt{4 - x^2}$, for any nonzero constant A . If the response has subsequent

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errors in simplification of the antiderivative or sign errors, the response will earn the second point but will not earn the third point.

Responses that use u -substitution and have incorrect limits of integration or do not change the limits of integration from x - to u -values are eligible for the second point.

The response is eligible for the third point only if it has earned the second point.

The third point is earned only for the answer 16 or equivalent. In the case where a response only presents an indefinite integral, the use of the correct limits of integration to evaluate the antiderivative must be shown to earn the third point.

The response cannot correct -16 to $+16$ in order to earn the third point; there is no possible reversal here.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Integrand
- ☐ Antiderivative
- ☐ Answer

Solution:

$$6x\sqrt{4-x^2} = 0 \Rightarrow x = 0, x = 2$$

$$\text{Area} = \int_0^2 6x\sqrt{4-x^2} \, dx$$

$$\text{Let } u = 4 - x^2.$$

$$du = -2x \, dx \Rightarrow -\frac{1}{2} du = x \, dx$$

$$x = 0 \Rightarrow u = 4 - 0^2 = 4$$

$$x = 2 \Rightarrow u = 4 - 2^2 = 0$$

$$\begin{aligned} \int_0^2 6x\sqrt{4-x^2} \, dx &= \int_4^0 6 \left(-\frac{1}{2}\right) \sqrt{u} \, du = -3 \int_4^0 u^{1/2} \, du = 3 \int_0^4 u^{1/2} \, du \\ &= 2u^{3/2} \Big|_{u=0}^{u=4} = 2 \cdot 8 = 16 \end{aligned}$$

The area of the region is 16 square inches.

Part B

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Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for setting $\frac{dy}{dx} = 0$, $\frac{c(4-2x^2)}{\sqrt{4-x^2}} = 0$, or $c(4-2x^2) = 0$.

An unsupported $x = \sqrt{2}$ does not earn the first point.

The second point can be earned without the first point but is earned only for the answer $c = 0.6$ with supporting work.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Sets $\frac{dy}{dx} = 0$
- ☐ Answer

Solution:

The cross-sectional circular slice with the largest radius occurs where $cx\sqrt{4-x^2}$ has its maximum on the interval $0 < x < 2$.

$$\frac{dy}{dx} = \frac{c(4-2x^2)}{\sqrt{4-x^2}} = 0 \Rightarrow x = \sqrt{2}$$

$$x = \sqrt{2} \Rightarrow y = c\sqrt{2}\sqrt{4 - (\sqrt{2})^2} = 2c$$

$$2c = 1.2 \Rightarrow c = 0.6$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for presenting an integrand of the form $A(x\sqrt{4-x^2})^2$ in a definite integral with any limits of integration (numeric or alphanumeric) and any nonzero constant A . Mishandling the c will result in the response being ineligible for the fourth point.

The second point can be earned without the first point. The second point is earned for the limits of integration, $x = 0$ and $x = 2$, and the constant π (but not for 2π) as part of an integral with a correct or incorrect integrand.

If an indefinite integral is presented with the correct constant π , the second point can be earned if the antiderivative (correct or incorrect) is evaluated using the correct limits of integration.

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A response that presents $2 = \int_0^2 \left(cx\sqrt{4-x^2} \right)^2 dx$ earns the first and second points.

The third point is earned for presenting a correct antiderivative of the presented integrand of the form $A(x\sqrt{4-x^2})^2$ for any nonzero A . If there are subsequent errors in simplification of the antiderivative, linkage errors, or sign errors, the response will not earn the fourth point.

The fourth point cannot be earned without the third point. The fourth point is earned only for the correct answer. The expression does not need to be simplified to earn the fourth point.



0	1	2	3	4
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The student response accurately includes **all four** of the criteria below.

- ☐ Form of the integrand
- ☐ Limits and constant
- ☐ Antiderivative
- ☐ Answer

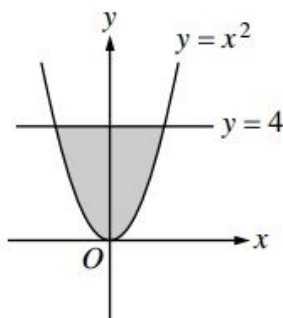
Solution:

$$\begin{aligned}
 \text{Volume} &= \int_0^2 \pi \left(cx\sqrt{4-x^2} \right)^2 dx = \pi c^2 \int_0^2 x^2 (4-x^2) dx \\
 &= \pi c^2 \int_0^2 (4x^2 - x^4) dx = \pi c^2 \left(\frac{4}{3}x^3 - \frac{1}{5}x^5 \right) \Big|_0^2 \\
 &= \pi c^2 \left(\frac{32}{3} - \frac{32}{5} \right) = \frac{64\pi c^2}{15}
 \end{aligned}$$

$$\frac{64\pi c^2}{15} = 2\pi \Rightarrow c^2 = \frac{15}{32} \Rightarrow c = \sqrt{\frac{15}{32}}$$

8-4b

60.



The shaded region, R , is bounded by the graph of $y = x^2$ and the line $y = 4$, as shown in the figure above.

(a) Find the area of R .

(b) Find the volume of the solid generated by revolving R about the x -axis.

(c) There exists a number k , $k > 4$, such that when R is revolved about the line $y = k$, the resulting solid has the same volume as the solid in part (b). Write, but do not solve, an equation involving an integral expression that can be used to find the value of k .

Part A

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned}
 \text{Area} &= \int_{-2}^2 (4 - x^2) dx \\
 &= 2 \int_0^2 (4 - x^2) dx \\
 &= 2 \left[4x - \frac{x^3}{3} \right]_0^2 \\
 &= \frac{32}{3} = 10.666 \text{ or } 10.667
 \end{aligned}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

8-4b

$$\begin{aligned}
 \text{Area} &= \int_{-2}^2 (4 - x^2) dx \\
 &= 2 \int_0^2 (4 - x^2) dx \\
 &= 2 \left[4x - \frac{x^3}{3} \right]_0^2 \\
 &= \frac{32}{3} = 10.666 \text{ or } 10.667
 \end{aligned}$$

Part B

1 point is earned for limits and constant

1 point is earned for integrand

1 point is earned for the answer

$$\begin{aligned}
 \text{Volume} &= \pi \int_{-2}^2 (4^2 - (x^2)^2) dx \\
 &= 2\pi \int_0^2 (16 - x^4) dx \\
 &= 2\pi \left[16x - \frac{x^5}{5} \right]_0^2 \\
 &= \frac{256\pi}{5} = 160.849 \text{ or } 160.850
 \end{aligned}$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for limits and constant

1 point is earned for integrand

1 point is earned for the answer

8-4b

$$\begin{aligned}
 \text{Volume} &= \pi \int_{-2}^2 (4^2 - (x^2)^2) dx \\
 &= 2\pi \int_0^2 (16 - x^4) dx \\
 &= 2\pi \left[16x - \frac{x^5}{5} \right]_0^2 \\
 &= \frac{256\pi}{5} = 160.849 \text{ or } 160.850
 \end{aligned}$$

Part C

1 point is earned for limits and constant

2 points are earned for integrand

<-1> each error

1 point is earned for the equation

$$\pi \int_{-2}^2 [(k - x^2)^2 - (k - 4)^2] dx = \frac{256\pi}{5}$$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for limits and constant

2 points are earned for integrand

<-1> each error

1 point is earned for the equation

$$\pi \int_{-2}^2 [(k - x^2)^2 - (k - 4)^2] dx = \frac{256\pi}{5}$$

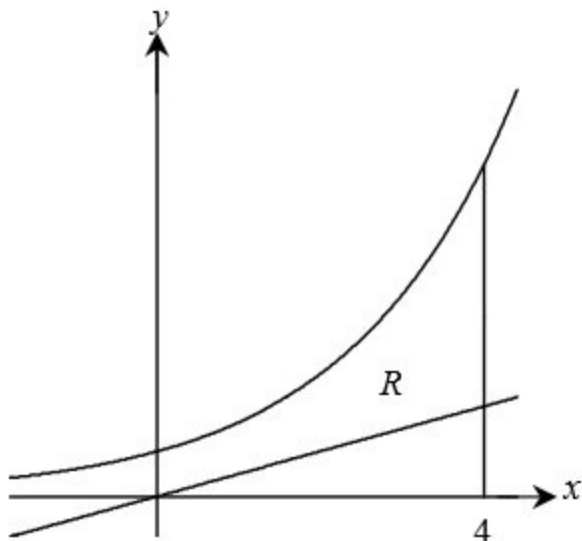
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61. Let R be the region enclosed by the graphs of $y = e^x$, $y = x$ and the lines $x = 0$ and $x = 4$.
- Find the area of R .
 - Find the volume of the solid generated when R is revolved about the x -axis.
 - Set up, but do not integrate, an integral expression in terms of a single variable for the volume of the solid generated when R is revolved about the y -axis.

Part A

Using Geometry (Area of Triangle) 3/3:

2 points are earned for the correct integral where one point is earned for the correct limits and one point is earned for the correct integrand



1 point is earned for the correct antidifferentiation and evaluation

0/1 if integrand is not $ex - x$ or $x - ex$

$$\begin{aligned}
 \text{Area} &= \int_0^4 e^x - x dx \\
 &= e^x - \frac{x^2}{2} \Big|_0^4 \\
 &= \left(e^4 - \frac{16}{2}\right) - (e^0 - 0) \\
 &= e^4 - 9 \\
 &\approx 45.598
 \end{aligned}$$

OR

Using y -axis 3/3:

8-4b

2 points are earned for the correct integral where one point is earned for the correct limits and one point is earned for answering all 3 integrands correct

Note: 0/2 if any limits incorrect

$$\int_0^1 y \, dy + \int_1^4 y - \ln y \, dy + \int_4^{e^4} 4 - \ln y \, dy$$

1 point is earned for the correct answer

≈ 45.598

Note: $k \int_0^4 e^x - x \, dx, k \neq 1$

Student scores 1 - 0 on first two points and is eligible for third point

Note: Student who believes $y = ex$ meets $y = x$

$$\int_0^k e^x - x \, dx + \int_k^4 x - e^x \, dx, 0/2 \text{ Integral}$$

Eligible for third point whether k is a number or parameter



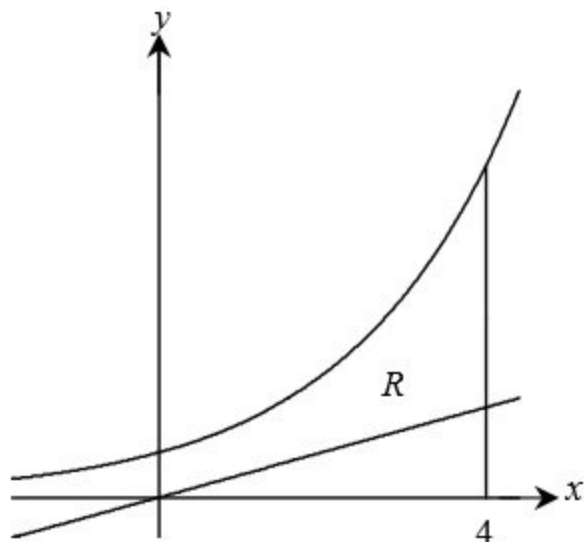
0	1	2	3
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The student response earns all of the following points:

Using Geometry (Area of Triangle) 3/3:

2 points are earned for the correct integral where one point is earned for the correct limits and one point is earned for the correct integrand

8-4b



1 point is earned for the correct antidifferentiation and evaluation

0/1 if integrand is not $ex - x$ or $x - ex$

$$\begin{aligned}
 \text{Area} &= \int_0^4 e^x - x dx \\
 &= e^x - \frac{x^2}{2} \Big|_0^4 \\
 &= \left(e^4 - \frac{16}{2}\right) - (e^0 - 0) \\
 &= e^4 - 9 \\
 &\approx 45.598
 \end{aligned}$$

OR

Using y-axis 3/3:

2 points are earned for the correct integral where one point is earned for the correct limits and one point is earned for answering all 3 integrands correct

Note: 0/2 if any limits incorrect

$$\int_0^1 y dy + \int_1^4 y - \ln y dy + \int_4^{e^4} 4 - \ln y dy$$

1 point is earned for the correct answer

$$\approx 45.598$$

8-4b

Note: $k \int_0^4 e^x - x \, dx, k \neq 1$

Student scores 1 - 0 on first two points and is eligible for third point

Note: Student who believes $y = ex$ meets $y = x$

$$\int_0^k e^x - x \, dx + \int_k^4 x - e^x \, dx, 0/2 \text{ Integral}$$

Eligible for third point whether k is a number or parameter

Part B

Using Geometry (Volume of Cone) 4/4:

3 points are earned for the correct integral where one point is earned for π and correct limits and two points are earned for correctly using $(e^x)^2 - (x)^2$

$$<-1> (e^x - x)^2$$

$$<-1> e^{x^2} - x^2$$

$$<-1> (x)^2 - (e^x)^2$$

$$<-1> e^{x^2}$$

$$\begin{aligned} V_x &= \pi \int_0^4 (e^x)^2 - (x)^2 \, dx \\ &= \pi \int_0^4 e^{2x} - x^2 \, dx \end{aligned}$$

1 point is earned for the correct antidifferentiation and evaluation

0/1 if student doesn't have correct cone volume or doesn't indicate using cone

8-4b

$$\begin{aligned}
 &= \pi \left(\frac{1}{2} e^{2x} - \frac{x^3}{3} \right) \Big|_0^4 \\
 &= \pi \left[\left(\frac{1}{2} e^8 - \frac{64}{3} \right) - \left(\frac{1}{2} e^0 - 0 \right) \right] \\
 &= \pi \left(\frac{1}{2} e^8 - \frac{131}{6} \right) \\
 &\approx 1468.646\pi \approx 4613.886
 \end{aligned}$$

OR

Using y-axis 4/4:

3 points are earned for the correct integral where one points is earned for 2π and correct limits and two points are earned for the correct integrands

<-1> each error

Note: 0/2 if any limits incorrect

$$2\pi \left[\int_0^1 y \cdot y \, dy + \int_1^4 y(y - \ln y) \, dy + \int_4^{e^4} y(4 - \ln y) \, dy \right]$$

1 point is earned for the correct answer

$$\approx 1468.646\pi \approx 4618.886$$

$$\text{Note: } \pi \left[\int_0^k e^{2x} - x^2 \, dx + \int_k^4 x^2 - e^{2x} \, dx \right]$$

Scores 1/3 for integral, and is eligible for fourth point



0	1	2	3	4
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The student response earns all of the following points:

Using Geometry (Volume of Cone) 4/4:

3 points are earned for the correct integral where one point is earned for π and correct limits and two points are earned for correctly using $(e^x)^2 - (x)^2$

$$<-1> (e^x - x)^2$$

8-4b

$$\langle -1 \rangle e^{x^2} - x^2$$

$$\langle -1 \rangle (x)^2 - (e^x)^2$$

$$\langle -1 \rangle e^{x^2}$$

$$\begin{aligned} V_x &= \pi \int_0^4 (e^x)^2 - (x)^2 dx \\ &= \pi \int_0^4 e^{2x} - x^2 dx \end{aligned}$$

1 point is earned for the correct antidifferentiation and evaluation

0/1 if student doesn't have correct cone volume or doesn't indicate using cone

$$\begin{aligned} &= \pi \left(\frac{1}{2} e^{2x} - \frac{x^3}{3} \right) \Big|_0^4 \\ &= \pi \left[\left(\frac{1}{2} e^8 - \frac{64}{3} \right) - \left(\frac{1}{2} e^0 - 0 \right) \right] \\ &= \pi \left(\frac{1}{2} e^8 - \frac{131}{6} \right) \\ &\approx 1468.646\pi \approx 4613.886 \end{aligned}$$

OR

Using y-axis 4/4:

3 points are earned for the correct integral where one point is earned for 2π and correct limits and two points are earned for the correct integrands

$\langle -1 \rangle$ each error

Note: 0/2 if any limits incorrect

$$2\pi \left[\int_0^1 y \cdot y \, dy + \int_1^4 y(y - \ln y) \, dy + \int_4^{e^4} y(4 - \ln y) \, dy \right]$$

1 point is earned for the correct answer

$$\approx 1468.646\pi \approx 4618.886$$

8-4b

Note: $\pi \left[\int_0^k e^{2x} - x^2 dx + \int_k^4 x^2 - e^{2x} dx \right]$

Scores 1/3 for integral, and is eligible for fourth point

Part C

1 point is earned for 2π and correct limits

1 point is earned for integrand

$$V_y = 2\pi \int_0^4 x(e^x - x) dx$$

OR

1 point is earned for $\pi \int_0^1 f(y) + \int_1^4 g(y) + \int_4^{e^4} h(y)$

1 point is earned for all three correct integrands

$$V_y = \pi \left[\int_0^1 y^2 dy + \int_1^4 y^2 - (\ln y)^2 dy + \int_4^{e^4} 16 - (\ln y)^2 dy \right]$$

Note: $2\pi \left[\int_0^k x(e^x - x) dx + \int_k^4 x(x - e^x) dx \right]$

Scores 1/2



0	1	2
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The student response earns all of the following points:

1 point is earned for 2π and correct limits

1 point is earned for integrand

8-4b

$$V_y = 2\pi \int_0^4 x(e^x - x)dx$$

OR

1 point is earned for $\pi \int_0^1 f(y) + \int_1^4 g(y) + \int_4^{e^4} h(y)$

1 point is earned for all three correct integrands

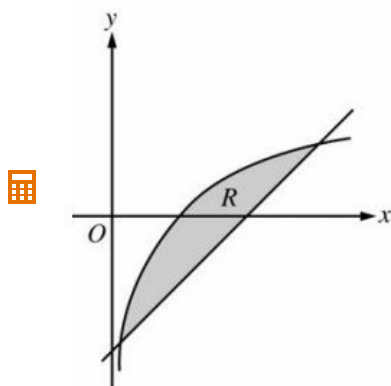
$$V_y = \pi \left[\int_0^1 y^2 dy + \int_1^4 y^2 - (\ln y)^2 dy + \int_4^{e^4} 16 - (\ln y)^2 dy \right]$$

Note: $2\pi \left[\int_0^k x(e^x - x)dx + \int_k^4 x(x - e^x)dx \right]$

Scores 1/2

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62.



Let R be the shaded region bounded by the graph of $y = \ln x$ and the line $y = x - 2$, as shown above.

(a) Find the area of R .

(b) Find the volume of the solid generated when R is rotated about the horizontal line $y = -3$.

(c) Write, but do not evaluate, an integral expression that can be used to find the volume of the solid generated when R is rotated about the y -axis.

Part A

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$\ln(x) = x - 2$ when $x = 0.15859$ and 3.14619 .

Let $S = 0.15859$ and $T = 3.14619$

$$\text{Area of } R = \int_S^T (\ln(x) - (x - 2)) dx = 1.949$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

8-4b

$\ln(x) = x - 2$ when $x = 0.15859$ and 3.14619 .

Let $S = 0.15859$ and $T = 3.14619$

$$\text{Area of } R = \int_S^T (\ln(x) - (x - 2)) dx = 1.949$$

Part B

2 points are earned for the integrand

1 point is earned for the limits, constant, and answer

$$\begin{aligned} \text{Volume} &= \pi \int_S^T ((\ln(x) + 3)^2 - (x - 2 + 3)^2) dx \\ &= 34.198 \text{ or } 34.199 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the limits, constant, and answer

$$\begin{aligned} \text{Volume} &= \pi \int_S^T ((\ln(x) + 3)^2 - (x - 2 + 3)^2) dx \\ &= 34.198 \text{ or } 34.199 \end{aligned}$$

Part C

2 points are earned for the integrand

1 point is earned for the limits and constant

$$\text{Volume} = \pi \int_{S-2}^{T-2} ((y + 2)^2 - (e^y)^2) dy$$

8-4b



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the limits and constant

$$\text{Volume} = \pi \int_{S-2}^{T-2} \left((y+2)^2 - (e^y)^2 \right) dy$$

8-4b

63.  Let f and g be the functions given by $f(x) = e^x$ and $g(x) = \ln x$.
- (a) Find the area of the region enclosed by the graphs of f and g between $x = \frac{1}{2}$ and $x = 1$.
- (b) Find the volume of the solid generated when the region enclosed by the graphs of f and g between $x = \frac{1}{2}$ and $x = 1$ is revolved about the line $y = 4$.
- (c) Let h be the function given by $h(x) = f(x) - g(x)$. Find the absolute minimum value of $h(x)$ on the closed interval $\frac{1}{2} \leq x \leq 1$, and find the absolute maximum value of $h(x)$ on the closed interval $\frac{1}{2} \leq x \leq 1$. Show the analysis that leads to your answers.

Part A

1 point is earned for integral

1 point is earned for the answer

$$\text{Area} = \int_{1/2}^1 (e^x - \ln x) dx = 1.222 \text{ or } 1.223$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

$$\text{Area} = \int_{1/2}^1 (e^x - \ln x) dx = 1.222 \text{ or } 1.223$$

Part B

1 point is earned for limits and constant

2 points are earned for integrand

<-1> each error

1 point is earned for the answer

8-4b

$$\begin{aligned}\text{Volume} &= \pi \int_{1/2}^1 \left((4 - \ln x)^2 - (4 - e^x)^2 \right) dx \\ &= 7.515\pi \text{ or } 23.609\end{aligned}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for limits and constant

2 points are earned for integrand

<-1> each error

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \pi \int_{1/2}^1 \left((4 - \ln x)^2 - (4 - e^x)^2 \right) dx \\ &= 7.515\pi \text{ or } 23.609\end{aligned}$$

Part C

1 point is earned for considering $h'(x) = 0$

1 point is earned for identifying critical point and endpoints as candidates

1 point is earned for the answers

$$\begin{aligned}h'(x) &= f'(x) - g'(x) = e^x - \frac{1}{x} = 0 \\ x &= 0.567143\end{aligned}$$

Absolute minimum value and absolute maximum value occur at the critical point or at the endpoints.

$$h(0.567143) = 2.330$$

$$h(0.5) = 2.3418$$

$$h(1) = 2.718$$

The absolute minimum is 2.330.

8-4b

The absolute maximum is 2.718.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $h'(x) = 0$

1 point is earned for identifying critical point and endpoints as candidates

1 point is earned for the answers

$$h'(x) = f'(x) - g'(x) = e^x - \frac{1}{x} = 0$$
$$x = 0.567143$$

Absolute minimum value and absolute maximum value occur at the critical point or at the endpoints.

$$h(0.567143) = 2.330$$

$$h(0.5) = 2.3418$$

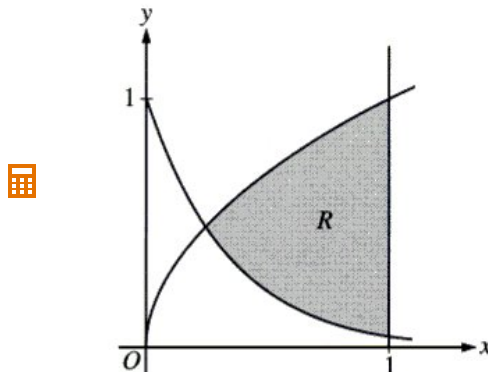
$$h(1) = 2.718$$

The absolute minimum is 2.330.

The absolute maximum is 2.718.

8-4b

64.



Let R be the shaded region bounded by the graphs of $y = \sqrt{x}$ and $y = e^{-3x}$ and the vertical line $x = 1$, as shown in the figure above.

- Find the area of R .
- Find the volume of the solid generated when R is revolved about the horizontal line $y = 1$.
- The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a rectangle whose height is 5 times the length of its base in region R . Find the volume of this solid.

Part A

1 point is earned for the correct integrand

$$\text{Area} = \int_T^1 (\sqrt{x} - e^{-3x}) dx$$

1 point is earned for the correct answer

$$= 0.442 \text{ or } 0.443$$

1 point is earned for correct limits in an integral in (a), (b), (c).

Points of intersection

$$e^{-3x} = \sqrt{x} \text{ at } (T, S) = (0.238734, 0.488604)$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct integrand

8-4b

$$\text{Area} = \int_T^1 (\sqrt{x} - e^{-3x}) dx$$

1 point is earned for the correct answer

$$= 0.442 \text{ or } 0.443$$

1 point is earned for correct limits in an integral in (a), (b), (c).

Points of intersection

$$e^{-3x} = \sqrt{x} \text{ at } (T, S) = (0.238734, 0.488604)$$

Part B

2 points are earned for the correct integrand

<-1> reversal

<-1> error with constant

<-1> omits 1 in one radius

<-2> other errors

$$\text{Volume} = \pi \int_1^T ((1 - e^{-3x})^2 - (1 - \sqrt{x})^2) dx$$

1 point is earned for the correct answer

$$= 0.453\pi \text{ or } 1.423 \text{ or } 1.424$$

1 point is earned for correct limits in an integral in (a), (b), (c).

Points of intersection

$$e^{-3x} = \sqrt{x} \text{ at } (T, S) = (0.238734, 0.488604)$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

8-4b

2 points are earned for the correct integrand

<-1> reversal

<-1> error with constant

<-1> omits 1 in one radius

<-2> other errors

$$\text{Volume} = \pi \int_1^T ((1 - e^{-3x})^2 - (1 - \sqrt{x})^2) dx$$

1 point is earned for the correct answer

$$= 0.453\pi \text{ or } 1.423 \text{ or } 1.424$$

1 point is earned for correct limits in an integral in (a), (b), (c).

Points of intersection

$$e^{-3x} = \sqrt{x} \text{ at } (T, S) = (0.238734, 0.488604)$$

Part C

2 points are earned for the correct integrand

<-1> incorrect but has $\sqrt{x} - e^{-3x}$ as a factor

$$\text{Length} = \sqrt{x} - e^{-3x}$$

$$\text{Height} = 5(\sqrt{x} - e^{-3x})$$

1 point is earned for the correct answer

$$\text{Volume} = \int_T^1 5(\sqrt{x} - e^{-3x})^2 dx = 1.554$$

1 point is earned for correct limits in an integral in (a), (b), (c).

Points of intersection

$$e^{-3x} = \sqrt{x} \text{ at } (T, S) = (0.238734, 0.488604)$$

8-4b



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for the correct integrand

<-1> incorrect but has $\sqrt{x} - e^{-3x}$ as a factor

$$\text{Length} = \sqrt{x} - e^{-3x}$$

$$\text{Height} = 5(\sqrt{x} - e^{-3x})$$

1 point is earned for the correct answer

$$\text{Volume} = \int_T^1 5(\sqrt{x} - e^{-3x})^2 dx = 1.554$$

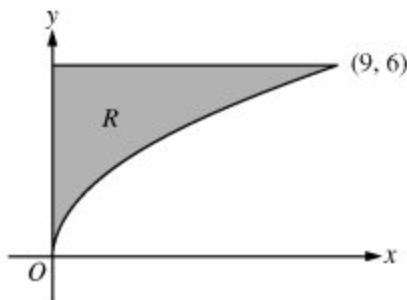
1 point is earned for correct limits in an integral in (a), (b), (c).

Points of intersection

$$e^{-3x} = \sqrt{x} \text{ at } (T, S) = (0.238734, 0.488604)$$

8-4b

65.



Let R be the region in the first quadrant bounded by the graph of $y = 2\sqrt{x}$, the horizontal line $y = 6$, and the y -axis, as shown in the figure above.

(a) Find the area of R .

(b) Write, but do not evaluate, an integral expression that gives the volume of the solid generated when R is rotated about the horizontal line $y = 7$.

(c) Region R is the base of a solid. For each y , where $0 \leq y \leq 6$, the cross section of the solid taken perpendicular to the y -axis is a rectangle whose height is 3 times the length of its base in region R . Write, but do not evaluate, an integral expression that gives the volume of the solid.

Part A

1 point is earned for integrand

1 point is earned for the antiderivative

1 point is earned for the answer

$$\text{Area} = \int_0^9 (6 - 2\sqrt{x}) dx = \left(6x - \frac{4}{3}x^{3/2} \right) \Big|_{x=0}^{x=9} = 18$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for the antiderivative

1 point is earned for the answer

$$\text{Area} = \int_0^9 (6 - 2\sqrt{x}) dx = \left(6x - \frac{4}{3}x^{3/2} \right) \Big|_{x=0}^{x=9} = 18$$

8-4b

Part B

2 points are earned for integrand

1 point is earned for limits and constant

$$\text{Volume} = \pi \int_0^9 \left((7 - 2\sqrt{x})^2 - (7 - 6)^2 \right) dx$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for integrand

1 point is earned for limits and constant

$$\text{Volume} = \pi \int_0^9 \left((7 - 2\sqrt{x})^2 - (7 - 6)^2 \right) dx$$

Part C

2 points are earned for integrand

1 point is earned for the answer

$$\text{Solving } y = 2\sqrt{x} \text{ for } x \text{ yields } x = \frac{y^2}{4}.$$

$$\text{Each rectangular cross section has area } \left(3\frac{y^2}{4} \right) \left(\frac{y^2}{4} \right) = \frac{3}{16}y^4.$$

$$\text{Volume} = \int_0^6 \frac{3}{16}y^4 dy$$

8-4b



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for integrand

1 point is earned for the answer

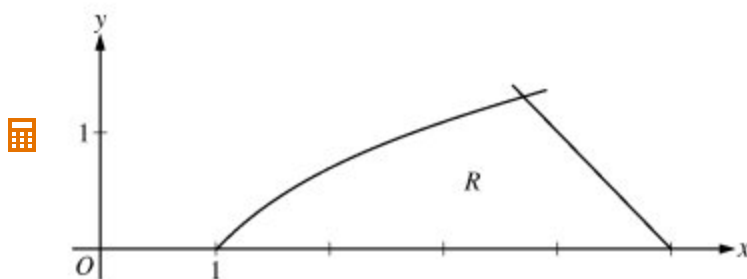
Solving $y = 2\sqrt{x}$ for x yields $x = \frac{y^2}{4}$.

Each rectangular cross section has area $\left(3\frac{y^2}{4}\right)\left(\frac{y^2}{4}\right) = \frac{3}{16}y^4$.

$$\text{Volume} = \int_0^6 \frac{3}{16}y^4 dy$$

8-4b

66.



Let R be the region in the first quadrant bounded by the x -axis and the graphs of $y = \ln x$ and $y = 5 - x$, as shown in the figure above.

(a) Find the area of R .

(b) Region R is the base of a solid. For the solid, each cross section perpendicular to the x -axis is a square. Write, but do not evaluate, an expression involving one or more integrals that gives the volume of the solid.

(c) The horizontal line $y = k$ divides R into two regions of equal area. Write, but do not solve, an equation involving one or more integrals whose solution gives the value of k .

Part A

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$\ln x = 5 - x \Rightarrow x = 3.69344$$

Therefore, the graphs of $y = \ln x$ and $y = 5 - x$ intersect in the first quadrant at the point $(A, B) = (3.69344, 1.30656)$.

$$\begin{aligned} \text{Area} &= \int_0^B (5 - y - e^y) dy \\ &= 2.986 \text{ (or } 2.985) \end{aligned}$$

OR

$$\begin{aligned} \text{Area} &= \int_1^A \ln x \, dx + \int_A^5 (5 - x) \, dx \\ &= 2.986 \text{ (or } 2.985) \end{aligned}$$



0	1	2	3
---	---	---	---

8-4b

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$\ln x = 5 - x \Rightarrow x = 3.69344$$

Therefore, the graphs of $y = \ln x$ and $y = 5 - x$ intersect in the first quadrant at the point $(A, B) = (3.69344, 1.30656)$.

$$\begin{aligned} \text{Area} &= \int_0^B (5 - y - e^y) dy \\ &= 2.986 \text{ (or } 2.985) \end{aligned}$$

OR

$$\begin{aligned} \text{Area} &= \int_1^A \ln x \, dx + \int_A^5 (5 - x) \, dx \\ &= 2.986 \text{ (or } 2.985) \end{aligned}$$

Part B

2 points are earned for the integrands

1 point is earned for expression for total volume

$$\text{Volume} = \int_1^A (\ln x)^2 \, dx + \int_A^5 (5 - x)^2 \, dx$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrands

1 point is earned for expression for total volume

8-4b

$$\text{Volume} = \int_1^A (\ln x)^2 dx + \int_A^5 (5 - x)^2 dx$$

Part C

1 point is earned for the integrand

1 points are earned the limit

1 point is earned for the equation

$$\int_0^k (5 - y - e^y) dy = \frac{1}{2} \cdot 2.986 \left(\text{or } \frac{1}{2} \cdot 2.985 \right)$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

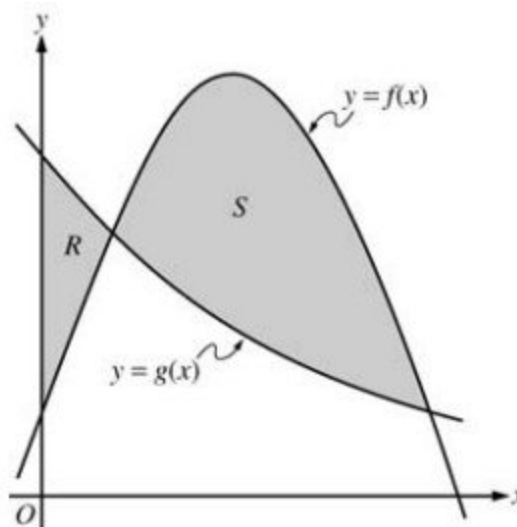
1 points are earned the limit

1 point is earned for the equation

$$\int_0^k (5 - y - e^y) dy = \frac{1}{2} \cdot 2.986 \left(\text{or } \frac{1}{2} \cdot 2.985 \right)$$

8-4b

67.



Let f and g be the functions given by $f(x) = \frac{1}{4} + \sin(\pi x)$ and $g(x) = 4^{-x}$. Let R be the shaded region in the first quadrant enclosed by the y -axis and the graphs of f and g , and let S be the shaded region in the first quadrant enclosed by the graphs of f and g , as shown in the figure above.

- Find the area of R .
- Find the area of S .
- Find the volume of the solid generated when S is revolved about the horizontal line $y = -1$.

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$f(x) = g(x) \text{ when } \frac{1}{4} + \sin(\pi x) = 4^{-x}.$$

f and g intersect when $x = 0.178218$ and when $x = 1$.

Let $a = 0.178218$

$$\int_0^a (g(x) - f(x)) dx = 0.064 \text{ or } 0.065$$



0	1	2	3
---	---	---	---

8-4b

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$f(x) = g(x) \text{ when } \frac{1}{4} + \sin(\pi x) = 4^{-x}.$$

f and g intersect when $x = 0.178218$ and when $x = 1$.

Let $a = 0.178218$

$$\int_0^a (g(x) - f(x)) dx = 0.064 \text{ or } 0.065$$

Part B

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\int_a^1 (f(x) - g(x)) dx = 0.410$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

8-4b

$$\int_a^1 (f(x) - g(x)) dx = 0.410$$

Part C

2 points are earned for the integrand

1 point is earned for the limits, constant, and answer

$$\pi \int_a^1 ((f(x) + 1)^2 - (g(x) + 1)^2) dx = 4.558 \text{ or } 4.559$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

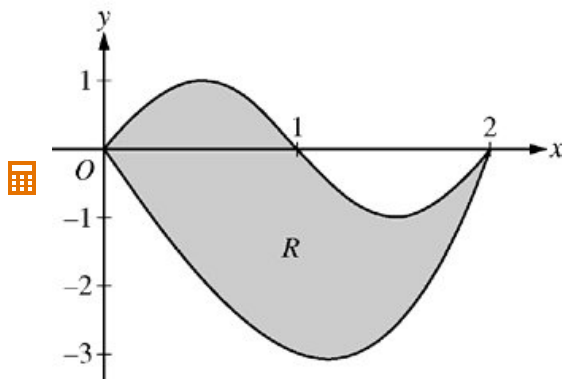
2 points are earned for the integrand

1 point is earned for the limits, constant, and answer

$$\pi \int_a^1 ((f(x) + 1)^2 - (g(x) + 1)^2) dx = 4.558 \text{ or } 4.559$$

8-4b

68.



Let R be the region bounded by the graphs of $y = \sin(\pi x)$ and $y = x^3 - 4x$, as shown in the figure above.

- Find the area of R .
- The horizontal line $y = -2$ splits the region R into two parts. Write, but do not evaluate, an integral expression for the area of the part of R that is below this horizontal line.
- The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a square. Find the volume of this solid.
- The region R models the surface of a small pond. At all points in R at a distance x from the y -axis, the depth of the water is given by $h(x) = 3 - x$. Find the volume of water in the pond.

Part A

The response can earn up to 3 points:

1 point is earned for the correct limits

$$\sin(\pi x) = x^3 - 4x \text{ at } x = 0 \text{ and } x = 2$$

1 point is earned for the correct integrand

1 point is earned for the correct answer

$$\text{Area} = \int_0^2 (\sin(\pi x) - (x^3 - 4x)) dx = 4$$



0	1	2	3
---	---	---	---

The response earns all of the following points:

1 point is earned for the correct limits

$$\sin(\pi x) = x^3 - 4x \text{ at } x = 0 \text{ and } x = 2$$

8-4b

1 point is earned for the correct integrand

1 point is earned for the correct answer

$$\text{Area} = \int_0^2 (\sin(\pi x) - (x^3 - 4x)) dx = 4$$

Part B

The response can earn up to 2 points:

1 point is earned for the correct limits

$$x^3 - 4x = -2 \text{ at } r = 0.5391889 \text{ and } s = 1.6751309$$

1 point is earned for the correct integrand

$$\text{The area of the stated region is } \int_r^s (-2 - (x^3 - 4x)) dx$$



0	1	2
---	---	---

The response earns all of the following points:

1 point is earned for the correct limits

$$x^3 - 4x = -2 \text{ at } r = 0.5391889 \text{ and } s = 1.6751309$$

1 point is earned for the correct integrand

$$\text{The area of the stated region is } \int_r^s (-2 - (x^3 - 4x)) dx$$

Part C

The response can earn up to 2 points:

1 point is earned for the integrand

1 point is earned for the correct answer

$$\text{Volume} = \int_0^2 (\sin(\pi x) - (x^3 - 4x))^2 dx = 9.978$$

8-4b

✓

0	1	2
---	---	---

The response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the correct answer

$$\text{Volume} = \int_0^2 (\sin(\pi x) - (x^3 - 4x))^2 dx = 9.978$$

Part D

The response can earn up to 2 points:

1 point is earned for the integrand

1 point is earned for the correct answer

$$\text{Volume} = \int_0^2 (3 - x) (\sin(\pi x) - (x^3 - 4x)) dx = 8.369 \text{ or } 8.370$$

✓

0	1	2
---	---	---

The response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the correct answer

$$\text{Volume} = \int_0^2 (3 - x) (\sin(\pi x) - (x^3 - 4x)) dx = 8.369 \text{ or } 8.370$$

69. The area of the region bounded by the curve $y = e^{2x}$, the x -axis, the y -axis, and the line $x = 2$ is equal to

(A) $e^4/2 - e$

(B) $e^4/2 - 1$

(C) $e^4/2 - 1/2$

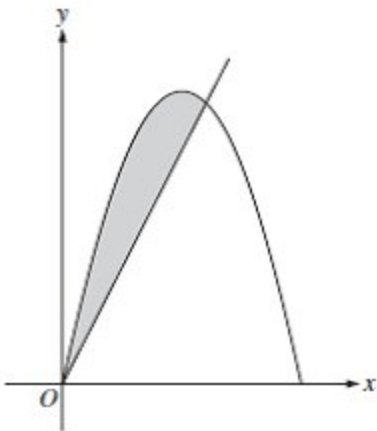
(D) $2e^4 - e$

(E) $2e^4 - 2$

✓

8-4b

70.



The figure above shows the graph of $y = 5x - x^2$ and the graph of the line $y = 2x$. What is the area of the shaded region?

(A) $\frac{25}{6}$

(B) $\frac{9}{2}$

(C) 9

(D) $\frac{27}{2}$

(E) $\frac{45}{2}$



8-4b

71. Consider the curve $y^2=4+x$ and chord AB joining the points $A(-4,0)$ and $B(0,2)$ on the curve. Find the area of the region R enclosed by the curve and the chord AB .

Part B

Method 1:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\begin{aligned}\int_{-4}^0 \left[(\sqrt{4+x} - (\frac{1}{2}x + 2)) \right] dx &= \frac{2}{3}(4+x)^{3/2} - \frac{1}{4}x^2 - 2x \Big|_{-4}^0 \\ &= \frac{2}{3}(4)^{3/2} - (-4 + 8) = \frac{16}{3} - 4 = \frac{4}{3}\end{aligned}$$

Method 2:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\begin{aligned}\int_0^2 [(2y - 4) - (y^2 - 4)] dy &= y^2 - \frac{y^3}{3} \Big|_0^2 \\ &= 4 - \frac{8}{3} = \frac{4}{3}\end{aligned}$$

Method 3:

1 point is earned for the correct bounds and constant

8-4b

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\int_{-4}^0 \sqrt{4+x} dx = \frac{16}{3}; \text{ Area of triangle} = 4$$

$$\text{Area of region} = \frac{16}{3} - 4 = \frac{4}{3}$$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.



0	1	2	3
---	---	---	---

The student response earns three of the following points:

Method 1:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\begin{aligned} \int_{-4}^0 \left[(\sqrt{4+x} - (\frac{1}{2}x + 2)) \right] dx &= \frac{2}{3}(4+x)^{3/2} - \frac{1}{4}x^2 - 2x \Big|_{-4}^0 \\ &= \frac{2}{3}(4)^{3/2} - (-4 + 8) = \frac{16}{3} - 4 = \frac{4}{3} \end{aligned}$$

Method 2:

1 point is earned for the correct bounds and constant

8-4b

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\int_0^2 [(2y - 4) - (y^2 - 4)] dy = y^2 - \frac{y^3}{3} \Big|_0^2$$
$$= 4 - \frac{8}{3} = \frac{4}{3}$$

Method 3:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

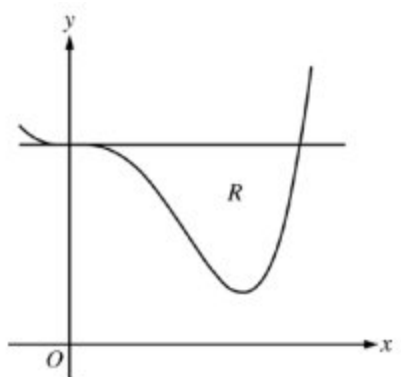
$$\int_{-4}^0 \sqrt{4+x} dx = \frac{16}{3}; \text{ Area of triangle} = 4$$

$$\text{Area of region} = \frac{16}{3} - 4 = \frac{4}{3}$$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.

8-4b

72.



Let R be the region enclosed by the graph of $f(x) = x^4 - 2.3x^3 + 4$ and the horizontal line $y = 4$, as shown in the figure above.

The vertical line $x = k$ divides R into two regions with equal areas. Write, but do not solve, an equation involving integral expressions whose solution gives the value k .

Part C

1 point is earned for the area of one region

1 point is earned for the equation

$$\int_0^k (4 - f(x)) dx = \int_k^{2.3} (4 - f(x)) dx$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the area of one region

1 point is earned for the equation

$$\int_0^k (4 - f(x)) dx = \int_k^{2.3} (4 - f(x)) dx$$

8-4b

73.

Distance from the river's edge (feet)	0	8	14	22	24
Depth of the water (feet)	0	7	8	2	0

A scientist measures the depth of the Doe River at Picnic Point. The river is 24 feet wide at this location. The measurements are taken in a straight line perpendicular to the edge of the river. The data are shown in the table above. The velocity of the water at Picnic Point, in feet per minute, is modeled by

$$v(t) = 16 + 2 \sin(\sqrt{t+10}) \text{ for } 0 \leq t \leq 120 \text{ minutes.}$$



The scientist proposes the function f , given by $f(x) = 8\sin(\pi x/24)$, as a model for the depth of the water, in feet, at Picnic Point x feet from the river's edge. Find the area of the cross section of the river at Picnic Point based on this model.

Part a

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^{24} 8 \sin\left(\frac{\pi x}{24}\right) dx$$

$$= 122.230 \text{ or } 122.231 \text{ ft}^2$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

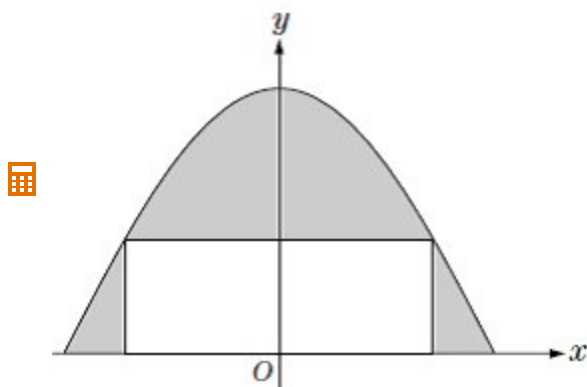
1 point is earned for the answer

$$\int_0^{24} 8 \sin\left(\frac{\pi x}{24}\right) dx$$

$$= 122.230 \text{ or } 122.231 \text{ ft}^2$$

8-4b

74.



A rectangle with one side on the x -axis has its upper vertices on the graph of $y = \cos x$, as shown in the figure above. What is the minimum area of the shaded region?

(A) 0.799

(B) 0.878

(C) 1.140

(D) 1.439

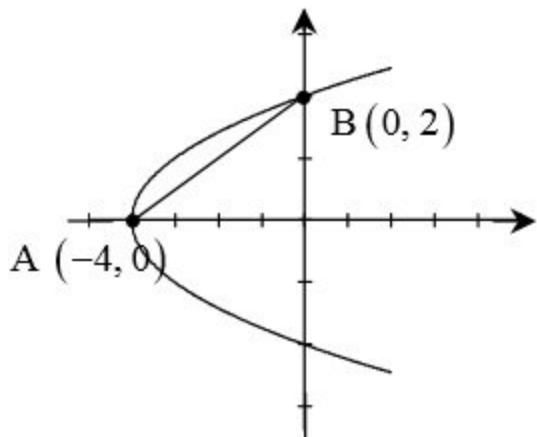
(E) 2.000



8-4b

75. Consider the curve $y^2 = 4 + x$ and chord AB joining the points $A(-4, 0)$ and $B(0, 2)$ on the curve.
- Find the x - and y -coordinates of the point on the curve where the tangent line is parallel to chord AB .
 - Find the area of the region R enclosed by the curve and the chord AB .
 - Find the volume of the solid generated when the region R , defined in part (b), is revolved about the x -axis.

Part A



$$\text{slope of } AB = \frac{2}{4} = \frac{1}{2}$$

Method 1:

1 point is earned for the correct differentiation

1 point is earned for correctly setting student's derivative = student's slope of AB

$$y = \sqrt{4 + x}; \frac{dy}{dx} = \frac{1}{2\sqrt{4 + x}}; \frac{1}{2\sqrt{4 + x}} = \frac{1}{2}$$

1 point is earned for the correct answer

$$\text{so } x = -3, y = 1$$

Method 2:

1 point is earned for the correct differentiation

1 point is earned for correctly setting student's derivative = student's slope of AB

8-4b

$$2y \frac{dy}{dx} = 1, \text{ so } 2y \left(\frac{1}{2} \right) = 1$$

1 point is earned for the correct answer

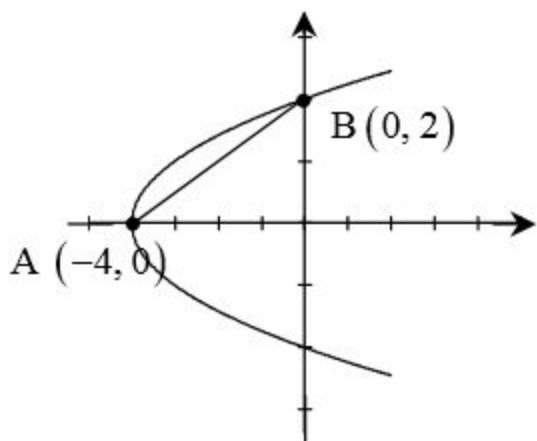
and $y = 1, x = -3$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.



0	1	2	3
---	---	---	---

The student response earns all of the following points:



$$\text{slope of } AB = \frac{2}{4} = \frac{1}{2}$$

Method 1:

1 point is earned for the correct differentiation

1 point is earned for correctly setting student's derivative = student's slope of AB

$$y = \sqrt{4+x}; \frac{dy}{dx} = \frac{1}{2\sqrt{4+x}}; \frac{1}{2\sqrt{4+x}} = \frac{1}{2}$$

1 point is earned for the correct answer

so $x = -3, y = 1$

Method 2:

8-4b

1 point is earned for the correct differentiation

1 point is earned for correctly setting student's derivative = student's slope of AB

$$2y \frac{dy}{dx} = 1, \text{ so } 2y \left(\frac{1}{2} \right) = 1$$

1 point is earned for the correct answer

and $y = 1, x = -3$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.

Part BMethod 1:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\begin{aligned} \int_{-4}^0 \left[(\sqrt{4+x} - \left(\frac{1}{2}x + 2 \right)) \right] dx &= \frac{2}{3}(4+x)^{3/2} - \frac{1}{4}x^2 - 2x \Big|_{-4}^0 \\ &= \frac{2}{3}(4)^{3/2} - (-4 + 8) = \frac{16}{3} - 4 = \frac{4}{3} \end{aligned}$$

Method 2:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

8-4b

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\int_0^2 [(2y - 4) - (y^2 - 4)] dy = y^2 - \frac{y^3}{3} \Big|_0^2$$

$$= 4 - \frac{8}{3} = \frac{4}{3}$$

Method 3:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\int_{-4}^0 \sqrt{4+x} dx = \frac{16}{3}; \text{ Area of triangle} = 4$$

$$\text{Area of region} = \frac{16}{3} - 4 = \frac{4}{3}$$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

Method 1:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

8-4b

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\begin{aligned}\int_{-4}^0 \left[(\sqrt{4+x} - \left(\frac{1}{2}x + 2 \right)) \right] dx &= \frac{2}{3}(4+x)^{3/2} - \frac{1}{4}x^2 - 2x \Big|_{-4}^0 \\ &= \frac{2}{3}(4)^{3/2} - (-4 + 8) = \frac{16}{3} - 4 = \frac{4}{3}\end{aligned}$$

Method 2:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\begin{aligned}\int_0^2 [(2y-4) - (y^2-4)] dy &= y^2 - \frac{y^3}{3} \Big|_0^2 \\ &= 4 - \frac{8}{3} = \frac{4}{3}\end{aligned}$$

Method 3:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

8-4b

1 point is earned for the correct answer

$$\int_{-4}^0 \sqrt{4+x} dx = \frac{16}{3}; \text{ Area of triangle} = 4$$

$$\text{Area of region} = \frac{16}{3} - 4 = \frac{4}{3}$$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.

Part CMethod 1:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0 of 1 if integral not of the form $\int_a^b ([f(x)]^2 - [g(x)]^2) dx$ or $\int_c^d y[f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\begin{aligned} & \pi \int_{-4}^0 \left((\sqrt{4+x})^2 - \left(\frac{1}{2}x + 2 \right)^2 \right) dx \\ &= \pi \int_{-4}^0 \left(4 + x - \left(\frac{1}{4}x^2 + 2x + 4 \right) \right) dx \end{aligned}$$

$$= \pi(8 - 16/3) = 8\pi/3 \approx 8.378$$

Method 2:

1 point is earned for the correct bounds and constant

8-4b

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0 of 1 if integral not of the form $\int_a^b ([f(x)]^2 - [g(x)]^2) dx$ or $\int_c^d y[f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\int_0^2 2\pi y [(2y - 4) - (y^2 - 4)] dy = \frac{8\pi}{3}$$

Method 3:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0 of 1 if integral not of the form $\int_a^b ([f(x)]^2 - [g(x)]^2) dx$ or $\int_c^d y[f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\pi \int_{-4}^0 (\sqrt{4+x})^2 dx = \pi \left(4x + \frac{x^2}{2} \right) \bigg|_{-4}^0$$

$$= 0 - \pi(-16 + 8) = 8\pi$$

8-4b

$$\text{Volume of cone} = 1/3 \pi (2)^2 (4) = 16\pi/3$$

$$\text{Volume} = 8\pi - 16\pi/3 = 8\pi/3$$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

Method 1:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0 of 1 if integral not of the form $\int_a^b ([f(x)]^2 - [g(x)]^2) dx$ or $\int_c^d y [f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\begin{aligned} & \pi \int_{-4}^0 \left((\sqrt{4+x})^2 - \left(\frac{1}{2}x + 2 \right)^2 \right) dx \\ &= \pi \int_{-4}^0 \left(4 + x - \left(\frac{1}{4}x^2 + 2x + 4 \right) \right) dx \end{aligned}$$

$$= \pi(8 - 16/3) = 8\pi/3 \approx 8.378$$

Method 2:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

8-4b

1 point is earned for the correct antidifferentiation and evaluation

Note: 0 of 1 if integral not of the form $\int_a^b ([f(x)]^2 - [g(x)]^2) dx$ or $\int_c^d y [f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\int_0^2 2\pi y [(2y - 4) - (y^2 - 4)] dy = \frac{8\pi}{3}$$

Method 3:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0 of 1 if integral not of the form $\int_a^b ([f(x)]^2 - [g(x)]^2) dx$ or $\int_c^d y [f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\pi \int_{-4}^0 (\sqrt{4+x})^2 dx = \pi \left(4x + \frac{x^2}{2} \right) \bigg|_{-4}^0$$

$$= 0 - \pi(-16 + 8) = 8\pi$$

$$\text{Volume of cone} = 1/3 \pi (2)^2 (4) = 16\pi/3$$

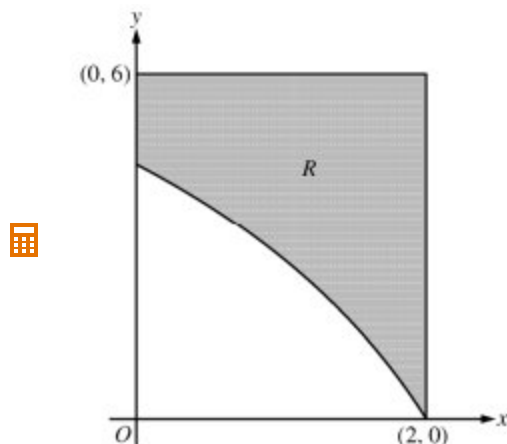
8-4b

$$\text{Volume} = 8\pi - 16\pi/3 = 8\pi/3$$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.

8-4b

76.



In the figure above, R is the shaded region in the first quadrant bounded by the graph of $y = 4\ln(3 - x)$, the horizontal line $y = 6$, and the vertical line $x = 2$.

- Find the area of R .
- Find the volume of the solid generated when R is revolved about the horizontal line $y = 8$.
- The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a square. Find the volume of the solid.

Part A

1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^2 (6 - 4\ln(3 - x)) dx = 6.816 \text{ or } 6.817$$

1 point is earned for correct limits in an integral in (a), (b), or (c)



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the answer

8-4b

$$\int_0^2 (6 - 4\ln(3 - x))dx = 6.816 \text{ or } 6.817$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

Part B

2 points are earned for the integrand

1 point is earned for the answer

$$\pi \int_0^2 \left((8 - 4\ln(3 - x))^2 - (8 - 6)^2 \right) dx$$

$$= 168.179 \text{ or } 168.180$$

1 point is earned for correct limits in an integral in (a), (b), or (c)



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the answer

$$\pi \int_0^2 \left((8 - 4\ln(3 - x))^2 - (8 - 6)^2 \right) dx$$

$$= 168.179 \text{ or } 168.180$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

Part C

2 points are earned for the integrand

1 point is earned for the answer

8-4b

$$\int_0^2 (6 - 4\ln(3 - x))^2 dx = 26.266 \text{ or } 26.267$$

1 point is earned for correct limits in an integral in (a), (b), or (c)



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the answer

$$\int_0^2 (6 - 4\ln(3 - x))^2 dx = 26.266 \text{ or } 26.267$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

8-4b

77. Let R be the region between the graphs of $y = 1 + \sin(\pi x)$ and $y = x^2$ from $x = 0$ to $x = 1$.
- (a) Find the area of R .
- (b) Set up, but do not integrate an integral expression in terms of a single variable for the volume of the solid generated when R is revolved about the x -axis.
- (c) Set up, but do not integrate an integral expression in terms of a single variable for the volume of the solid generated when R is revolved about the y -axis.

Part A

2 points are earned for integral

1 point is **deducted** for limit error

2 points are **deducted** for integrand error

2 points are **deducted** for reversed integral

3 points are earned for antiderivative and evaluate

3 points are **deducted** for constant term

2 points are **deducted** for chain rule error

1 point is **deducted** for each other error

$$\begin{aligned}
 A &= \int_0^1 1 + \sin \pi x - x^2 dx \\
 &= \left(x - \frac{1}{\pi} \cos \pi x - \frac{1}{3} x^3 \right) \Big|_0^1 \\
 &= \left(1 - \frac{1}{\pi}(-1) - \frac{1}{3} \right) - \left(0 - \frac{1}{\pi} - 0 \right) \\
 &= \frac{2}{3} + \frac{2}{\pi}
 \end{aligned}$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

2 points are earned for integral

1 point is **deducted** for limit error

2 points are **deducted** for integrand error

8-4b

2 points are **deducted** for reversed integral

3 points are earned for antiderivative and evaluate

3 points are **deducted** for constant term

2 points are **deducted** for chain rule error

1 point is **deducted** for each other error

$$\begin{aligned}
 A &= \int_0^1 1 + \sin \pi x - x^2 dx \\
 &= \left(x - \frac{1}{\pi} \cos \pi x - \frac{1}{3} x^3 \right) \Big|_0^1 \\
 &= \left(1 - \frac{1}{\pi}(-1) - \frac{1}{3} \right) - \left(0 - \frac{1}{\pi} - 0 \right) \\
 &= \frac{2}{3} + \frac{2}{\pi}
 \end{aligned}$$

Part B

2 points are earned for integral expression

1 point is **deducted** for incorrect limits

$$V = \pi \int_0^1 (1 + \sin \pi x)^2 - x^4 dx$$

or

$$2\pi \int_0^1 y^{3/2} dy + 2\pi \int_1^2 y \left(1 - \frac{2}{\pi} \arcsin(y-1) \right) dy$$



0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for integral expression

1 point is **deducted** for incorrect limits

8-4b

$$V = \pi \int_0^1 (1 + \sin \pi x)^2 - x^4 dx$$

or

$$2\pi \int_0^1 y^{3/2} dy + 2\pi \int_1^2 y \left(1 - \frac{2}{\pi} \arcsin(y-1)\right) dy$$

Part C

2 points are earned for integral

1 point is **deducted** for π or 2π error2 points are **deducted** for integrand error (including reverse)1 point is **deducted** for one or more limit error

$$V = 2\pi \int_0^1 x(1 + \sin(\pi x) - x^2) dx$$

or

$$\pi \int_0^1 y dy + \pi \int_1^2 \left(1 - \frac{1}{\pi} \arcsin(y-1)\right)^2 - \left(\frac{1}{\pi} \arcsin(y-1)\right)^2 dy$$



0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for integral

1 point is **deducted** for π or 2π error2 points are **deducted** for integrand error (including reverse)1 point is **deducted** for one or more limit error

8-4b

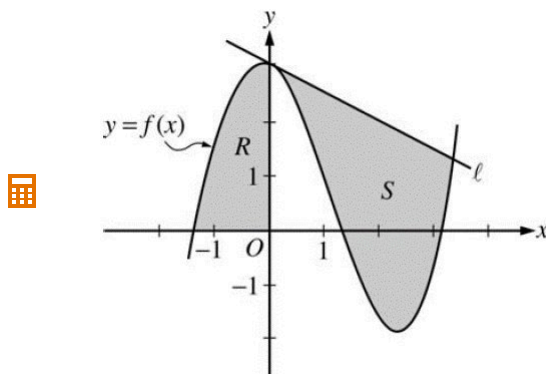
$$V = 2\pi \int_0^1 x(1 + \sin(\pi x) - x^2) dx$$

or

$$\pi \int_0^1 y dy + \pi \int_1^2 \left(1 - \frac{1}{\pi} \arcsin(y-1)\right)^2 - \left(\frac{1}{\pi} \arcsin(y-1)\right)^2 dy$$

8-4b

78.



Let f be the function given by $f(x) = \frac{x^3}{4} - \frac{x^2}{3} - \frac{x}{2} + 3 \cos x$. Let R be the shaded region in the second quadrant bounded by the graph of f , and let S be the shaded region bounded by the graph of f and line ℓ , the line tangent to the graph of f at $x = 0$, as shown above.

- Find the area of R .
- Find the volume of the solid generated when R is rotated about the horizontal line $y = -2$.
- Write, but do not evaluate, an integral expression that can be used to find the area of S .

Part A

1 point is earned for integral

1 point is earned for the answer

For $x < 0$, $f(x) = 0$ when $x = -1.37312$

Let $P = -1.37312$.

$$\text{Area of } R = \int_P^0 f(x) dx = 2.903$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

For $x < 0$, $f(x) = 0$ when $x = -1.37312$

Let $P = -1.37312$.

8-4b

$$\text{Area of } R = \int_P^0 f(x)dx = 2.903$$

Part B

1 point is earned for limits and constant

2 points are earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_P^0 \left((f(x) + 2)^2 - 4 \right) dx = 59.361$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for limits and constant

2 points are earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_P^0 \left((f(x) + 2)^2 - 4 \right) dx = 59.361$$

Part C

1 point is earned for the tangent line

1 point is earned for the integrand

1 point is earned for the limits

The equation of the tangent line ℓ is $y = 3 - \frac{1}{2}x$.

The graph of f and line ℓ intersect at $A = 3.38987$.

8-4b

$$\text{Area of } S = \int_0^A \left(\left(3 - \frac{1}{2}x \right) - f(x) \right) dx$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the tangent line

1 point is earned for the integrand

1 point is earned for the limits

The equation of the tangent line ℓ is $y = 3 - \frac{1}{2}x$.

The graph of f and line ℓ intersect at $A = 3.38987$.

$$\text{Area of } S = \int_0^A \left(\left(3 - \frac{1}{2}x \right) - f(x) \right) dx$$

8-4b

79. Let R be the region enclosed by the graphs of $y = e^x$, $y = (x - 1)^2$, and the line $x = 1$.
- Find the area of R .
 - Find the volume of the solid generated when R is revolved about the x -axis.
 - Set up, but do not integrate, an integral expression in terms of a single variable for the volume of the solid generated when R is revolved about the y -axis.

Part A

1 point is earned for definite integral

1 point is **deducted** for incorrect limits

1 point is **deducted** for reversal

1 point is earned for anti derivative

1 point is earned for evaluation

$$\begin{aligned}
 A &= \int_0^1 e^x - (x - 1)^2 dx \\
 &= \int_0^1 e^x - x^2 + 2x - 1 dx \\
 &= e^x \Big|_0^1 - \frac{1}{3}(x - 1)^3 \Big|_0^1 \\
 &= (e - 1) - \frac{1}{3} = e - \frac{4}{3}
 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for definite integral

1 point is **deducted** for incorrect limits

1 point is **deducted** for reversal

1 point is earned for anti derivative

1 point is earned for evaluation

8-4b

$$\begin{aligned}
 A &= \int_0^1 e^x - (x-1)^2 dx \\
 &= \int_0^1 e^x - x^2 + 2x - 1 dx \\
 &= e^x \Big|_0^1 - \frac{1}{3}(x-1)^3 \Big|_0^1 \\
 &= (e-1) - \frac{1}{3} = e - \frac{4}{3}
 \end{aligned}$$

Part B

2 points are earned for definite integral

2 points are **deducted** for faulty f, g, or u

1 point is earned for correct antiderivative

1 point is earned for evaluation of anti derivative

$$\begin{aligned}
 V &= \pi \int_0^1 e^{2x} - (x-1)^4 dx \\
 &= \pi \left[\frac{e^{2x}}{2} \right]_0^1 - \pi \left[\frac{1}{5}(x-1)^5 \right]_0^1 \\
 &= \pi \left[\left(\frac{e^2}{2} - \frac{1}{2} \right) - \frac{1}{5} \right] = \pi \left(\frac{e^2}{2} - \frac{7}{10} \right)
 \end{aligned}$$

or

$$\begin{aligned}
 V &= 2\pi \int_0^1 y[1 - (1 - \sqrt{y})] dy + 2\pi \int_1^e y(1 - \ln y) dy \\
 &= 2\pi \cdot \frac{2}{5} y^{5/2} \Big|_0^1 + 2\pi \left[\frac{1}{2} y^2 - \left(\frac{1}{2} y^2 \ln y - \frac{1}{4} y^2 \right) \right]_1^e \\
 &= \frac{4}{5} \pi + 2\pi \left[\frac{1}{4} e^2 - \frac{3}{4} \right] = \pi \left(\frac{e^2}{2} - \frac{7}{10} \right)
 \end{aligned}$$



0	1	2	3
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The student response earns all of the following points:

2 points are earned for definite integral

8-4b

2 points are **deducted** for faulty f, g, or u

1 point is earned for correct antiderivative

1 point is earned for evaluation of anti derivative

$$\begin{aligned} V &= \pi \int_0^1 e^{2x} - (x-1)^4 dx \\ &= \pi \left[\frac{e^{2x}}{2} \right]_0^1 - \pi \left[\frac{1}{5} (x-1)^5 \right]_0^1 \\ &= \pi \left[\left(\frac{e^2}{2} - \frac{1}{2} \right) - \frac{1}{5} \right] = \pi \left(\frac{e^2}{2} - \frac{7}{10} \right) \end{aligned}$$

or

$$\begin{aligned} V &= 2\pi \int_0^1 y [1 - (1 - \sqrt{y})] dy + 2\pi \int_1^e y (1 - \ln y) dy \\ &= 2\pi \cdot \frac{2}{5} y^{5/2} \Big|_0^1 + 2\pi \left[\frac{1}{2} y^2 - \left(\frac{1}{2} y^2 \ln y - \frac{1}{4} y^2 \right) \right]_1^e \\ &= \frac{4}{5} \pi + 2\pi \left[\frac{1}{4} e^2 - \frac{3}{4} \right] = \pi \left(\frac{e^2}{2} - \frac{7}{10} \right) \end{aligned}$$

Part C

2 points are earned for Definite integral

2 points are **deducted** for f, g, or u error

1 point is **deducted** for limit error, K error and integrand reversal

$$V = 2\pi \int_0^1 x [e^x - (x-1)^2] dx$$

or

$$V = \pi \int_0^1 1 - (1 - \sqrt{y})^2 dy + \pi \int_1^e 1 - (\ln y)^2 dy$$



0	1	2
---	---	---

8-4b

The student response earns all of the following points:

2 points are earned for Definite integral

2 points are **deducted** for f, g, or u error

1 point is **deducted** for limit error, K error and integrand reversal

$$V = 2\pi \int_0^1 x \left[e^x - (x-1)^2 \right] dx$$

or

$$V = \pi \int_0^1 1 - (1 - \sqrt{y})^2 dy + \pi \int_1^e 1 - (\ln y)^2 dy$$

8-4b

80.  Let R be the region in the first and second quadrants bounded above by the graph of $y = \frac{20}{1+x^2}$ and below by the horizontal line $y = 2$.

- (a) Find the area of R .
- (b) Find the volume of the solid generated when R is rotated about the x -axis.
- (c) The region R is the base of a solid. For this solid, the cross sections perpendicular to the x -axis are semicircles. Find the volume of this solid.

Part A

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Area} = \int_{-3}^3 \left(\frac{20}{1+x^2} - 2 \right) dx = 37.961 \text{ or } 37.962$$

1 point is earned for the correct limits in an integral in (a), (b), or (c)

$$\frac{20}{1+x^2} = 2 \text{ when } x = \pm 3$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Area} = \int_{-3}^3 \left(\frac{20}{1+x^2} - 2 \right) dx = 37.961 \text{ or } 37.962$$

1 point is earned for the correct limits in an integral in (a), (b), or (c)

$$\frac{20}{1+x^2} = 2 \text{ when } x = \pm 3$$

Part B

8-4b

2 points are earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_{-3}^3 \left(\left(\frac{20}{1+x^2} \right)^2 - 2^2 \right) dx = 1871.190$$

1 point is earned for the correct limits in an integral in (a), (b), or (c)

$$\frac{20}{1+x^2} = 2 \text{ when } x = \pm 3$$



0	1	2	3	4
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The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_{-3}^3 \left(\left(\frac{20}{1+x^2} \right)^2 - 2^2 \right) dx = 1871.190$$

1 point is earned for the correct limits in an integral in (a), (b), or (c)

$$\frac{20}{1+x^2} = 2 \text{ when } x = \pm 3$$

Part C

2 points are earned for the integrand

1 point is earned for the answer

8-4b

$$\begin{aligned}\text{Volume} &= \frac{\pi}{2} \int_{-3}^3 \left(\frac{1}{2} \left(\frac{20}{1+x^2} - 2 \right) \right)^2 dx \\ &= \frac{\pi}{8} \int_{-3}^3 \left(\frac{20}{1+x^2} - 2 \right)^2 dx = 174.268\end{aligned}$$

1 point is earned for the correct limits in an integral in (a), (b), or (c)

$$\frac{20}{1+x^2} = 2 \text{ when } x = \pm 3$$



0	1	2	3	4
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The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \frac{\pi}{2} \int_{-3}^3 \left(\frac{1}{2} \left(\frac{20}{1+x^2} - 2 \right) \right)^2 dx \\ &= \frac{\pi}{8} \int_{-3}^3 \left(\frac{20}{1+x^2} - 2 \right)^2 dx = 174.268\end{aligned}$$

1 point is earned for the correct limits in an integral in (a), (b), or (c)

$$\frac{20}{1+x^2} = 2 \text{ when } x = \pm 3$$

8-4b

81. Let R be the region enclosed by the graph of $y = x^2 / x^2 + 1$, the line $x = 1$, and the x -axis.

(a) Find the area of R .

(b) Find the volume of the solid generated when R is rotated about the y -axis.

Part A

1 point is earned for bounds

1 point is earned for integrand

2 points are earned for antiderivative

2 points **deducted** for trig calculation error

1 point is earned for evaluation

$$\begin{aligned}\text{Area} &= \int_0^1 \frac{x^2}{x^2 + 1} dx \\ &= \int_0^1 1 - \frac{1}{x^2 + 1} dx \\ &= x - \arctan x \Big|_0^1 \\ &= 1 - \frac{\pi}{4}\end{aligned}$$



0	1	2	3	4	5
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The student response earns all of the following points:

1 point is earned for bounds

1 point is earned for integrand

2 points are earned for antiderivative

2 points **deducted** for trig calculation error

1 point is earned for evaluation

8-4b

$$\begin{aligned}
 \text{Area} &= \int_0^1 \frac{x^2}{x^2 + 1} dx \\
 &= \int_0^1 1 - \frac{1}{x^2 + 1} dx \\
 &= x - \arctan x \Big|_0^1 \\
 &= 1 - \frac{\pi}{4}
 \end{aligned}$$

Part B

1 point is earned for setup

2 point are earned for antiderivative

1 point is earned for evaluation

$$\begin{aligned}
 \text{Volume} &= 2\pi \int_0^1 x \left(\frac{x^2}{x^2 + 1} \right) dx \\
 &= 2\pi \int_0^1 x - \frac{x}{x^2 + 1} dx \\
 &= 2\pi \left(\frac{x^2}{2} - \frac{1}{2} \ln |x^2 + 1| \right) \Big|_0^1 \\
 &= \pi (1 - \ln 2)
 \end{aligned}$$

or

$$\begin{aligned}
 \text{Volume} &= \pi \int_0^{\frac{1}{2}} x \left(1 - \frac{y}{1 - y} \right) dy \\
 &= \pi (2y + \ln |y - 1|) \Big|_0^{\frac{1}{2}} \\
 &= \pi (1 - \ln 2)
 \end{aligned}$$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for setup

2 point are earned for antiderivative

8-4b

1 point is earned for evaluation

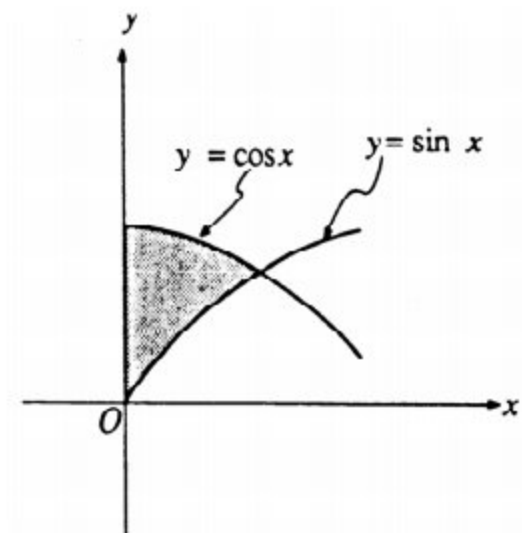
$$\begin{aligned}\text{Volume} &= 2\pi \int_0^1 x \left(\frac{x^2}{x^2 + 1} \right) dx \\ &= 2\pi \int_0^1 x - \frac{x}{x^2 + 1} dx \\ &= 2\pi \left(\frac{x^2}{2} - \frac{1}{2} \ln |x^2 + 1| \right) \Big|_0^1 \\ &= \pi (1 - \ln 2)\end{aligned}$$

or

$$\begin{aligned}\text{Volume} &= \pi \int_0^{\frac{1}{2}} x \left(1 - \frac{y}{1 - y} \right) dy \\ &= \pi (2y + \ln |y - 1|) \Big|_0^{\frac{1}{2}} \\ &= \pi (1 - \ln 2)\end{aligned}$$

8-4b

82.



Let R be the shaded region in the first quadrant enclosed by the y -axis and the graphs of $y = \sin x$ and $y = \cos x$, as shown in the figure above.

- Find the area of R .
- Find the volume of the solid generated when R is revolved about the x -axis.
- Find the volume of the solid whose base is R and whose cross sections cut by planes perpendicular to the x -axis are squares.

Part A

1 point is earned for limits

1 point is earned for integrand

1 point is earned for antiderivative and evaluation

$$\begin{aligned}
 \text{Area} &= \int_0^{\pi/4} \cos x - \sin x \, dx \\
 &= (\sin x + \cos x) \Big|_0^{\pi/4} \\
 &= \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \right) - (0 + 1) \\
 &= \sqrt{2} - 1
 \end{aligned}$$



0	1	2	3
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The student response earns all of the following points:

8-4b

1 point is earned for limits

1 point is earned for integrand

1 point is earned for antiderivative and evaluation

$$\begin{aligned}
 \text{Area} &= \int_0^{\pi/4} \cos x - \sin x \, dx \\
 &= (\sin x + \cos x) \Big|_0^{\pi/4} \\
 &= \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \right) - (0 + 1) \\
 &= \sqrt{2} - 1
 \end{aligned}$$

Part B

1 point is earned for limits

1 point is earned for integrand

1 point is **deducted** for constant error

1 point is earned for antiderivative and evaluation

$$\begin{aligned}
 V &= \pi \int_0^{\pi/4} \cos^2 x - \sin^2 x \, dx \\
 &= \pi \int_0^{\pi/4} \cos 2x \, dx \\
 &= \frac{\pi}{2} \sin 2x \Big|_0^{\pi/4} \\
 &= \frac{\pi}{2} (1 - 0) = \frac{\pi}{2}
 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for limits

1 point is earned for integrand

1 point is **deducted** for constant error

8-4b

1 point is earned for antiderivative and evaluation

$$\begin{aligned}
 V &= \pi \int_0^{\pi/4} \cos^2 x - \sin^2 x \, dx \\
 &= \pi \int_0^{\pi/4} \cos 2x \, dx \\
 &= \frac{\pi}{2} \sin 2x \Big|_0^{\pi/4} \\
 &= \frac{\pi}{2} (1 - 0) = \frac{\pi}{2}
 \end{aligned}$$

Part C

2 points are earned for integral

1 point is **deducted** for constant or limit error

1 point is earned for antiderivative and evaluation

$$\begin{aligned}
 V &= \int_0^{\pi/4} (\cos x - \sin x)^2 \, dx \\
 &= \int_0^{\pi/4} 1 - 2 \sin x \cos x \, dx \\
 &= (x - \sin^2 x) \Big|_0^{\pi/4} \\
 &= \frac{\pi}{4} - \frac{1}{2} - (0 - 0) \\
 &= \frac{\pi}{4} - \frac{1}{2}
 \end{aligned}$$



0	1	2	3
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The student response earns all of the following points:

2 points are earned for integral

1 point is **deducted** for constant or limit error

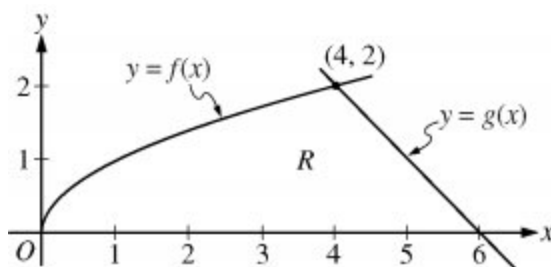
1 point is earned for antiderivative and evaluation

8-4b

$$\begin{aligned} V &= \int_0^{\pi/4} (\cos x - \sin x)^2 dx \\ &= \int_0^{\pi/4} 1 - 2 \sin x \cos x \, dx \\ &= (x - \sin^2 x) \Big|_0^{\pi/4} \\ &= \frac{\pi}{4} - \frac{1}{2} - (0 - 0) \\ &= \frac{\pi}{4} - \frac{1}{2} \end{aligned}$$

8-4b

83.



The functions f and g are given by $f(x) = \sqrt{x}$ and $g(x) = 6 - x$. Let R be the region bounded by the x -axis and the graphs of f and g , as shown in the figure above.

(a) Find the area of R .

(b) The region R is the base of a solid. For each y , where $0 \leq y \leq 2$, the cross section of the solid taken perpendicular to the y -axis is a rectangle whose base lies in R and whose height is $2y$. Write, but do not evaluate, an integral expression that gives the volume of the solid.

(c) There is a point P on the graph of f at which the line tangent to the graph of f is perpendicular to the graph of g . Find the coordinates of point P .

Part A

1 point is earned for the integral

1 point is earned for the antiderivative

1 point is earned for the answer

$$\text{Area} = \int_0^4 \sqrt{x} \, dx + \frac{1}{2} \cdot 2 \cdot 2 = \frac{2}{3} x^{3/2} \Big|_{x=0}^{x=4} + 2 = \frac{22}{3}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the antiderivative

1 point is earned for the answer

$$\text{Area} = \int_0^4 \sqrt{x} \, dx + \frac{1}{2} \cdot 2 \cdot 2 = \frac{2}{3} x^{3/2} \Big|_{x=0}^{x=4} + 2 = \frac{22}{3}$$

8-4b

Part B

2 points are earned for integrand

1 point is earned for the answer

$$y = \sqrt{x} \Rightarrow x = y^2$$

$$y = 6 - x \Rightarrow x = 6 - y$$

$$\text{Width} = (6 - y) - y^2$$

$$\text{Volume} = \int_0^2 2y(6 - y - y^2) dy$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for integrand

1 point is earned for the answer

$$y = \sqrt{x} \Rightarrow x = y^2$$

$$y = 6 - x \Rightarrow x = 6 - y$$

$$\text{Width} = (6 - y) - y^2$$

$$\text{Volume} = \int_0^2 2y(6 - y - y^2) dy$$

Part C

1 point is earned for $f'(x)$

1 point is earned for the equation

1 point is earned for the answer

$$g'(x) = -1$$

Thus a line perpendicular to the graph of g has slope 1.

8-4b

$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$\frac{1}{2\sqrt{x}} = 1 \Rightarrow x = \frac{1}{4}$$

The point P has coordinates $\left(\frac{1}{4}, \frac{1}{2}\right)$.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for $f'(x)$

1 point is earned for the equation

1 point is earned for the answer

$$g'(x) = -1$$

Thus a line perpendicular to the graph of g has slope 1.

$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$\frac{1}{2\sqrt{x}} = 1 \Rightarrow x = \frac{1}{4}$$

The point P has coordinates $\left(\frac{1}{4}, \frac{1}{2}\right)$.

8-4b

84. Let R be the region in the first quadrant enclosed by the graph of $y = \sqrt{6x + 4}$, the line $y = 2x$, and the y -axis.

(a) Find the area of R .

(b) Set up, but do not integrate, an integral expression in terms of a single variable for the volume of the solid generated when R is revolved about the x -axis.

(c) Set up, but do not integrate, an integral expression in terms of a single variable for the volume of the solid generated when R is revolved about the y -axis.

Part A

1 point is earned for limit

2 point are earned for integrand

1 point is earned for antiderivative

1 point is earned for $x = 3$

$$\begin{aligned} \text{Area} &= \int_0^2 \sqrt{6x + 4} - 2x \, dx \\ &= \frac{1}{6} \cdot \frac{2}{3} (6x + 4)^{3/2} - x^2 \Big|_0^2 \\ &= \left(\frac{64}{9} - 4 \right) - \frac{8}{9} = \boxed{\frac{20}{9}} \end{aligned}$$



0	1	2	3	4	5
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The student response earns all of the following points:

1 point is earned for limit

2 point are earned for integrand

1 point is earned for antiderivative

1 point is earned for $x = 3$

8-4b

$$\begin{aligned}
 \text{Area} &= \int_0^2 \sqrt{6x+4} - 2x \, dx \\
 &= \frac{1}{6} \cdot \frac{2}{3} (6x+4)^{3/2} - x^2 \Big|_0^2 \\
 &= \left(\frac{64}{9} - 4 \right) - \frac{8}{9} = \boxed{\frac{20}{9}}
 \end{aligned}$$

Part B

1 point is earned for integrand

1 point is earned for limits and constant

2 points **deducted** for incorrect integrand1 point **deducted** for wrong limits, wrong constant or reversal

Volume about x-axis

$$\begin{aligned}
 \text{Disks} &= \pi \int_0^2 (6x+4) - 4x^2 \, dx \\
 &\quad \text{or} \\
 &= \pi \int_0^2 (6x+4) \, dx - \frac{32\pi}{3}
 \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for limits and constant

1 point **deducted** for wrong limits, wrong constant or reversal

Volume about x-axis

8-4b

$$\text{Disks} = \pi \int_0^2 (6x + 4) - 4x^2 \, dx$$

or

$$= \pi \int_0^2 (6x + 4) \, dx - \frac{32\pi}{3}$$

Part C

1 point is earned for integrand

1 point is earned for limits and constant

1 point **deducted** for wrong limits, wrong constant or reversalVolume about y -axis

$$\text{Shells} = 2\pi \int_0^2 x (\sqrt{6x + 4} - 2x) \, dx$$

$$\text{Disks} = \pi \int_0^4 \left(\frac{y}{2}\right)^2 dy - \pi \int_2^4 \left(\frac{y^2 - 4}{6}\right)^2 dy$$



0	1	2
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The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for limits and constant

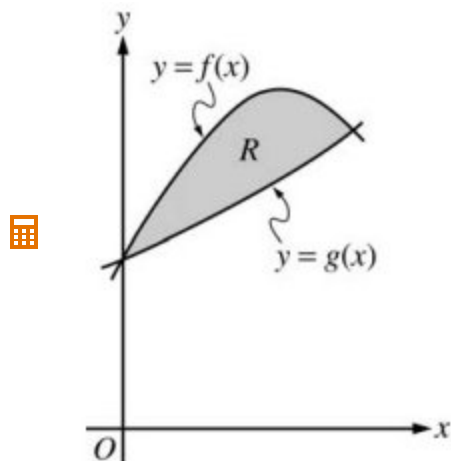
1 point **deducted** for wrong limits, wrong constant or reversalVolume about y -axis

$$\text{Shells} = 2\pi \int_0^2 x (\sqrt{6x + 4} - 2x) \, dx$$

$$\text{Disks} = \pi \int_0^4 \left(\frac{y}{2}\right)^2 dy - \pi \int_2^4 \left(\frac{y^2 - 4}{6}\right)^2 dy$$

8-4b

85.



Let f and g be the functions given by $f(x) = 1 + \sin(2x)$ and $g(x) = e^{x/2}$. Let R be the shaded region in the first quadrant enclosed by the graphs of f and g as shown in the figure above.

- Find the area of R .
- Find the volume of the solid generated when R is revolved about the x -axis.
- The region R is the base of a solid. For this solid, the cross sections perpendicular to the x -axis are semicircles with diameters extending from $y = f(x)$ to $y = g(x)$. Find the volume of this solid.

Part A

1 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned}
 \text{Area} &= \int_0^S (f(x) - g(x)) dx \\
 &= \int_0^S (1 + \sin(2x) - e^{x/2}) dx \\
 &= 0.429
 \end{aligned}$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

The graphs of f and g intersect in the first quadrant at $(S, T) = (1.13569, 1.76446)$.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the integrand

8-4b

1 point is earned for the answer

$$\begin{aligned}\text{Area} &= \int_0^S (f(x) - g(x))dx \\ &= \int_0^S (1 + \sin(2x) - e^{x/2})dx \\ &= 0.429\end{aligned}$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

The graphs of f and g intersect in the first quadrant at $(S, T) = (1.13569, 1.76446)$.

Part B

2 point is earned for the integrand

<-1> for each error

Note: 0/2 if integral not of form $c \int_a^b (R^2(x) - r^2(x))dx$

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \pi \int_0^S ((f(x))^2 - (g(x))^2)dx \\ &= \pi \int_0^S ((1 + \sin(2x))^2 - (e^{x/2})^2)dx \\ &= 4.266 \text{ or } 4.267\end{aligned}$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

The graphs of f and g intersect in the first quadrant at $(S, T) = (1.13569, 1.76446)$.



0	1	2	3	4
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The student response earns all of the following points:

2 point is earned for the integrand

<-1> for each error

8-4b

Note: 0/2 if integral not of form $c \int_a^b (R^2(x) - r^2(x))dx$

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \pi \int_0^S \left((f(x))^2 - (g(x))^2 \right) dx \\ &= \pi \int_0^S \left((1 + \sin(2x))^2 - (e^{x/2})^2 \right) dx \\ &= 4.266 \text{ or } 4.267\end{aligned}$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

The graphs of f and g intersect in the first quadrant at $(S, T) = (1.13569, 1.76446)$.

Part C

2 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \int_0^S \frac{\pi}{2} \left(\frac{f(x) - g(x)}{2} \right)^2 dx \\ &= \int_0^S \frac{\pi}{2} \left(\frac{1 + \sin(2x) - e^{x/2}}{2} \right)^2 dx \\ &= 0.077 \text{ or } 0.078\end{aligned}$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

The graphs of f and g intersect in the first quadrant at $(S, T) = (1.13569, 1.76446)$.



0	1	2	3	4
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The student response earns all of the following points:

2 point is earned for the integrand

1 point is earned for the answer

8-4b

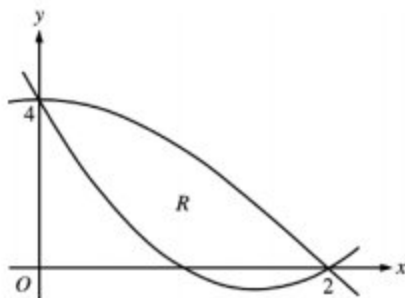
$$\begin{aligned}\text{Volume} &= \int_0^S \frac{\pi}{2} \left(\frac{f(x) - g(x)}{2} \right)^2 dx \\ &= \int_0^S \frac{\pi}{2} \left(\frac{1 + \sin(2x) - e^{x/2}}{2} \right)^2 dx \\ &= 0.077 \text{ or } 0.078\end{aligned}$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

The graphs of f and g intersect in the first quadrant at $(S, T) = (1.13569, 1.76446)$.

8-4b

86.



Let $f(x) = 2x^2 - 6x + 4$ and $g(x) = 4\cos(1/4 \pi x)$. Let R be the region bounded by the graphs of f and g , as shown in the figure above.

- Find the area of R .
- Write, but do not evaluate, an integral expression that gives the volume of the solid generated when R is rotated about the horizontal line $y = 4$.
- The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a square. Write, but do not evaluate, an integral expression that gives the volume of the solid.

Part A

1 point is earned for the integrand

2 points are earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned}
 \text{Area} &= \int_0^2 [g(x) - f(x)] dx \\
 &= \int_0^2 \left[4 \cos\left(\frac{\pi}{4}x\right) - (2x^2 - 6x + 4) \right] dx \\
 &= \left[4 \cdot \frac{4}{\pi} \sin\left(\frac{\pi}{4}x\right) - \left(\frac{2x^3}{3} - 3x^2 + 4x\right) \right]_0^2 \\
 &= \frac{16}{\pi} - \left(\frac{16}{3} - 12 + 8\right) = \frac{16}{\pi} - \frac{4}{3}
 \end{aligned}$$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for the integrand

2 points are earned for the antiderivative

8-4b

1 point is earned for the answer

$$\begin{aligned}
 \text{Area} &= \int_0^2 [g(x) - f(x)] dx \\
 &= \int_0^2 \left[4 \cos\left(\frac{\pi}{4}x\right) - (2x^2 - 6x + 4) \right] dx \\
 &= \left[4 \cdot \frac{4}{\pi} \sin\left(\frac{\pi}{4}x\right) - \left(\frac{2x^3}{3} - 3x^2 + 4x\right) \right]_0^2 \\
 &= \frac{16}{\pi} - \left(\frac{16}{3} - 12 + 8\right) = \frac{16}{\pi} - \frac{4}{3}
 \end{aligned}$$

Part B

2 points are earned for the integrand.

1 point is earned for the limits and constant.

$$\begin{aligned}
 \text{Volume} &= \pi \int_0^2 \left[(4 - f(x))^2 - (4 - g(x))^2 \right] dx \\
 &= \pi \int_0^2 \left[(4 - (2x^2 - 6x + 4))^2 - \left(4 - 4 \cos\left(\frac{\pi}{4}x\right)\right)^2 \right] dx
 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand.

1 point is earned for the limits and constant.

$$\begin{aligned}
 \text{Volume} &= \pi \int_0^2 \left[(4 - f(x))^2 - (4 - g(x))^2 \right] dx \\
 &= \pi \int_0^2 \left[(4 - (2x^2 - 6x + 4))^2 - \left(4 - 4 \cos\left(\frac{\pi}{4}x\right)\right)^2 \right] dx
 \end{aligned}$$

Part C

1 point is earned for the integrand

8-4b

1 point is earned for the limits and constant

$$\begin{aligned}\text{Volume} &= \int_0^2 [g(x) - f(x)] \, dx \\ &= \int_0^2 \left[4 \cos\left(\frac{\pi}{4}x\right) - (2x^2 - 6x + 4) \right] \, dx\end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

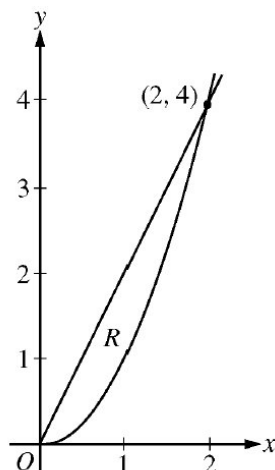
1 point is earned for the integrand

1 point is earned for the limits and constant

$$\begin{aligned}\text{Volume} &= \int_0^2 [g(x) - f(x)] \, dx \\ &= \int_0^2 \left[4 \cos\left(\frac{\pi}{4}x\right) - (2x^2 - 6x + 4) \right] \, dx\end{aligned}$$

8-4b

87.



Let R be the region in the first quadrant enclosed by the graphs of $y = 2x$ and $y = x^2$, as shown in the figure above.

- Find the area of R .
- The region R is the base of a solid. For this solid, at each x the cross section perpendicular to the x -axis has area $A(x) = \sin(\pi/2x)$. Find the volume of the solid.
- Another solid has the same base R . For this solid, the cross sections perpendicular to the y -axis are squares. Write, but do not evaluate, an integral expression for the volume of the solid.

Part A

1 point is earned for the integrand

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned}
 \text{Area} &= \int_0^2 (2x - x^2) dx \\
 &= x^2 - \frac{1}{3}x^3 \Big|_{x=0}^{x=2} \\
 &= \frac{4}{3}
 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the antiderivative

8-4b

1 point is earned for the answer

$$\begin{aligned}\text{Area} &= \int_0^2 (2x - x^2) dx \\ &= x^2 - \frac{1}{3}x^3 \Big|_{x=0}^{x=2} \\ &= \frac{4}{3}\end{aligned}$$

Part B

1 point is earned for the integrand

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \int_0^2 \sin\left(\frac{\pi}{2}x\right) dx \\ &= -\frac{2}{\pi} \cos\left(\frac{\pi}{2}x\right) \Big|_{x=0}^{x=2} \\ &= \frac{4}{\pi}\end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \int_0^2 \sin\left(\frac{\pi}{2}x\right) dx \\ &= -\frac{2}{\pi} \cos\left(\frac{\pi}{2}x\right) \Big|_{x=0}^{x=2} \\ &= \frac{4}{\pi}\end{aligned}$$

Part C

8-4b

2 points are earned for the integrand

1 point is earned for the limit

$$\text{Volume} = \int_0^4 \left(\sqrt{y} - \frac{y}{2} \right)^2 dy$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

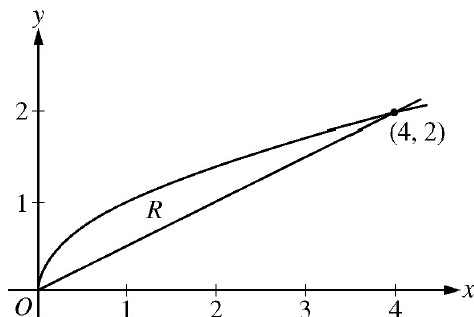
2 points are earned for the integrand

1 point is earned for the limit

$$\text{Volume} = \int_0^4 \left(\sqrt{y} - \frac{y}{2} \right)^2 dy$$

8-4b

88.



Let R be the region bounded by the graphs of $y = \sqrt{x}$ and $y = \frac{x}{2}$, as shown in the figure above.

(a) Find the area of R .

(b) The region R is the base of a solid. For this solid, the cross sections perpendicular to the x -axis are squares. Find the volume of this solid.

(c) Write, but do not evaluate, an integral expression for the volume of the solid generated when R is rotated about the horizontal line $y = 2$.

Part A

1 point is earned for integrand

1 point is earned for antiderivative

1 point is earned for answer

$$\text{Area} = \int_0^4 \left(\sqrt{x} - \frac{x}{2} \right) dx = \frac{2}{3} x^{3/2} - \frac{x^2}{4} \Big|_{x=0}^{x=4} = \frac{4}{3}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for antiderivative

1 point is earned for answer

$$\text{Area} = \int_0^4 \left(\sqrt{x} - \frac{x}{2} \right) dx = \frac{2}{3} x^{3/2} - \frac{x^2}{4} \Big|_{x=0}^{x=4} = \frac{4}{3}$$

Part B

1 point is earned for integrand

8-4b

1 point is earned for antiderivative

1 point is earned for answer

$$\begin{aligned}\text{Volume} &= \int_0^4 \left(\sqrt{x} - \frac{x}{2} \right)^2 dx = \int_0^4 \left(x - x^{3/2} + \frac{x^2}{4} \right) dx \\ &= \frac{x^2}{2} - \frac{2x^{5/2}}{5} + \frac{x^3}{12} \Big|_{x=0}^{x=4} = \frac{8}{15}\end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for antiderivative

1 point is earned for answer

$$\begin{aligned}\text{Volume} &= \int_0^4 \left(\sqrt{x} - \frac{x}{2} \right)^2 dx = \int_0^4 \left(x - x^{3/2} + \frac{x^2}{4} \right) dx \\ &= \frac{x^2}{2} - \frac{2x^{5/2}}{5} + \frac{x^3}{12} \Big|_{x=0}^{x=4} = \frac{8}{15}\end{aligned}$$

Part C

1 point is earned for the limits and constant

2 points are earned for the integrand

$$\text{Volume} = \pi \int_0^4 \left(\left(2 - \frac{x}{2} \right)^2 - (2 - \sqrt{x})^2 \right) dx$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits and constant

8-4b

2 points are earned for the integrand

$$\text{Volume} = \pi \int_0^4 \left(\left(2 - \frac{x}{2} \right)^2 - (2 - \sqrt{x})^2 \right) dx$$

8-4b

89.  Let R be the region in the first quadrant bounded by the graphs of $y = \sqrt{x}$ and $y = \frac{x}{3}$.
- (a) Find the area of R .
- (b) Find the volume of the solid generated when R is rotated about the vertical line $x = -1$.
- (c) The region R is the base of a solid. For this solid, the cross sections perpendicular to the y -axis are squares. Find the volume of this solid.

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^9 \left(\sqrt{x} - \frac{x}{3} \right) dx = 4.5$$

OR

$$\int_0^3 (3y - y^2) dy = 4.5$$

The graphs of $y = \sqrt{x}$ and $y = \frac{x}{3}$ intersect at the points $(0, 0)$ and $(9, 3)$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^9 \left(\sqrt{x} - \frac{x}{3} \right) dx = 4.5$$

OR

$$\int_0^3 (3y - y^2) dy = 4.5$$

The graphs of $y = \sqrt{x}$ and $y = \frac{x}{3}$ intersect at the points $(0, 0)$ and $(9, 3)$.

8-4b

Part B

1 point is earned for constant and limits

2 points are earned for integrand

1 point is earned for the answer

$$\pi \int_0^3 \left((3y+1)^2 - (y^2+1)^2 \right) dy$$

$$= \frac{207\pi}{5} = 130.061 \text{ or } 130.062$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for constant and limits

2 points are earned for integrand

1 point is earned for the answer

$$\pi \int_0^3 \left((3y+1)^2 - (y^2+1)^2 \right) dy$$

$$= \frac{207\pi}{5} = 130.061 \text{ or } 130.062$$

Part C

1 point is earned for integrand

1 point is earned for limits and answer

$$\int_0^3 (3y - y^2)^2 dy = 8.1$$



0	1	2
---	---	---

8-4b

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for limits and answer

$$\int_0^3 (3y - y^2)^2 dy = 8.1$$

8-4b

90. Let f be the function given by $f(x) = e^{-x}$, and let g be the function given by $g(x) = kx$, where k is the nonzero constant such that the graph of f is tangent to the graph of g .
- (a) Find the x -coordinate of the point of tangency and the value of k .
- (b) Let R be the region enclosed by the y -axis and the graphs of f and g . Using the results found in part (a), determine the area of R .
- (c) Set up, but do not integrate, an integral expression in terms of a single variable for the volume of the solid generated by revolving the region R , given in part (b), about the x -axis.

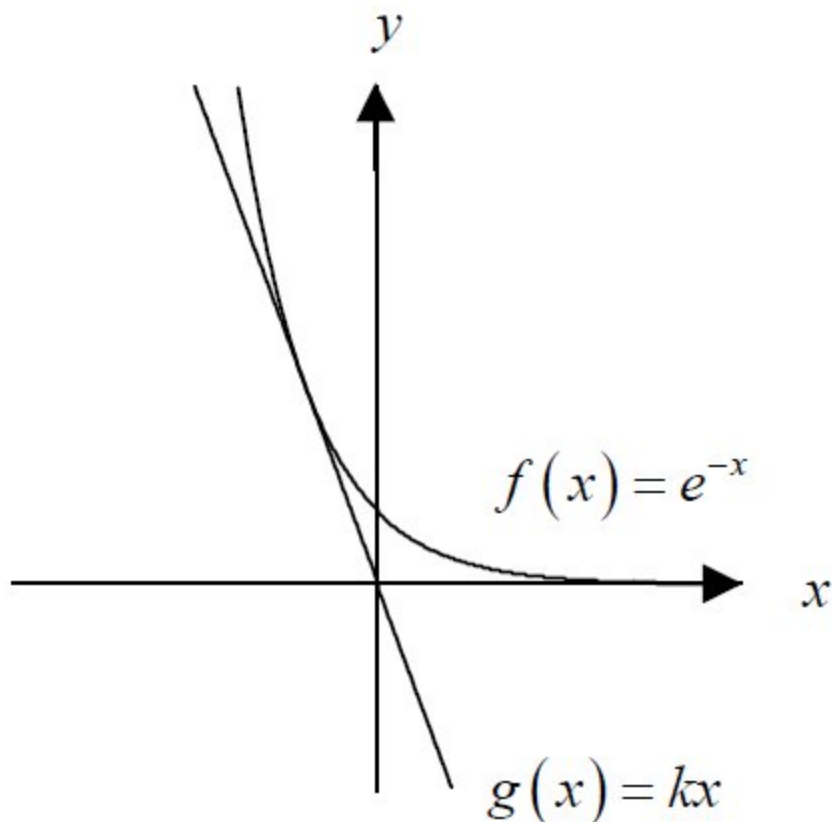
Part A

1 point is earned if finds $f'(x)$ and $g'(x)$ explicitly

1 point is earned for setting student's $f'(x)$ equal to student's $g'(x)$ explicitly

1 point is earned for setting $f(x) = g(x)$

1 point is earned for solving for x and k



$$f'(x) = -e^{-x}; \quad g'(x) = k$$

$$-e^{-x} = k \leftarrow +2$$

$$e^{-x} = kx$$

$$x = -1 \text{ and } k = -e$$

8-4b



0	1	2	3	4
---	---	---	---	---

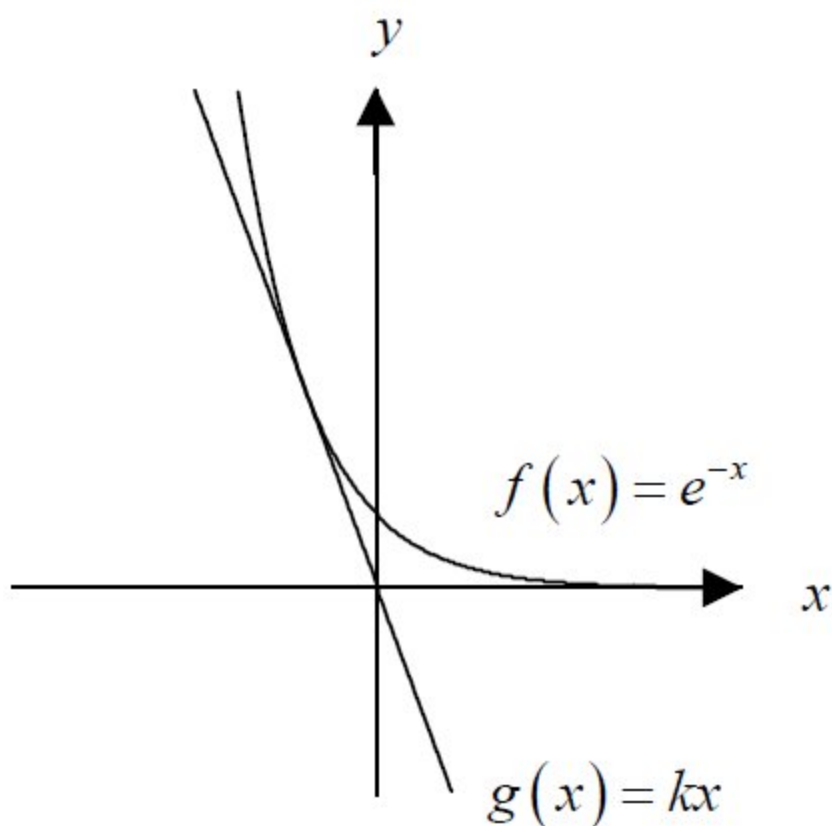
The student response earns all of the following points:

1 point is earned if finds $f'(x)$ and $g'(x)$ explicitly

1 point is earned for setting student's $f'(x)$ equal to student's $g'(x)$ explicitly

1 point is earned for setting $f(x) = g(x)$

1 point is earned for solving for x and k



$$f'(x) = -e^{-x}; g'(x) = k$$

$$-e^{-x} = k \leftarrow +2$$

$$e^{-x} = kx$$

$$x = -1 \text{ and } k = -e$$

Part B

2 points are earned for integral

8-4b

1 point is earned if anti-differentiates

$$\begin{aligned}\int_{-1}^0 (e^{-x} - (-ex)) dx &= \int_{-1}^0 (e^{-x} + ex) dx \\ &= \left(-e^{-x} + \frac{ex^2}{2} \right) \Big|_{-1}^0 \\ &= (-1 + 0) - \left(-e + \frac{e}{2} \right) \\ &= \frac{e}{2} - 1\end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for integral

1 point is earned if anti-differentiates

$$\begin{aligned}\int_{-1}^0 (e^{-x} - (-ex)) dx &= \int_{-1}^0 (e^{-x} + ex) dx \\ &= \left(-e^{-x} + \frac{ex^2}{2} \right) \Big|_{-1}^0 \\ &= (-1 + 0) - \left(-e + \frac{e}{2} \right) \\ &= \frac{e}{2} - 1\end{aligned}$$

Part C

2 points are earned for setting up integral with correct integrand

1 point **deducted** for incorrect limits/error involving π

$$\pi \int_{-1}^0 \left((e^{-x})^2 - (-ex)^2 \right) dx$$



0	1	2
---	---	---

The student response earns all of the following points:

8-4b

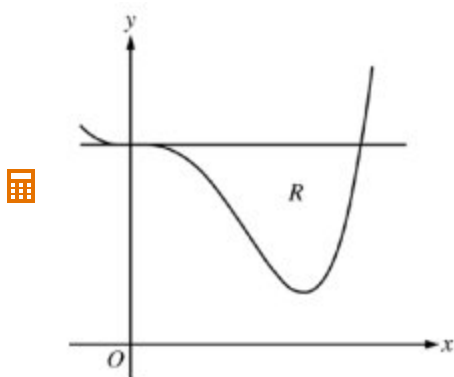
2 points are earned for setting up integral with correct integrand

1 point **deducted** for incorrect limits/error involving π

$$\pi \int_{-1}^0 \left((e^{-x})^2 - (-ex)^2 \right) dx$$

8-4b

91.



Let R be the region enclosed by the graph of $f(x) = x^4 - 2.3x^3 + 4$ and the horizontal line $y = 4$, as shown in the figure above.

- (a) Find the volume of the solid generated when R is rotated about the horizontal line $y = -2$.
- (b) Region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is an isosceles right triangle with a leg in R . Find the volume of the solid.
- (c) The vertical line $x = k$ divides R into two regions with equal areas. Write, but do not solve, an equation involving integral expressions whose solution gives the value k .

Part A

2 points are earned for the integrand

1 point is earned for limits

1 point is earned for the answer

$$f(x) = 4 \Rightarrow x = 0, 2.3$$

$$\begin{aligned} \text{Volume} &= \pi \int_0^{2.3} \left[(4 + 2)^2 - (f(x) + 2)^2 \right] dx \\ &= 98.868 \text{ (or } 98.867) \end{aligned}$$



0	1	2	3	4
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The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for limits

1 point is earned for the answer

8-4b

$$f(x) = 4 \Rightarrow x = 0, 2.3$$

$$\begin{aligned}\text{Volume} &= \pi \int_0^{2.3} \left[(4+2)^2 - (f(x)+2)^2 \right] dx \\ &= 98.868 \text{ (or } 98.867)\end{aligned}$$

Part B

2 points are earned for integrand

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \int_0^{2.3} \frac{1}{2} (4 - f(x))^2 dx \\ &= 3.574 \text{ (or } 3.573)\end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for integrand

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \int_0^{2.3} \frac{1}{2} (4 - f(x))^2 dx \\ &= 3.574 \text{ (or } 3.573)\end{aligned}$$

Part C

1 point is earned for the area of one region

1 point is earned for the equation

$$\int_0^k (4 - f(x)) dx = \int_k^{2.3} (4 - f(x)) dx$$

8-4b



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the area of one region

1 point is earned for the equation

$$\int_0^k (4 - f(x)) dx = \int_k^{2.3} (4 - f(x)) dx$$

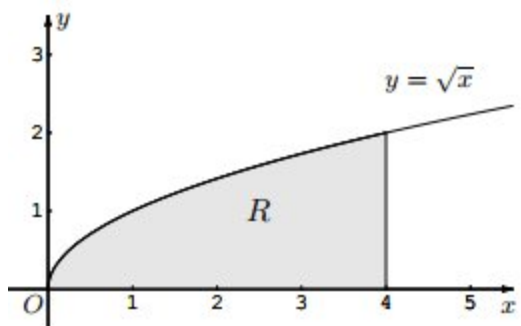
8-4b

92. Let R be the region bounded by the x -axis, the graph of $y = \sqrt{x}$, and the line $x = 4$.
- Find the area of the region R .
 - Find the value of h such that the vertical line $x = h$ divides the region R into two regions of equal area.
 - Find the volume of the solid generated when R is revolved about the x -axis.
 - The vertical line $x = k$ divides the region R into two regions such that when these two regions are revolved about the x -axis, they generate solids with equal volumes. Find the value of k .

Part A

1 point is earned for equation of area $A = \int_0^4 \sqrt{x} dx$

1 point is earned for solving equation



$$A = \int_0^4 \sqrt{x} dx = \frac{2}{3} x^{3/2} \Big|_0^4 = \frac{16}{3} \text{ or } 5.333$$



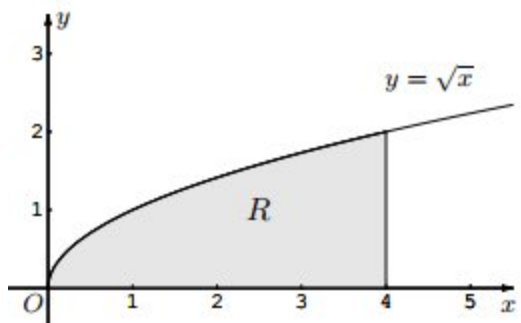
0	1	2
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The student response earns all of the following points:

1 point is earned for equation of area $A = \int_0^4 \sqrt{x} dx$

1 point is earned for solving equation

8-4b



$$A = \int_0^4 \sqrt{x} \, dx = \frac{2}{3} x^{3/2} \Big|_0^4 = \frac{16}{3} \text{ or } 5.333$$

Part B

1 point is earned for equation in h

1 point is earned for answer

$$\int_0^h \sqrt{x} \, dx = \frac{8}{3} \quad \int_0^h \sqrt{x} \, dx = \int_h^4 \sqrt{x} \, dx$$

or

$$\frac{2}{3} h^{3/2} = \frac{8}{3} \quad \frac{2}{3} h^{3/2} = \frac{16}{3} - \frac{2}{3} h^{3/2}$$

$$h = \sqrt[3]{16} \text{ or } 2.520 \text{ or } 2.519$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for equation in h

1 point is earned for answer

$$\int_0^h \sqrt{x} \, dx = \frac{8}{3} \quad \int_0^h \sqrt{x} \, dx = \int_h^4 \sqrt{x} \, dx$$

or

$$\frac{2}{3} h^{3/2} = \frac{8}{3} \quad \frac{2}{3} h^{3/2} = \frac{16}{3} - \frac{2}{3} h^{3/2}$$

$$h = \sqrt[3]{16} \text{ or } 2.520 \text{ or } 2.519$$

8-4b

Part C

1 point is earned for limits and constant

1 point is earned for integrand

1 point is earned for answer

$$V = \pi \int_0^4 (\sqrt{x})^2 dx = \pi \frac{x^2}{2} \Big|_0^4 = 8\pi$$

or 25.133 or 25.132



0	1	2	3
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The student response earns all of the following points:

1 point is earned for limits and constant

1 point is earned for integrand

1 point is earned for answer

$$V = \pi \int_0^4 (\sqrt{x})^2 dx = \pi \frac{x^2}{2} \Big|_0^4 = 8\pi$$

or 25.133 or 25.132

Part D

1 point is earned for equation in k

1 point is earned for the answer

$$\pi \int_0^k (\sqrt{x})^2 dx = 4\pi \quad \pi \int_0^k (\sqrt{x})^2 dx = \pi \int_k^4 (\sqrt{x})^2 dx$$

or

$$\pi \frac{k^2}{2} = 4\pi \quad \pi \frac{k^2}{2} = 8\pi - \pi \frac{k^2}{2}$$

$$k = \sqrt{8} \quad \text{or} \quad 2.828$$

8-4b



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for equation in k

1 point is earned for the answer

$$\pi \int_0^k (\sqrt{x})^2 dx = 4\pi \quad \pi \int_0^k (\sqrt{x})^2 dx = \pi \int_k^4 (\sqrt{x})^2 dx$$

or

$$\pi \frac{k^2}{2} = 4\pi \quad \pi \frac{k^2}{2} = 8\pi - \pi \frac{k^2}{2}$$

$$k = \sqrt{8} \quad \text{or} \quad 2.828$$

8-4b

93.  Let R be the region enclosed by the graph of $y = \sqrt{x - 1}$, the vertical line $x = 10$, and the x -axis.
- (a) Find the area of R .
- (b) Find the volume of the solid generated when R is revolved about the horizontal line $y = 3$.
- (c) Find the volume of the solid generated when R is revolved about the vertical line $x = 10$.

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Area} = \int_1^{10} \sqrt{x - 1} \, dx = 18$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Area} = \int_1^{10} \sqrt{x - 1} \, dx = 18$$

Part B

1 point is earned for the limits and constant

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_1^{10} \left(9 - \left(3 - \sqrt{x - 1} \right)^2 \right) dx$$

$$= 212.057 \text{ or } 212.058$$

8-4b



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the limits and constant

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_1^{10} \left(9 - \left(3 - \sqrt{x-1} \right)^2 \right) dx$$

$$= 212.057 \text{ or } 212.058$$

Part C

1 point is earned for the limits and constant

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_0^3 \left(10 - (y^2 + 1) \right)^2 dy$$

$$= 407.150$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the limits and constant

1 point is earned for the integrand

1 point is earned for the answer

8-4b

$$\text{Volume} = \pi \int_0^3 (10 - (y^2 + 1))^2 dy$$

$$= 407.150$$

8-4b

94.  Let R be the region in the first quadrant under the graph of $y = \frac{1}{\sqrt{x}}$ for $4 \leq x \leq 9$.

- (a) Find the area of R .
- (b) If the line $x = k$ divides the region R into two regions of equal area, what is the value of k ?
- (c) Find the volume of the solid whose base is the region R and whose cross sections cut by planes perpendicular to the x -axis are squares.

Part A

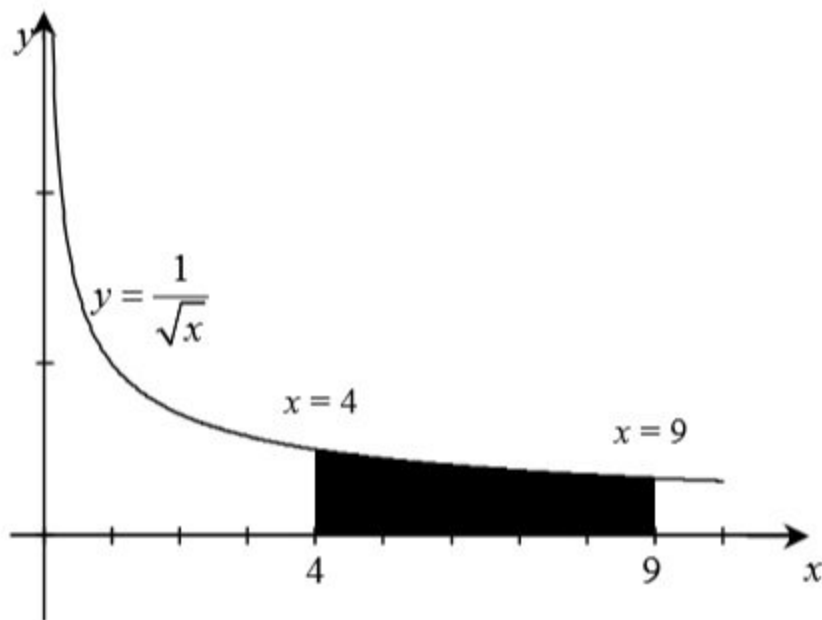
1 point is earned for the correct limits in x

1 point is earned for the correct integrand including constant must be $\frac{1}{\sqrt{x}}$

1 point is earned for the correct answer

$$\int_4^9 \frac{dx}{\sqrt{x}} = 2$$

0/1 if the integrand is incorrect, wrong limits are read



✓

0	1	2	3
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The student response earns all of the following points:

1 point is earned for the correct limits in x

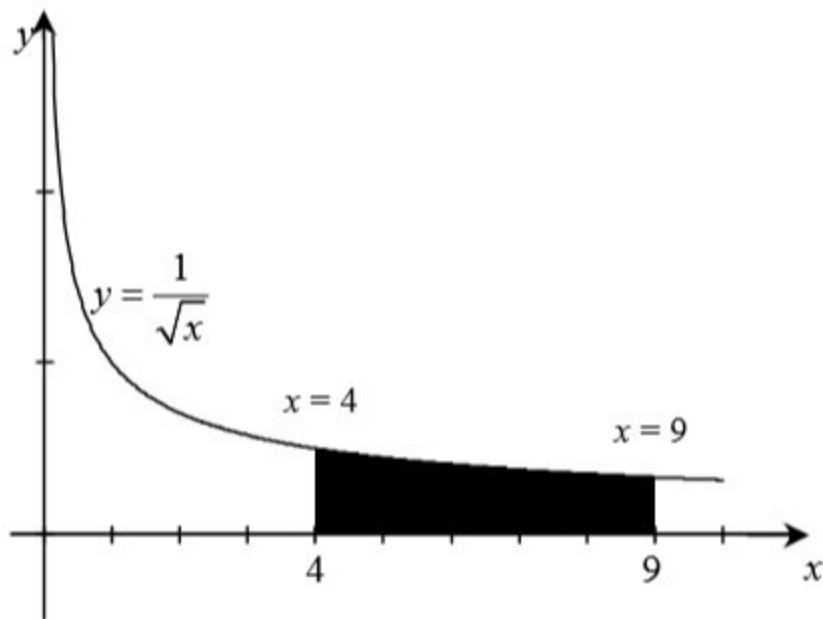
8-4b

1 point is earned for the correct integrand including constant must be $\frac{1}{\sqrt{x}}$

1 point is earned for the correct answer

$$\int_4^9 \frac{dx}{\sqrt{x}} = 2$$

0/1 if the integrand is incorrect, wrong limits are read

**Part B**

1 point is earned for the correct demonstration of variable limit using k as upper or lower limit with 4 or 9 (or imported limit)

$$\int_4^k \frac{dx}{\sqrt{x}} \text{ or } \int_k^9 \frac{dx}{\sqrt{x}}$$

1 point is earned for the correct equation involving the two halves of R that when solved will give answer with demonstrated algorithm that knows when to stop

1 point is earned for the correct answer

0/1 if answer from equation not involving relevant areas

1/3 bald answer

8-4b

$$\int_4^k \frac{dx}{\sqrt{x}} = 1$$

$$2\sqrt{x}]_4^k = 1$$

$$2\sqrt{k} - 2\sqrt{4} = 1$$

$$k = \frac{25}{4}$$

$$\left(\text{or } \int_k^9 \frac{dx}{\sqrt{x}} = 1 \text{ or } \int_4^k \frac{dx}{\sqrt{x}} = \int_k^9 \frac{dx}{\sqrt{x}} \right)$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct demonstration of variable limit using k as upper or lower limit with 4 or 9 (or imported limit)

$$\int_4^k \frac{dx}{\sqrt{x}} \text{ or } \int_k^9 \frac{dx}{\sqrt{x}}$$

1 point is earned for the correct equation involving the two halves of R that when solved will give answer with demonstrated algorithm that knows when to stop

1 point is earned for the correct answer

0/1 if answer from equation not involving relevant areas

1/3 bald answer

$$\int_4^k \frac{dx}{\sqrt{x}} = 1$$

$$2\sqrt{x}]_4^k = 1$$

$$2\sqrt{k} - 2\sqrt{4} = 1$$

$$k = \frac{25}{4}$$

$$\left(\text{or } \int_k^9 \frac{dx}{\sqrt{x}} = 1 \text{ or } \int_4^k \frac{dx}{\sqrt{x}} = \int_k^9 \frac{dx}{\sqrt{x}} \right)$$

Part C

1 point is earned for the correct indefinite limits using x

1 point is earned for the correct integrand including constant

8-4b

1 point is earned for the correct answer

0/1 if integrand is incorrect

$$\begin{aligned}\text{Volume} &= \int_4^9 \left(\frac{1}{\sqrt{x}} \right)^2 dx \\ &= \int_4^9 \frac{dx}{x} = \ln x \Big|_4^9 = \ln \frac{9}{4}\end{aligned}$$

(or 0.811)



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the correct indefinite limits using x

1 point is earned for the correct integrand including constant

1 point is earned for the correct answer

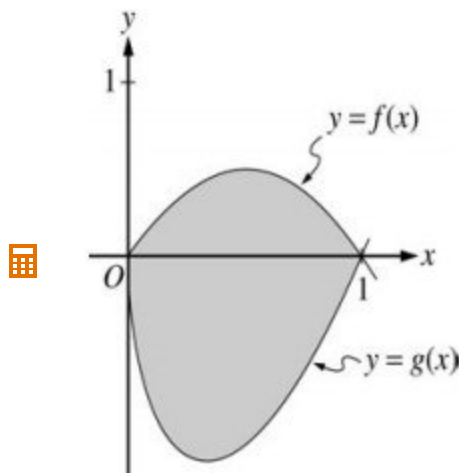
0/1 if integrand is incorrect

$$\begin{aligned}\text{Volume} &= \int_4^9 \left(\frac{1}{\sqrt{x}} \right)^2 dx \\ &= \int_4^9 \frac{dx}{x} = \ln x \Big|_4^9 = \ln \frac{9}{4}\end{aligned}$$

(or 0.811)

8-4b

95.



Let f and g be the functions given by $f(x) = 2x(1 - x)$ and $g(x) = 3(x - 1)\sqrt{x}$ for $0 \leq x \leq 1$. The graphs of f and g are shown in the figure above.

- (a) Find the area of the shaded region enclosed by the graphs of f and g .
- (b) Find the volume of the solid generated when the shaded region enclosed by the graphs of f and g is revolved about the horizontal line $y = 2$.
- (c) Let h be the function given by $h(x) = kx(1 - x)$ for $0 \leq x \leq 1$. For each $k > 0$, the region (not shown) enclosed by the graphs of h and g is the base of a solid with square cross sections perpendicular to the x -axis. There is a value of k for which the volume of this solid is equal to 15. Write, but do not solve, an equation involving an integral expression that could be used to find the value of k .

Part A

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^1 (f(x) - g(x)) dx \\ &= \int_0^1 (2x(1 - x) - 3(x - 1)\sqrt{x}) dx = 1.133 \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

8-4b

1 point is earned for the answer

$$\begin{aligned}\text{Area} &= \int_0^1 (f(x) - g(x)) dx \\ &= \int_0^1 (2x(1-x) - 3(x-1)\sqrt{x}) dx = 1.133\end{aligned}$$

Part B

1 point is earned for the limits and constant

2 points are earned for integrand

<-1> each error

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \pi \int_0^1 \left((2 - g(x))^2 - (2 - f(x))^2 \right) dx \\ &= \pi \int_0^1 \left((2 - 3(x-1)\sqrt{x})^2 - (2 - 2x(1-x))^2 \right) dx \\ &= 16.179\end{aligned}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits and constant

2 points are earned for integrand

<-1> each error

1 point is earned for the answer

8-4b

$$\begin{aligned}
 \text{Volume} &= \pi \int_0^1 \left((2 - g(x))^2 - (2 - f(x))^2 \right) dx \\
 &= \pi \int_0^1 \left((2 - 3(x-1)\sqrt{x})^2 - (2 - 2x(1-x))^2 \right) dx \\
 &= 16.179
 \end{aligned}$$

Part C

2 points are earned for the integrand

1 point is earned for the answer

$$\begin{aligned}
 \text{Volume} &= \int_0^1 (h(x) - g(x))^2 dx \\
 \int_0^1 (kx(1-x) - 3(x-1)\sqrt{x})^2 dx &= 15
 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

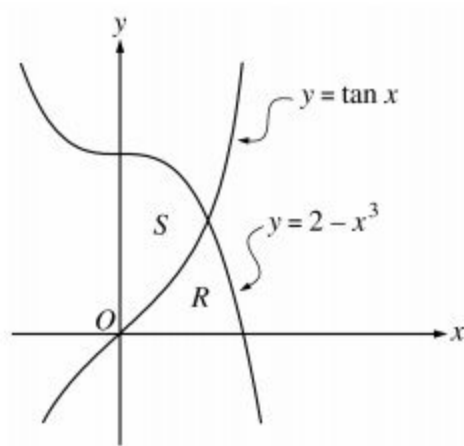
2 points are earned for the integrand

1 point is earned for the answer

$$\begin{aligned}
 \text{Volume} &= \int_0^1 (h(x) - g(x))^2 dx \\
 \int_0^1 (kx(1-x) - 3(x-1)\sqrt{x})^2 dx &= 15
 \end{aligned}$$

8-4b

96.



Let R and S be the regions in the first quadrant shown in the figure above. The region R is bounded by the x -axis and the graphs of $y = 2 - x^3$ and $y = \tan x$. The region S is bounded by the y -axis and the graphs of $y = 2 - x^3$ and $y = \tan x$.

- Find the area of R .
- Find the area of S .
- Find the volume of the solid generated when S is revolved about the x -axis.

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

Points of intersection

$$2 - x^3 = \tan x \text{ at } (A, B) = (0.902155, 1.265751)$$

$$\text{Area } R = \int_0^A \tan x \, dx + \int_A^{\sqrt[3]{2}} (2 - x^3) \, dx = 0.729$$

or

$$\text{Area } R = \int_0^B \left((2 - y)^{1/3} - \tan^{-1} y \right) dy = 0.729$$

or

$$\text{Area } R = \int_0^{\sqrt[3]{2}} (2 - x^3) \, dx - \int_0^A (2 - x^3 - \tan x) \, dx = 0.729$$



0	1	2	3
---	---	---	---

8-4b

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

Points of intersection

$$2 - x^3 = \tan x \text{ at } (A, B) = (0.902155, 1.265751)$$

$$\text{Area } R = \int_0^A \tan x \, dx + \int_A^{\sqrt[3]{2}} (2 - x^3) \, dx = 0.729$$

or

$$\text{Area } R = \int_0^B \left((2 - y)^{1/3} - \tan^{-1} y \right) dy = 0.729$$

or

$$\text{Area } R = \int_0^{\sqrt[3]{2}} (2 - x^3) \, dx - \int_0^A (2 - x^3 - \tan x) \, dx = 0.729$$

Part B

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Area } S = \int_0^A (2 - x^3 - \tan x) \, dx = 1.160 \text{ or } 1.161$$

or

$$\text{Area } S = \int_0^B \tan^{-1} y \, dy + \int_B^2 (2 - y)^{1/3} \, dy = 1.160 \text{ or } 1.161$$

or

Area S

$$\begin{aligned} &= \int_0^2 (2 - y)^{1/3} \, dy - \int_0^B \left((2 - y)^{1/3} - \tan^{-1} y \right) dy \\ &= 1.160 \text{ or } 1.161 \end{aligned}$$

8-4b



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Area } S = \int_0^A (2 - x^3 - \tan x) dx = 1.160 \text{ or } 1.161$$

or

$$\text{Area } S = \int_0^B \tan^{-1} y dy + \int_B^2 (2 - y)^{1/3} dy = 1.160 \text{ or } 1.161$$

or

$$\begin{aligned} \text{Area } S &= \int_0^2 (2 - y)^{1/3} dy - \int_0^B \left((2 - y)^{1/3} - \tan^{-1} y \right) dy \\ &= 1.160 \text{ or } 1.161 \end{aligned}$$

Part C

1 point is earned for the limits and constant

1 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \pi \int_0^A \left((2 - x^3)^2 - \tan^2 x \right) dx \\ &= 2.652\pi \text{ or } 8.331 \text{ or } 8.332 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits and constant

8-4b

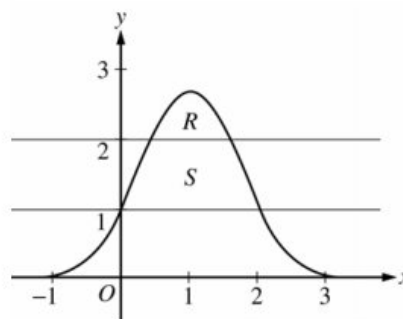
1 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \pi \int_0^A \left((2 - x^3)^2 - \tan^2 x \right) dx \\ &= 2.652\pi \text{ or } 8.331 \text{ or } 8.332\end{aligned}$$

8-4b

97.



Let R be the region bounded by the graph of $y = e^{2x-x^2}$ and the horizontal line $y = 2$, and let S be the region bounded by the graph of $y = e^{2x-x^2}$ and the horizontal lines $y = 1$ and $y = 2$, as shown above.

- Find the area of R .
- Find the area of S .
- Write, but do not evaluate, an integral expression that gives the volume of the solid generated when R is rotated about the horizontal line $y = 1$.

Part A

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$e^{2x-x^2} = 2 \text{ when } x = 0.446057, 1.553943$$

Let $P = 0.446057$ and $Q = 1.553943$

$$\text{Area of } R = \int_P^Q (e^{2x-x^2} - 2) dx = 0.514$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

8-4b

$$e^{2x-x^2} = 2 \text{ when } x = 0.446057, 1.553943$$

Let $P = 0.446057$ and $Q = 1.553943$

$$\text{Area of } R = \int_P^Q (e^{2x-x^2} - 2)dx = 0.514$$

Part B

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$e^{2x-x^2} = 2 \text{ when } x = 0.446057, 1.553943$$

Let $P = 0.446057$ and $Q = 1.553943$

$$e^{2x-x^2} = 1 \text{ when } x = 0, 2$$

$$\text{Area of } S = \int_0^2 (e^{2x-x^2} - 1)dx - \text{Area of } R$$

$$= 2.06016 - \text{Area of } R = 1.546$$

OR

$$\int_0^P (e^{2x-x^2} - 1)dx + (Q - P) \cdot 1 + \int_Q^2 (e^{2x-x^2} - 1)dx$$

$$= 0.219064 + 1.107886 + 0.219064 = 1.546$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

8-4b

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$e^{2x-x^2} = 2 \text{ when } x = 0.446057, 1.553943$$

$$\text{Let } P = 0.446057 \text{ and } Q = 1.553943$$

$$e^{2x-x^2} = 1 \text{ when } x = 0, 2$$

$$\text{Area of } S = \int_0^2 (e^{2x-x^2} - 1)dx - \text{Area of } R$$

$$= 2.06016 - \text{Area of } R = 1.546$$

OR

$$\int_0^P (e^{2x-x^2} - 1)dx + (Q - P) \cdot 1 + \int_Q^2 (e^{2x-x^2} - 1)dx$$

$$= 0.219064 + 1.107886 + 0.219064 = 1.546$$

Part C

2 points are earned for the integrand

1 point is earned for the constant and limits

$$e^{2x-x^2} = 2 \text{ when } x = 0.446057, 1.553943$$

$$\text{Let } P = 0.446057 \text{ and } Q = 1.553943$$

$$\text{Volume} = \pi \int_P^Q \left((e^{2x-x^2} - 1)^2 - (2 - 1)^2 \right) dx$$

8-4b



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the constant and limits

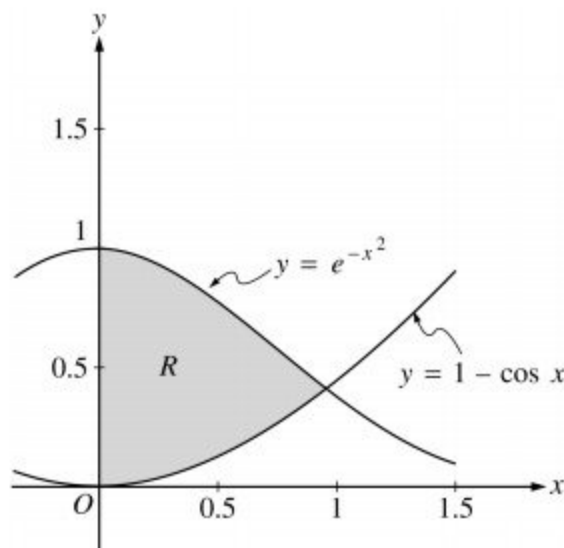
$$e^{2x-x^2} = 2 \text{ when } x = 0.446057, 1.553943$$

Let $P = 0.446057$ and $Q = 1.553943$

$$\text{Volume} = \pi \int_P^Q \left((e^{2x-x^2} - 1)^2 - (2 - 1)^2 \right) dx$$

8-4b

98.



Let R be the shaded region in the first quadrant enclosed by the graphs of $y = e^{-x^2}$, $y = 1 - \cos x$, and the y -axis, as shown in the figure above.

- Find the area of the region R .
- Find the volume of the solid generated when the region R is revolved about the x -axis.
- The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a square. Find the volume of this solid.

Part A

1 point is earned for the correct limits in an integral in (a), (b), or (c).

Region R

$$e^{-x^2} = 1 - \cos x \text{ at } x = 0.941944 = A$$

1 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^A (e^{-x^2} - (1 - \cos x)) dx \\ &= 0.590 \text{ or } 0.591 \end{aligned}$$



0	1	2	3
---	---	---	---

8-4b

The student response earns all of the following points:

1 point is earned for the correct limits in an integral in (a), (b), or (c).

Region R

$$e^{-x^2} = 1 - \cos x \text{ at } x = 0.941944 = A$$

1 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^A (e^{-x^2} - (1 - \cos x)) dx \\ &= 0.590 \text{ or } 0.591 \end{aligned}$$

Part B

2 points are earned for the integrand and constant

<-1> each error

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \pi \int_0^A \left((e^{-x^2})^2 - (1 - \cos x)^2 \right) dx \\ &= 0.55596\pi = 1.746 \text{ or } 1.747 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand and constant

<-1> each error

1 point is earned for the answer

8-4b

$$\begin{aligned}\text{Volume} &= \pi \int_0^A \left((e^{-x^2})^2 - (1 - \cos x)^2 \right) dx \\ &= 0.55596\pi = 1.746 \text{ or } 1.747\end{aligned}$$

Part C

2 points are earned for the integrand

<-1> each error

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \int_0^A \left(e^{-x^2} - (1 - \cos x) \right)^2 dx \\ &= 0.461\end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

<-1> each error

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \int_0^A \left(e^{-x^2} - (1 - \cos x) \right)^2 dx \\ &= 0.461\end{aligned}$$

8-4b

99.

 Distance from the river's edge (feet)	0	8	14	22	24
Depth of the water (feet)	0	7	8	2	0

A scientist measures the depth of the Doe River at Picnic Point. The river is 24 feet wide at this location. The measurements are taken in a straight line perpendicular to the edge of the river. The data are shown in the table above. The velocity of the water at Picnic Point, in feet per minute, is modeled by

$$v(t) = 16 + 2 \sin\left(\sqrt{t+10}\right) \text{ for } 0 \leq t \leq 120 \text{ minutes.}$$

(a) Use a trapezoidal sum with the four subintervals indicated by the data in the table to approximate the area of the cross section of the river at Picnic Point, in square feet. Show the computations that lead to your answer.

(b) The volumetric flow at a location along the river is the product of the cross-sectional area and the velocity of the water at that location. Use your approximation from part (a) to estimate the average value of the volumetric flow at Picnic Point, in cubic feet per minute, from $t = 0$ to $t = 120$ minutes.

(c) The scientist proposes the function f , given by $f(x) = 8\sin(\pi x/24)$, as a model for the depth of the water, in feet, at Picnic Point x feet from the river's edge. Find the area of the cross section of the river at Picnic Point based on this model.

(d) Recall that the volumetric flow is the product of the cross-sectional area and the velocity of the water at a location. To prevent flooding, water must be diverted if the average value of the volumetric flow at Picnic Point exceeds 2100 cubic feet per minute for a 20-minute period. Using your answer from part (c), find the average value of the volumetric flow during the time interval $40 \leq t \leq 60$ minutes. Does this value indicate that the water must be diverted?

Part A

1 point is earned for trapezoidal approximation

$$\begin{aligned} & \frac{(0+7)}{2} \cdot .8 + \frac{(7+8)}{2} \cdot .6 + \frac{(8+2)}{2} \cdot .8 + \frac{(2+0)}{2} \cdot .2 \\ & = 115 \text{ ft}^2 \end{aligned}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for trapezoidal approximation

$$\begin{aligned} & \frac{(0+7)}{2} \cdot .8 + \frac{(7+8)}{2} \cdot .6 + \frac{(8+2)}{2} \cdot .8 + \frac{(2+0)}{2} \cdot .2 \\ & = 115 \text{ ft}^2 \end{aligned}$$

8-4b

Part B

1 point is earned for the limits and average value of constant

1 point is earned for the integrand

1 point is earned for the answer

$$\frac{1}{120} \int_0^{120} 115v(t) dt$$

$$= 1807.169 \text{ or } 1807.170 \text{ ft}^3/\text{min}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits and average value of constant

1 point is earned for the integrand

1 point is earned for the answer

$$\frac{1}{120} \int_0^{120} 115v(t) dt$$

$$= 1807.169 \text{ or } 1807.170 \text{ ft}^3/\text{min}$$

Part C

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^{24} 8 \sin\left(\frac{\pi x}{24}\right) dx$$

$$= 122.230 \text{ or } 122.231 \text{ ft}^2$$

8-4b



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^{24} 8 \sin\left(\frac{\pi x}{24}\right) dx$$

$$= 122.230 \text{ or } 122.231 \text{ ft}^2$$

Part D

1 point is earned for the volumetric flow integral

1 point is earned for the average volumetric flow

1 point is earned for answer with reason

Let C be the cross-sectional area approximation from part (c). The average volumetric flow is

$$\frac{1}{20} \int_{10}^{60} C \cdot v(t) dt = 2181.912 \text{ or } 2181.913 \text{ ft}^3/\text{min}.$$

Yes, water must be diverted since the average volumetric flow for this 20-minute period exceeds $2100 \text{ ft}^3/\text{min}$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the volumetric flow integral

1 point is earned for the average volumetric flow

1 point is earned for answer with reason

Let C be the cross-sectional area approximation from part (c). The average volumetric flow is

$$\frac{1}{20} \int_{10}^{60} C \cdot v(t) dt = 2181.912 \text{ or } 2181.913 \text{ ft}^3/\text{min}.$$

8-4b

Yes, water must be diverted since the average volumetric flow for this 20-minute period exceeds $2100 \text{ ft}^3 / \text{min}$.